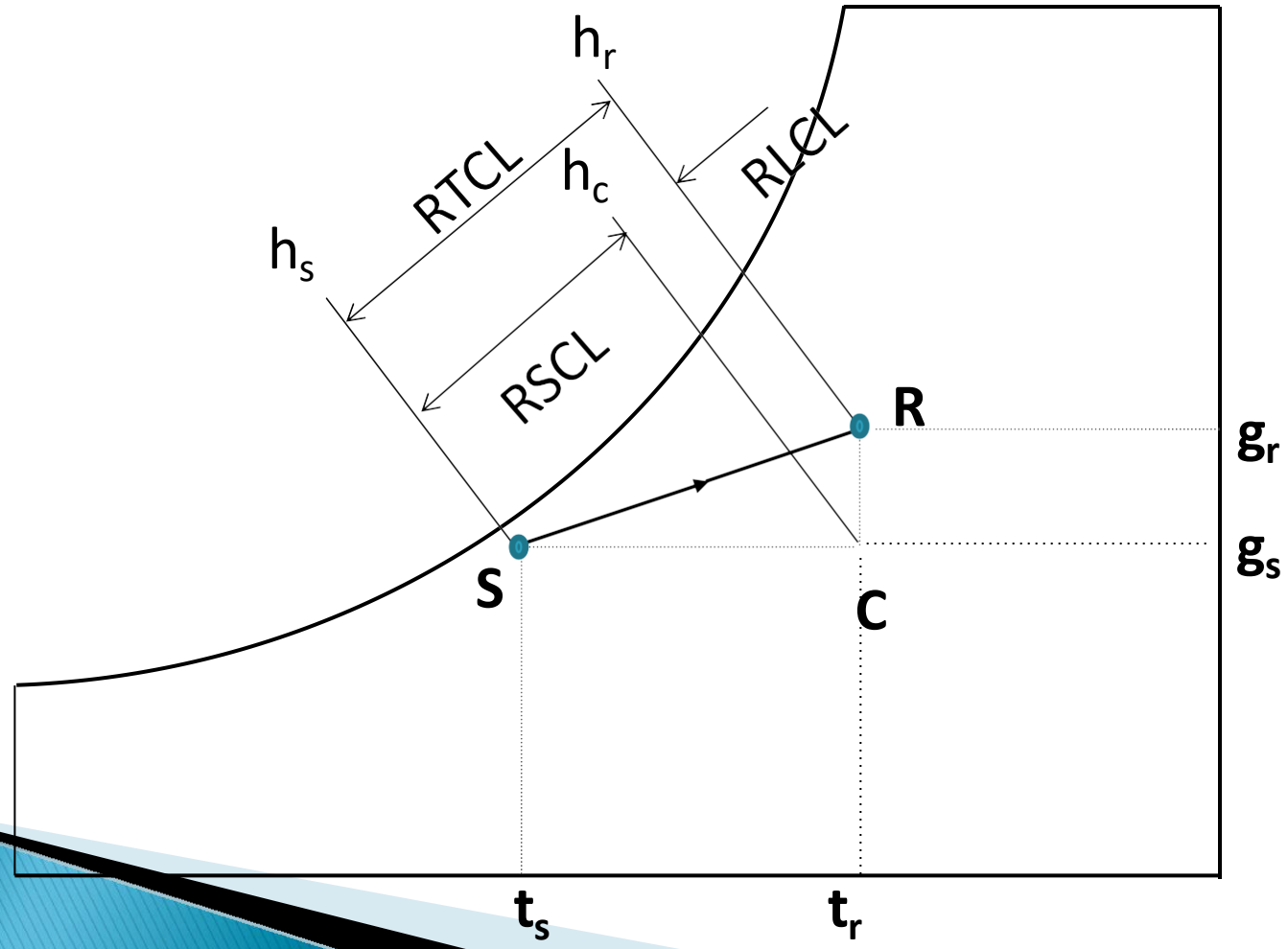
# Lesson 13 & 14

# PSYCHROMETRICS CAV, VAV & FCU

# Supply Air Condition

Chosen such that room condition will be maintained at indoor design condition.



# Supply Air Condition

- Room Sensible Cooling Load,

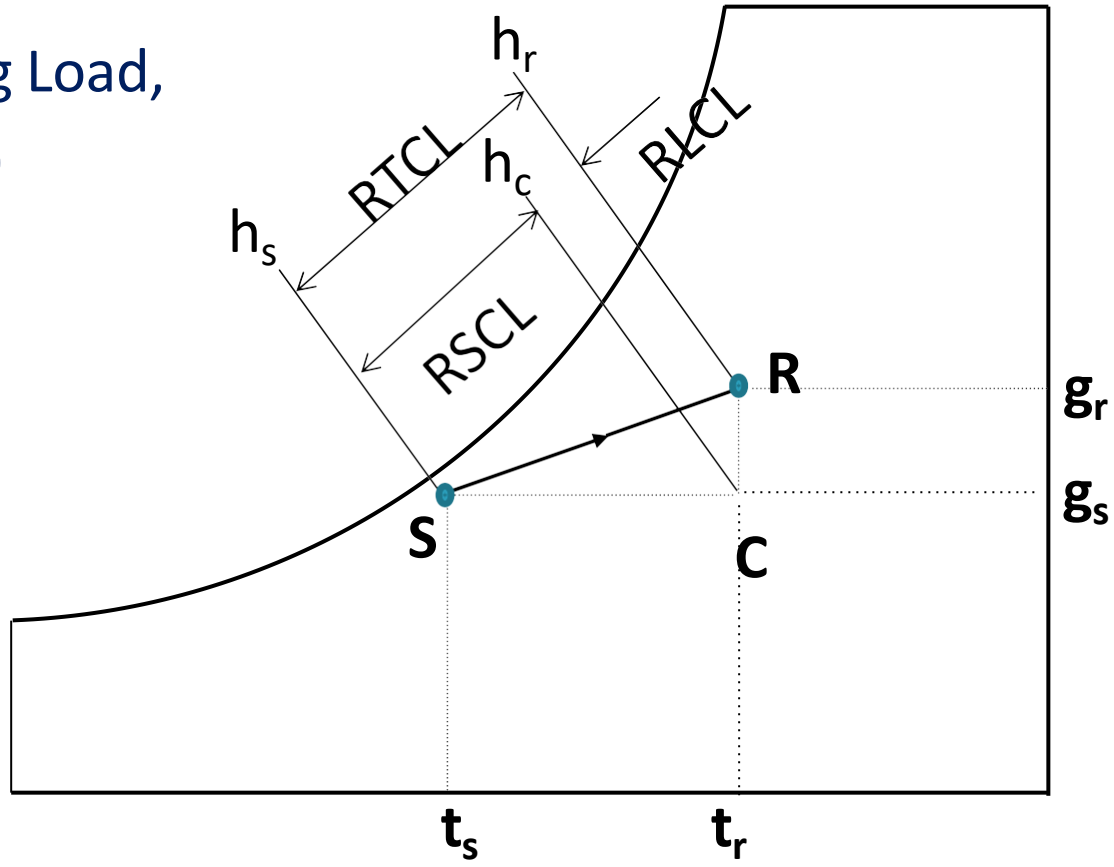
  RSCL = $\dot{m}_s . c_{pa} . (t_r - t_s)$

  $= \dot{m}_s . (h_c - h_s)$

- Room Latent Cooling Load,

  RLCL = $\dot{m}_s . h_{fg} . (g_r - g_s)$

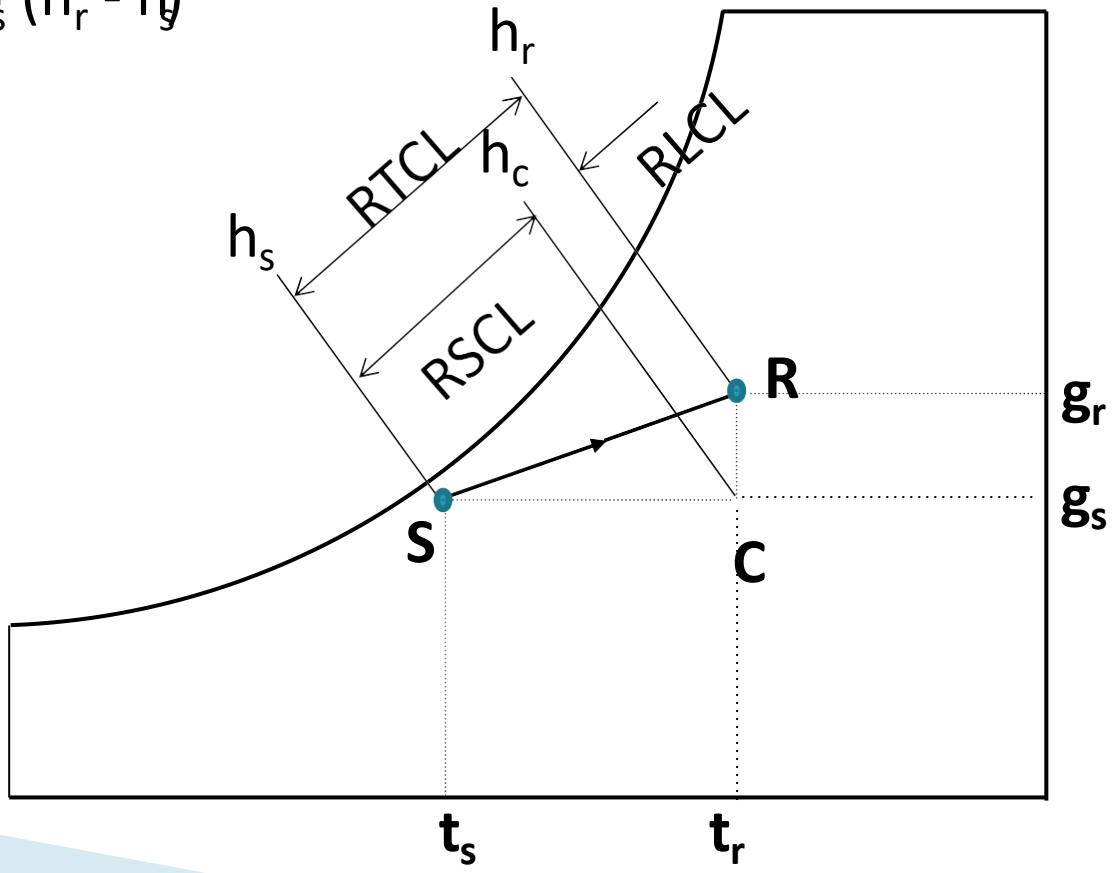
  $= \dot{m}_s . (h_r - h_c)$



# Supply Air Condition

- Room Sensible Heat Factor (RSHF)

$$= \frac{\text{RSCL}}{\text{RSCL + RLCL}} = \frac{\dot{m}_s (h_c - h_s)}{\dot{m}_s (h_r - h_s)}$$



# Supply Air Condition

$$RSCL = \dot{m}_s . C_{p,a} . (t_r - t_s)$$

- To save fan power, $t_s$ should be as low as possible.
- $t_s$ should preferably not be lower than dew point of indoor design condition to prevent condensation on air diffusers.
- $t_s$ about 8 to 11°C below indoor design temp. (i.e. $t_s$ = 14 to 17°C)

Sub-clause 7.3.2 of SS553 stipulates that the cool air leaving the supply diffuser should be designed at a temperature less than 2°C below the room dew point to prevent moisture condensing on the diffuser surface.

# Example 5.1

From cooling load estimate, the room sensible cooling load and room latent cooling load are 70 kW and 30 kW respectively. The room is to be maintained at 23°C db, 55% saturation. If the room-to-supply air temperature difference is 11 °C.

(i) Find the moisture content of the supply air using the Room Sensible Heat Factor.

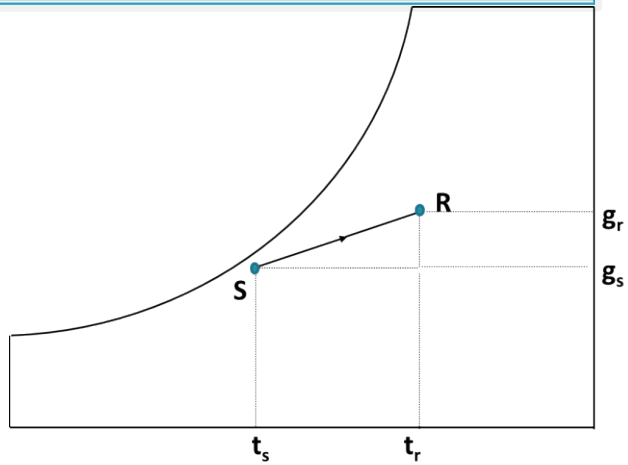
(ii) Find the mass flow rate of supply air.

# Example 5.1

**Solution**

(i) Plot Point R

$$\text{RSHF} = \frac{\text{SCL}}{\text{SCL + LCL}} = \frac{70}{70 + 30} = 0.70$$



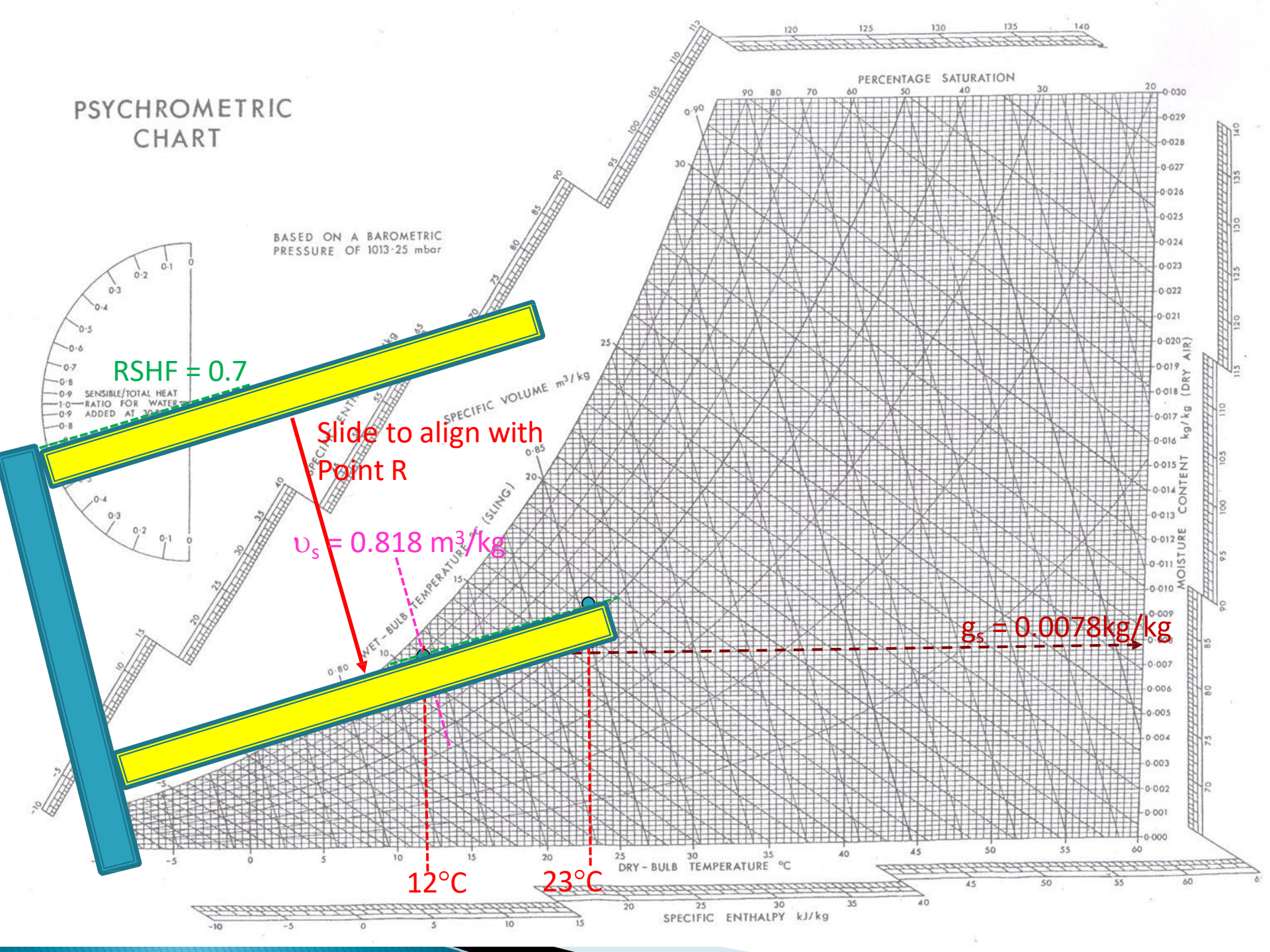

Given $t_r - t_s = 11°C$. Therefore, $t_s = t_r - 11 = 23 - 11 = 12°C$

Plot RSHF through Point R to intersect with $t_s = 12°C$ to obtain Point S.

From plot, $g_s = 0.0078$ kg/kg and $v_s = 0.818$ m³/kg

(ii) $\dot{m}_s = \frac{\text{SCL}}{c_{p,a}(t_r - t_s)} = \frac{70}{1.012 \times 11} = 6.29$ kg/s

$\dot{V}_s = \dot{m}_s \times v_s = 6.29 \times 0.818 =$ 5.15 m³/s


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE (SLING)
MOISTURE CONTENT kg/kg (DRY AIR)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
RSHF = 0.7
Slide to align with Point R
$\upsilon_s$ = 0.818 m³/kg
$g_s$ = 0.0078kg/kg
12°C
23°C

# Example 5.1

**Solution**

(i) Plot Point R

$$RSHF = \frac{SCL}{SCL + LCL} = \frac{70}{70 + 30} = 0.70$$

Alternative method of finding Point S without plotting RSHF.

$$RLCL = \dot{m}_s \times h_{fg} (g_r - g_s)$$

$$(g_r - g_s) = \frac{RLCL}{\dot{m}_s \times h_{fg}}$$

$$g_s = g_r - \frac{RLCL}{m_s \times h_{fg}}$$

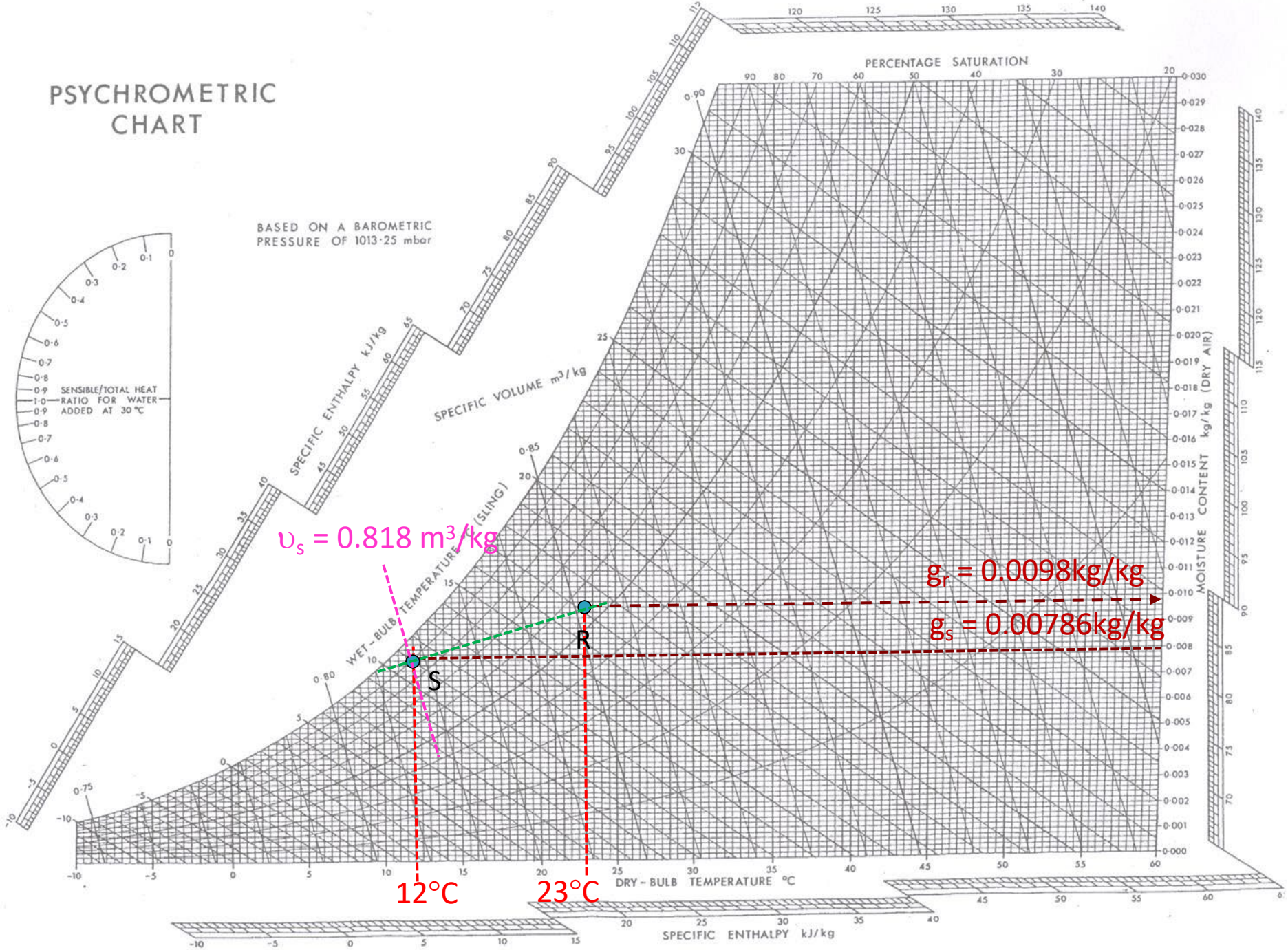
# Example 5.1

**Solution**

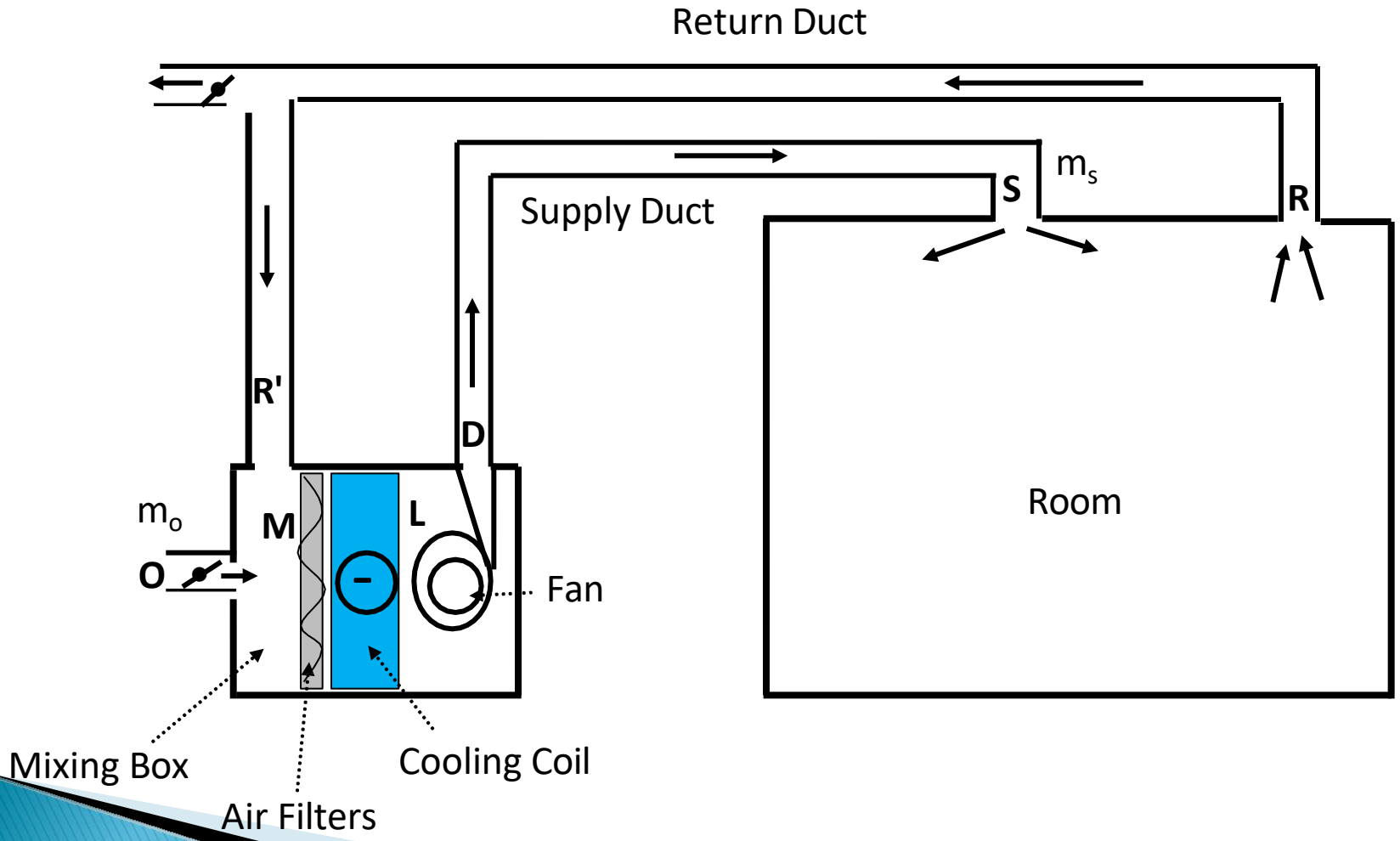
From plot $g_r = 0.0098$ kg/kg

$$g_s = 0.0098 - \frac{30}{6.29 \times 2454} = 0.0098 - 0.00194 = 0.00786 \text{ kg/kg}$$

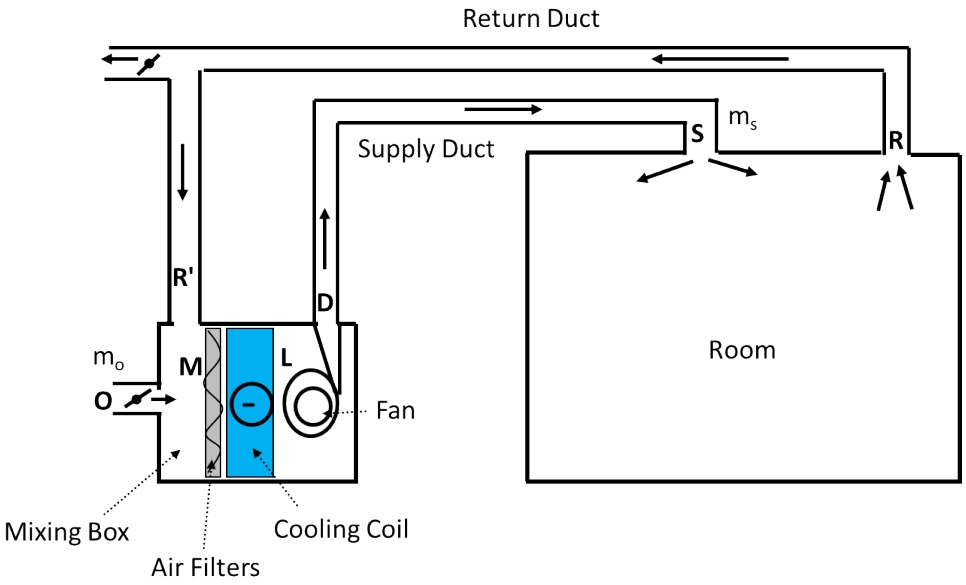
Plot $g_s = 0.00786$ kg/kg to intersect with $t_s$ to obtain Point S.

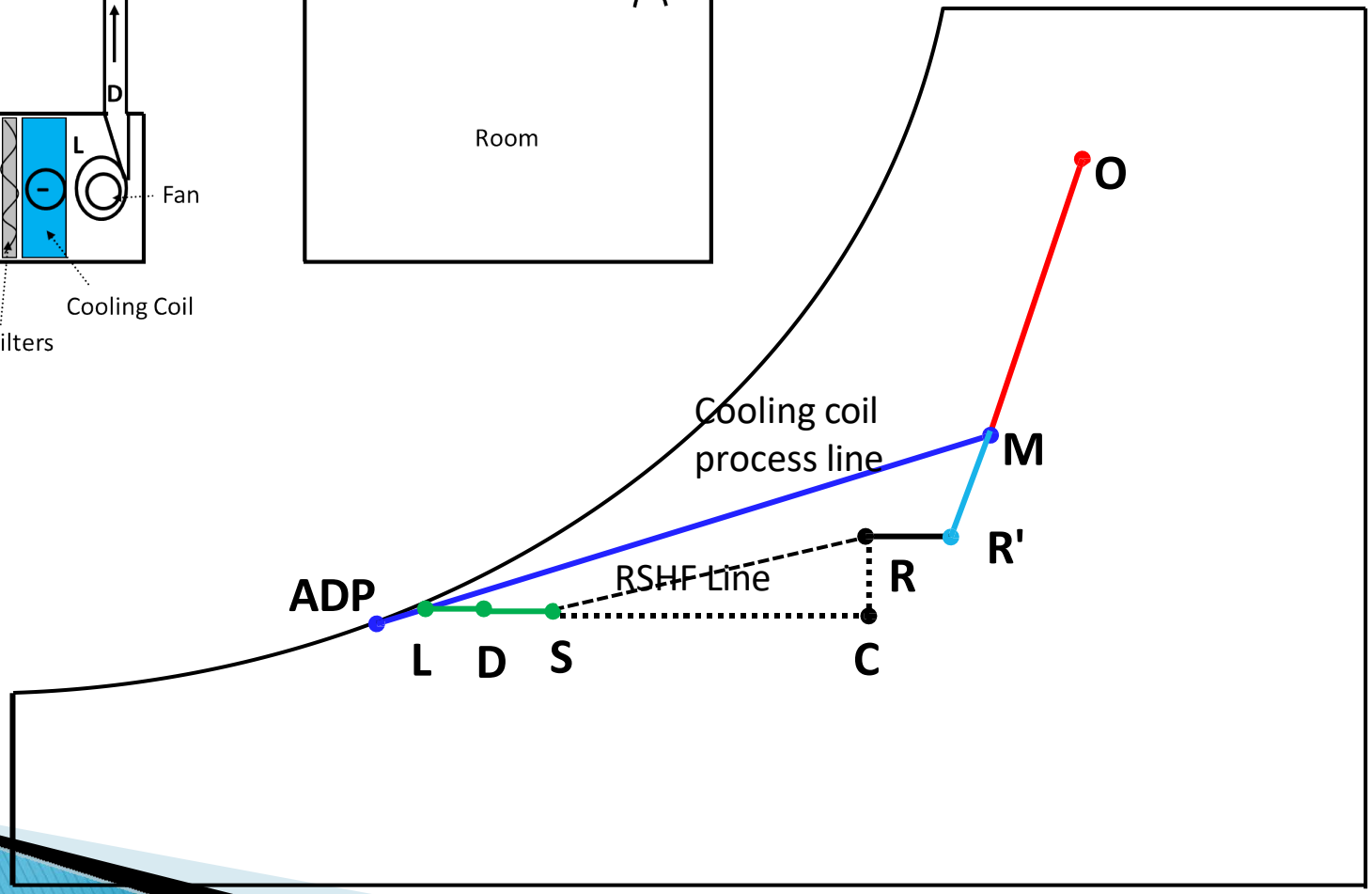

PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30 °C
SPECIFIC ENTHALPY kJ/kg
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE (SLING)
PERCENTAGE SATURATION
MOISTURE CONTENT kg/kg (DRY AIR)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
υs = 0.818 m³/kg
gr = 0.0098kg/kg
gs = 0.00786kg/kg
R
S
12°C
23°C

# Single Zone CAV



# Single Zone CAV



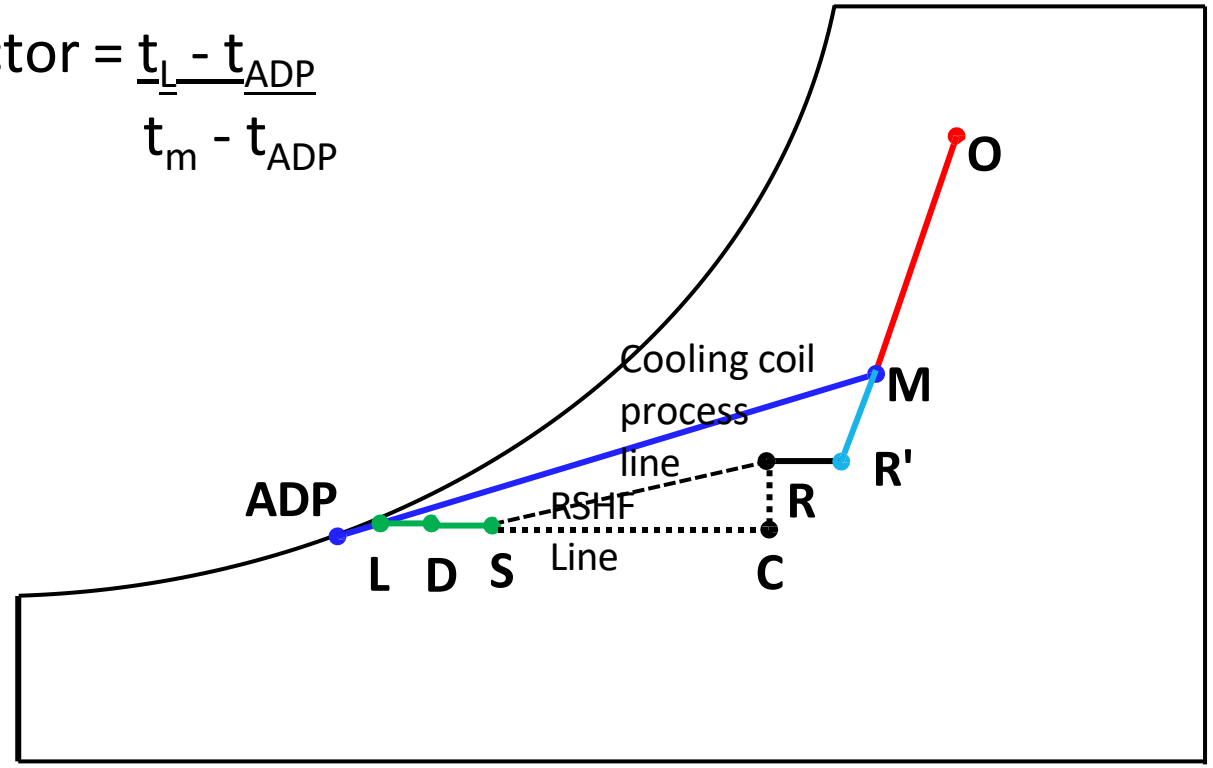



# Mixing of Two Air Streams

- Temperature of mixed air,

$$t_m = (\dot{m}_o/\dot{m}_s).t_o + (\dot{m}_r/\dot{m}_s).t_r \quad ℃$$

- Bypass Factor = $\frac{t_L - t_{ADP}}{t_m - t_{ADP}}$



# Components of Cooling Coil Load

- Room Sensible Cooling Load,

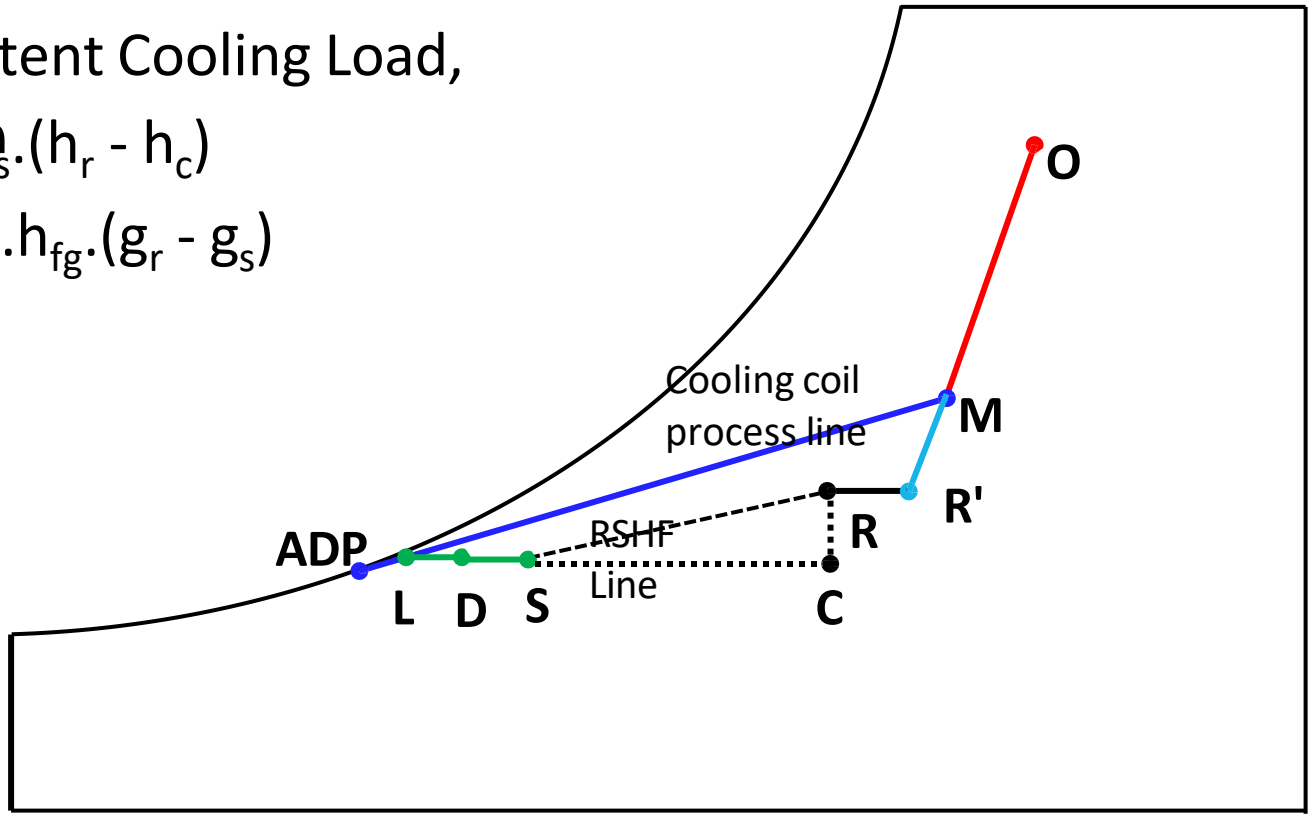
  SCL = $\dot{m}_s.(h_c - h_s)$

  $= \dot{m}_s.c_{p,a}.(t_r - t_s)$

- Room Latent Cooling Load,

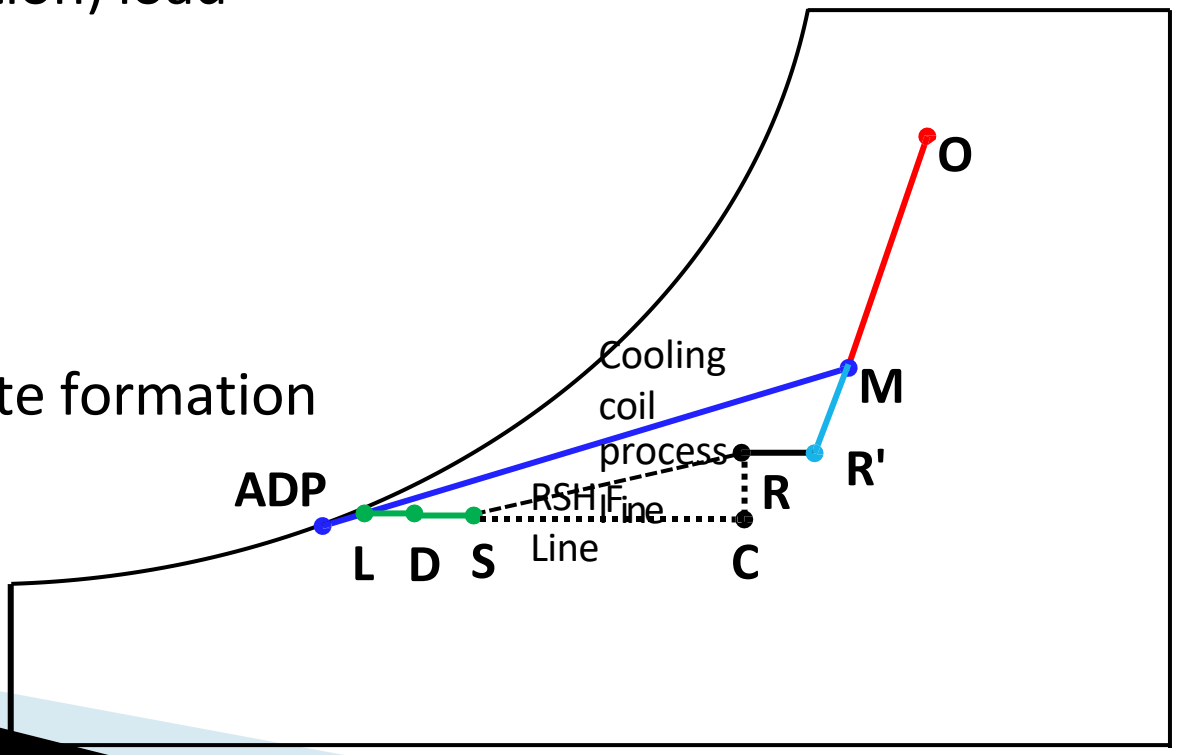
  LCL = $\dot{m}_s.(h_r - h_c)$

  $= \dot{m}_s.h_{fg}.(g_r - g_s)$



# Components of Cooling Coil Load

- Room Fan Heat Gain (sensible)
  $= \dot{m}_s . c_{p,a} . (t_d - t_l)$ or $\dot{m}_s (h_d - h_l)$ kW
- Supply Duct Heat Gain (sensible)
  $= \dot{m}_s . c_{p,a} . (t_s - t_d)$ or $\dot{m}_s (h_s - h_d)$ kW
- Fresh air (Ventilation) load
  $= \dot{m}_o (h_o - h_r)$ kW
- Cooling coil load
  $= \dot{m}_s (h_m - h_l)$ kW
- Rate of condensate formation
  $= \dot{m}_s (g_m - g_l)$ kg/s



# Partial Load of CAV Single-Zone System

- Equipment selected for peak load operation.
- Operation mostly at partial load conditions.
- Reduction in sensible or latent loads
- To simplify discussion, heat gains due to fans and air ducts are ignored in the partial load analysis of CAV systems.

# Supply Conditions

- Reduction in cooling load

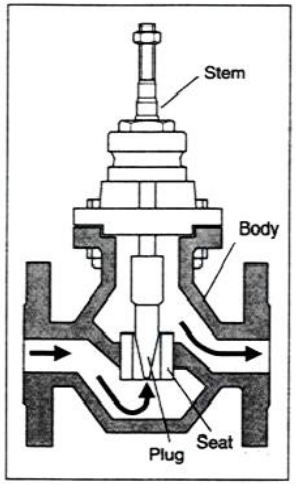
  - room temp maintained by supplying air at a higher dry-bulb temp.

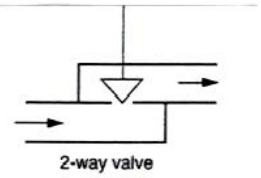
- Methods to change supply air temperature.

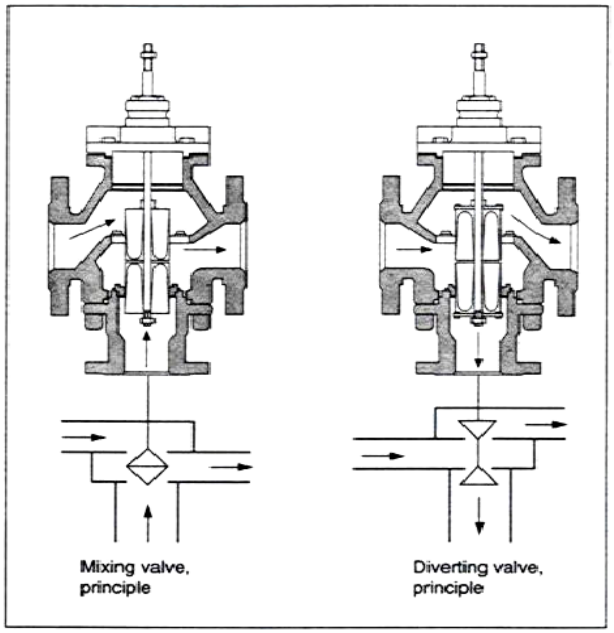
  - Control of apparatus dew point (ADP) of coil
  - Bypassing mixed air
  - Bypassing room air
  - Reheating after cooling

# Control of ADP

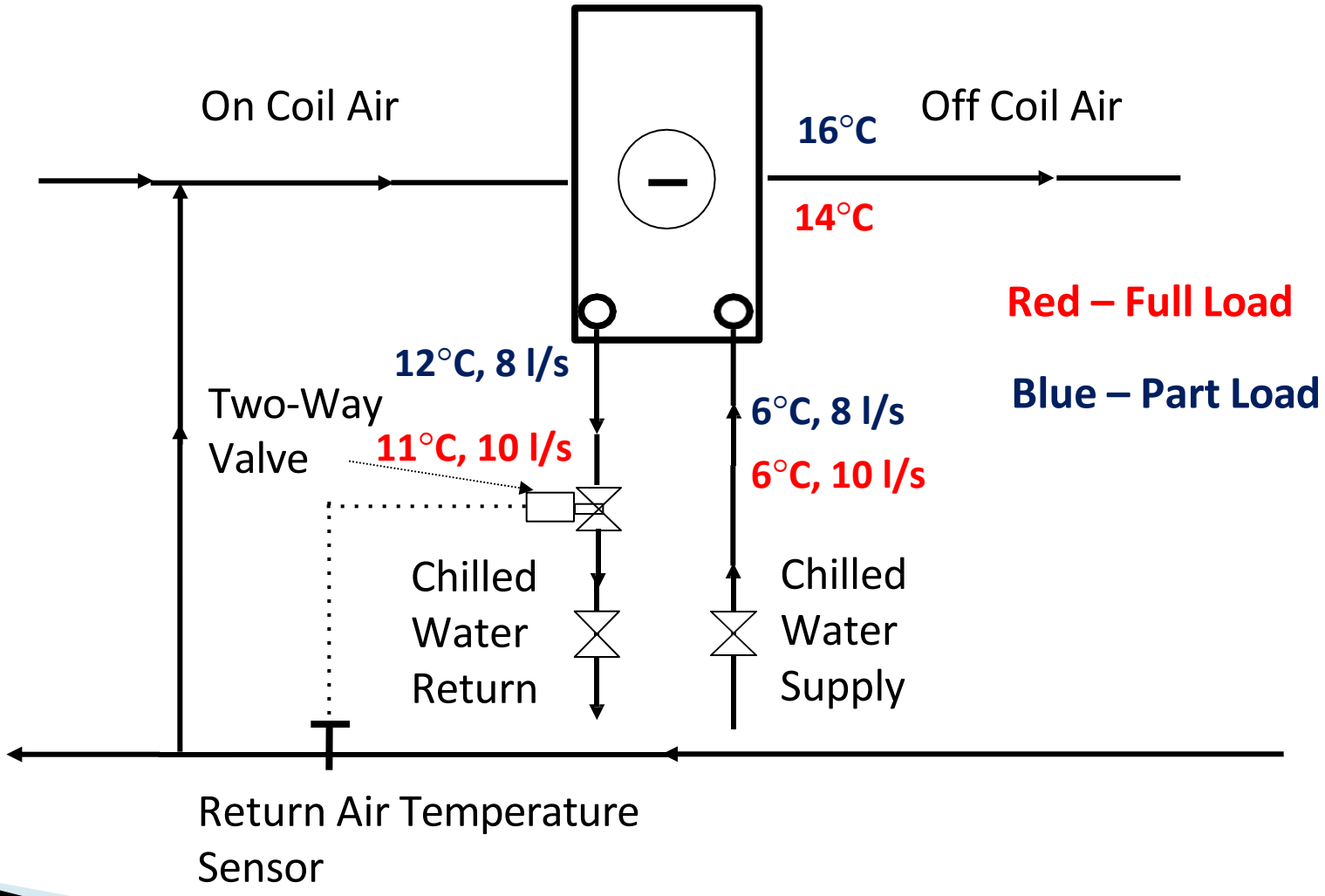
- 2-way modulating valve reduces flow through coil by throttling flow.
- 3-way valve bypasses part of chilled water around coil.



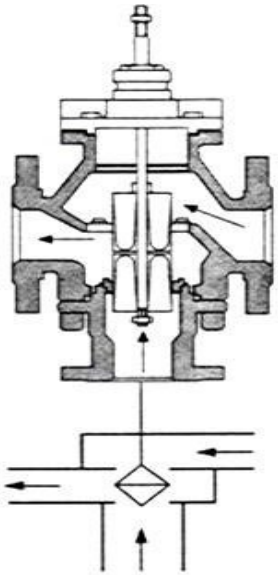



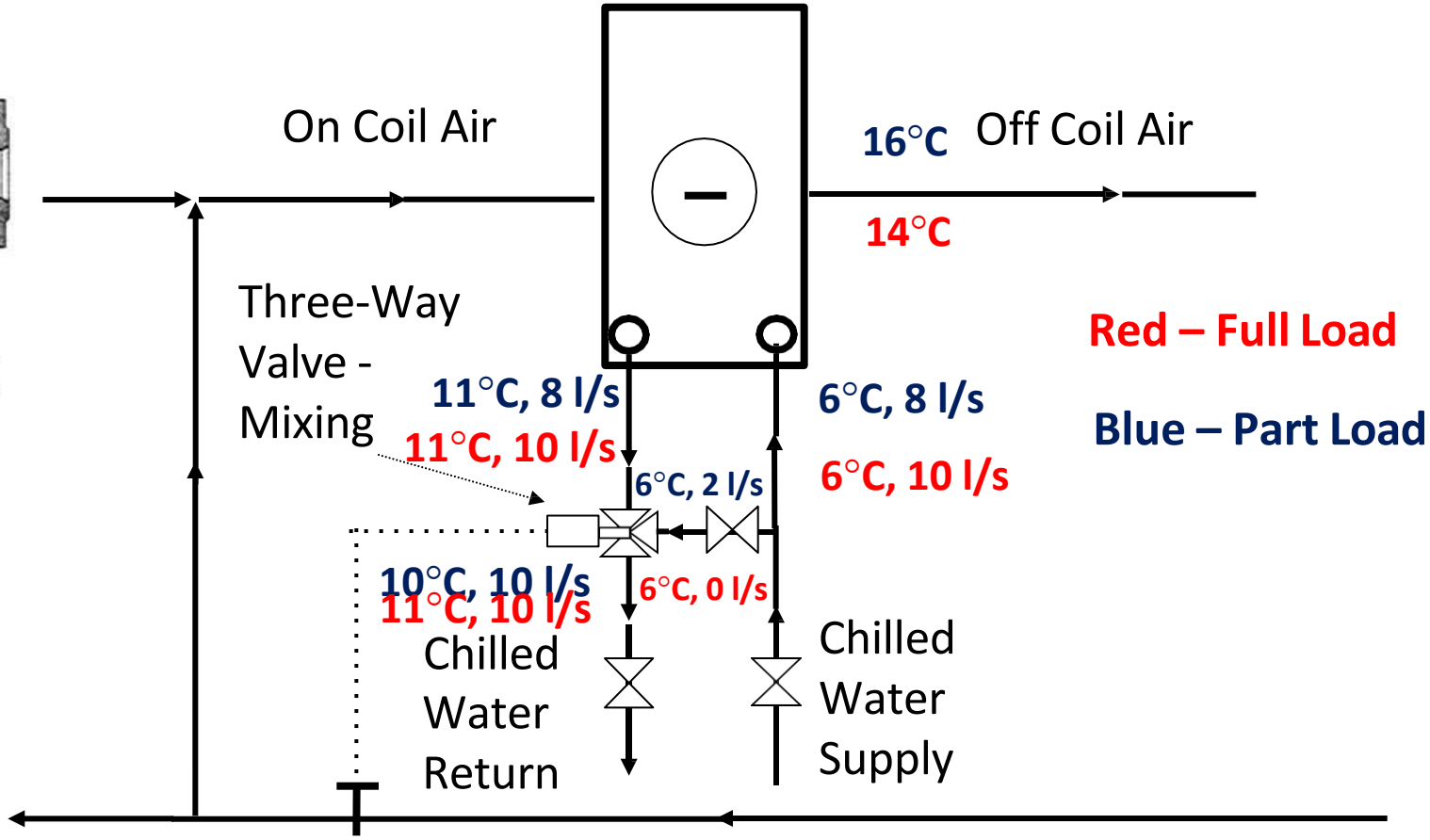



# Two-way Valve Control

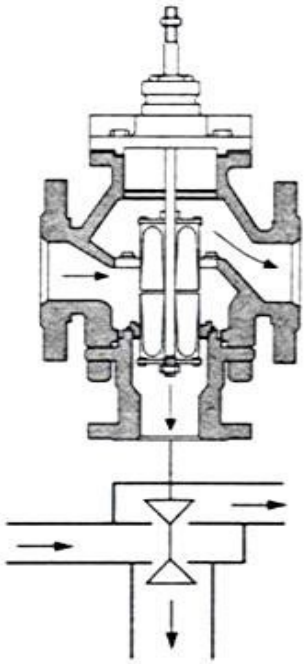


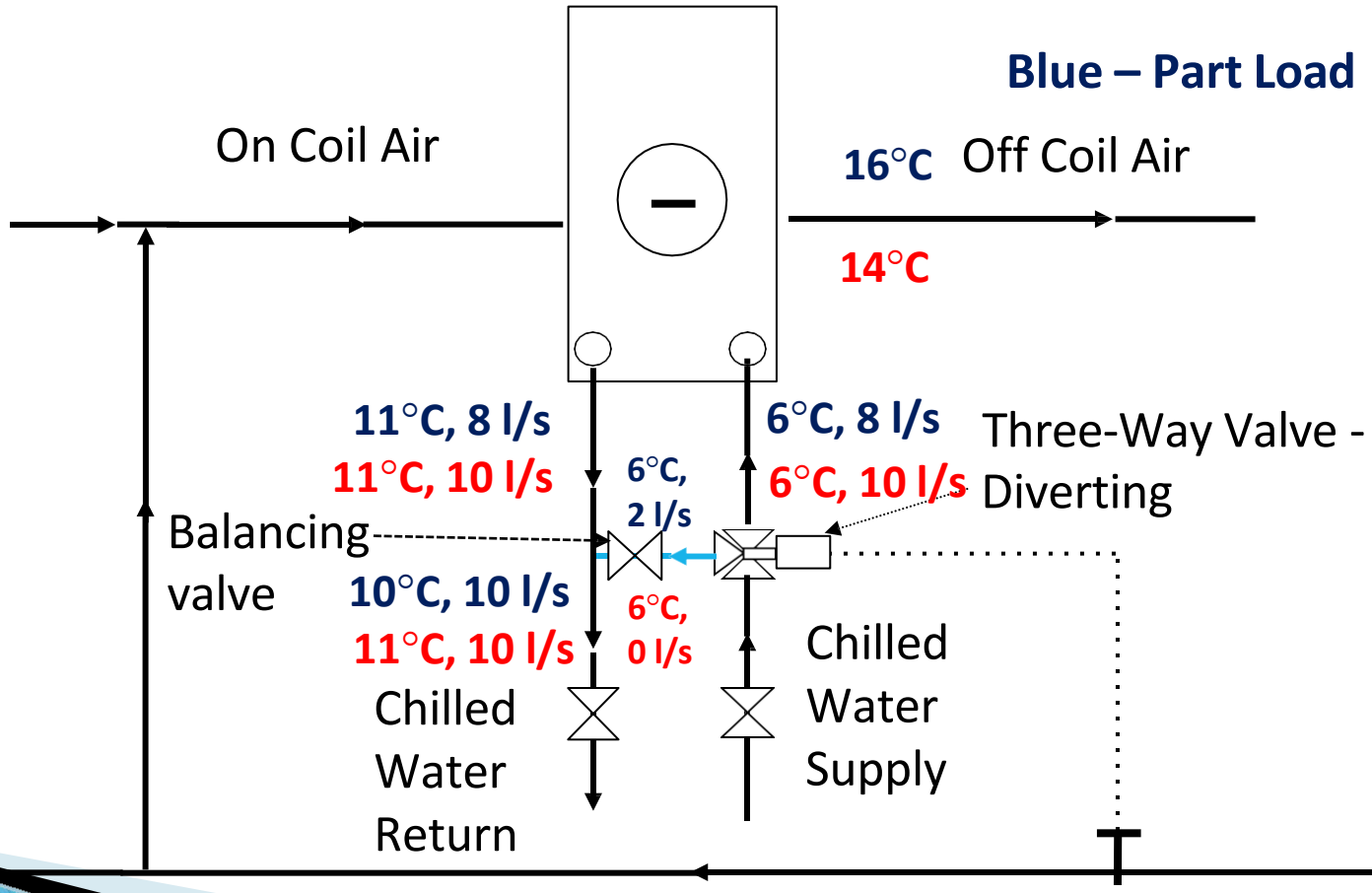
# Three-way Mixing Valve Control





# Three-way Diverting Valve Control

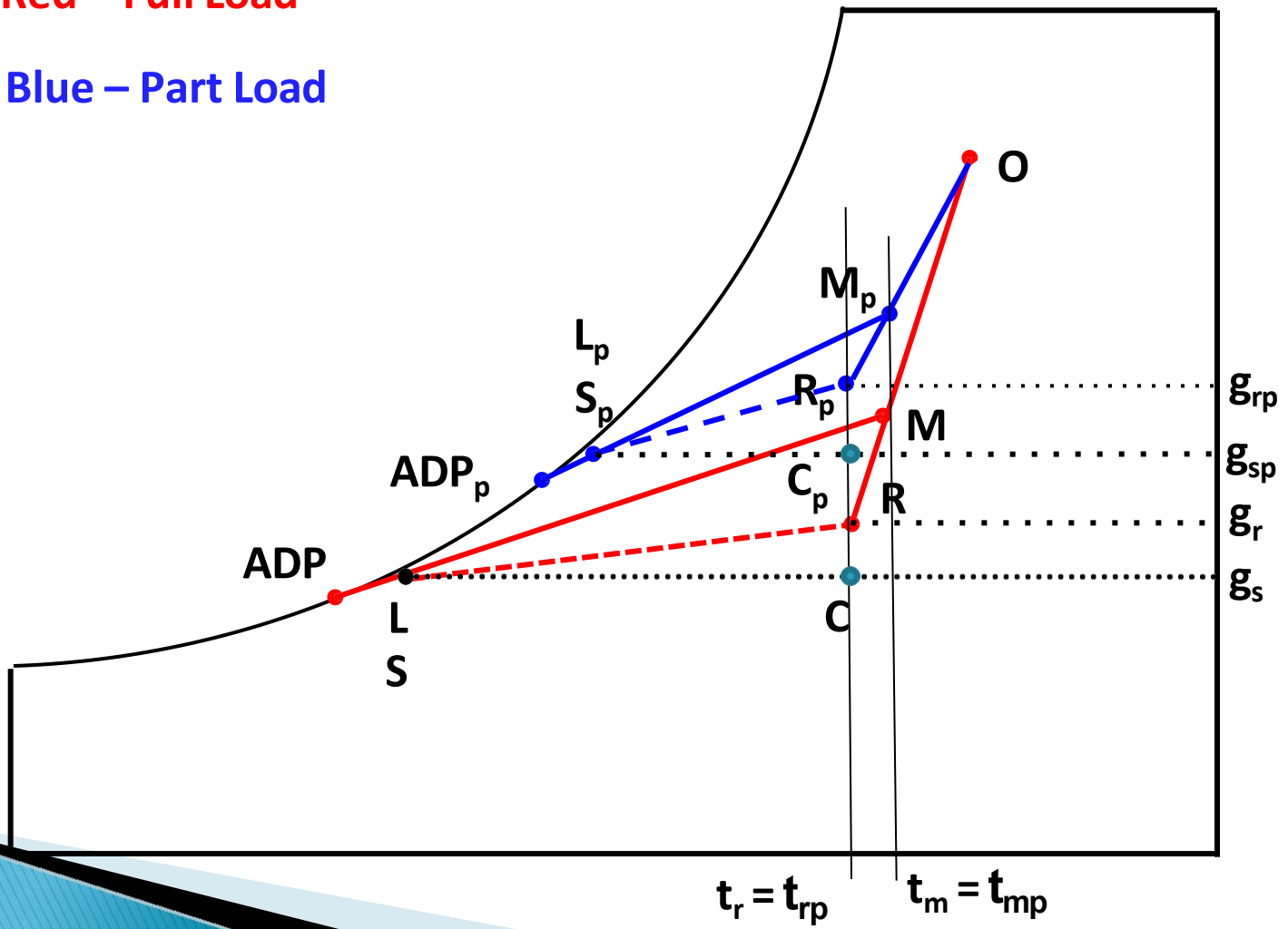




# Control of ADP

Red – Full Load

Blue – Part Load



# Control of ADP

- At Partial Load:

- Room Sensible Cooling Load, $SCL_p$

$= \dot{m}_s . c_{pa} . (t_{rp} - t_{sp}) = \dot{m}_s (h_{cp} - h_{sp})$

- Room Latent Cooling Load, $LCL_p$

$= \dot{m}_s . h_{fg} . (g_{rp} - g_{sp}) = \dot{m}_s (h_{rp} - h_{cp})$

- Refrigeration load required to produce chilled water will be reduced as flow rate of chilled water decreases. Energy will be saved during partial load.

# Example 5.2

A zone in a building is conditioned by a single zone constant air volume air handling unit using a two-way chilled water valve controlled by a return air temperature sensor to modulate the chilled water flow through the cooling coil.

The air handling unit has the following peak condition.

Outdoor air condition : 33°C db, 27°C wb
Room air condition : 24°C db, 50% satn
Peak room sensible cooling load : 75 kW
Room latent cooling load : 30 kW
Room-supply air temp differential : 12°C
Mass ratio of outside air to supply air : 0.15
Fan and duct heat gains equivalent to : 1 K

# Example 5.2

(i) Sketch a schematic of the system.

Determine:

(ii) the mass flowrate of supply air.

(iii) on-coil air dry bulb temperature.

(iv) moisture content of supply air,

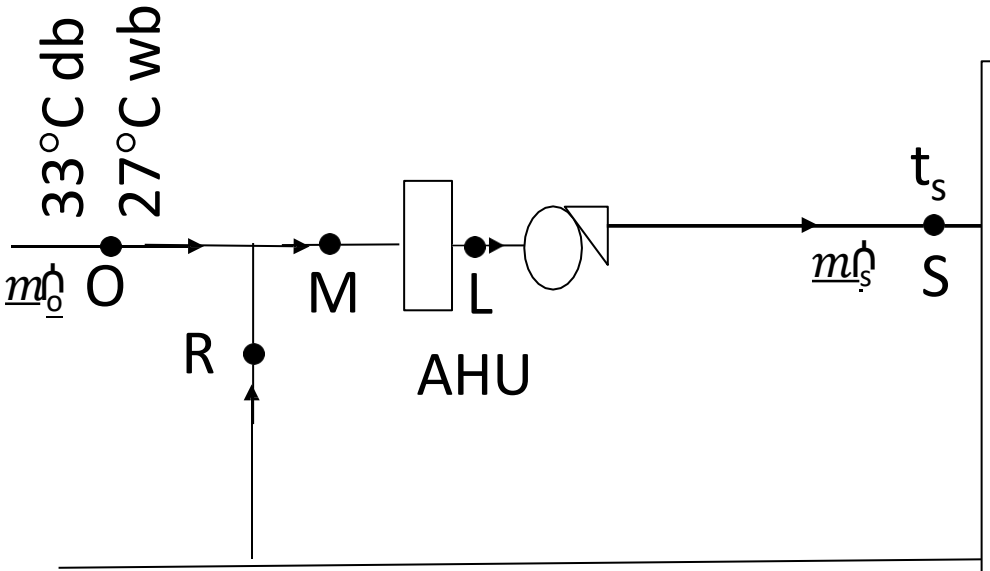
(v) apparatus dew point, bypass factor,

(vi) peak cooling capacity of each typical floor, air handling unit and

(vii) rate of condensate formation.

(viii) the supply air temperature when room sensible cooling load reduces by 30% from the peak condition with the room latent cooling load remaining unchanged.

# Example 5.2




RSCL = 75 kW

RLCL = 30 kW

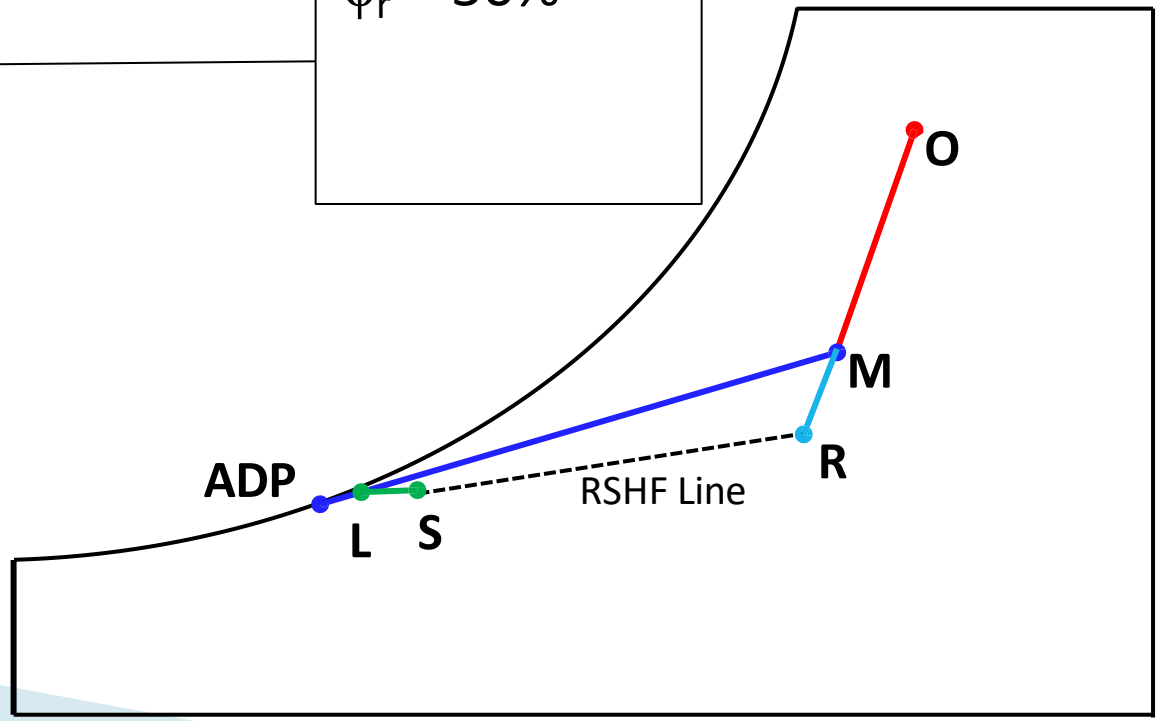
$t_r$ = 24°C

$\phi_r$ = 50%

$t_r - t_s = 12°C,$

$t_s - t_l = 1°C,$

$\frac{\dot{m}_o}{\dot{m}_s}$ = = 0.15

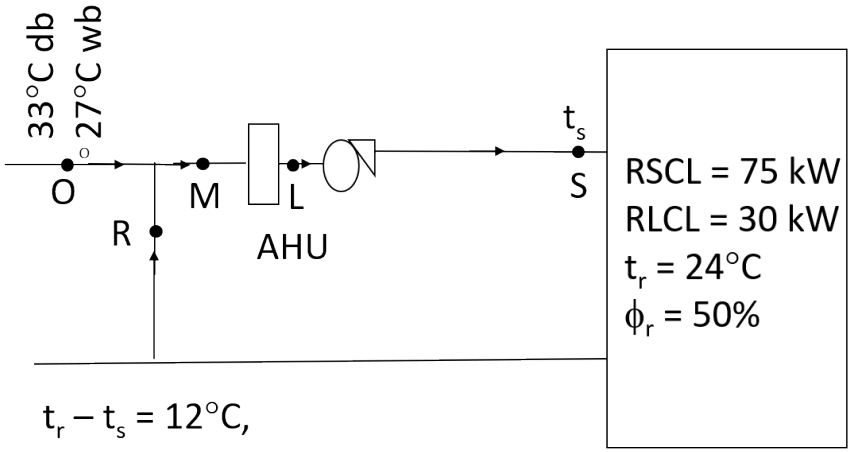


# Example 5.2

(ii) $\dot{m}_s$

RSCL = $\dot{m}_s . c_{p,a}.(t_r - t_s)$

$$\dot{m}_s = \frac{RSCL}{c_{p,a}(t_r - t_s)} = \frac{75}{1.012(12)} = 6.18 \text{ kg/s}$$




$t_r - t_s = 12°C$,

$t_s - t_l = 1°C$,

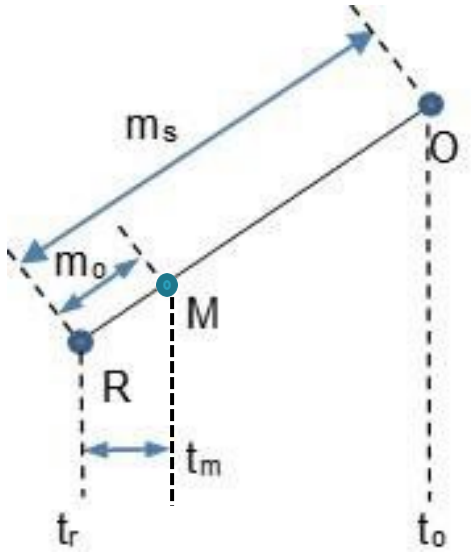
$\frac{\dot{m}_o}{\dot{m}_s} = = 0.15$

# Example 5.2

(iii) $\frac{\dot{m}_o}{\dot{m}_s} = \frac{t_m - t_r}{t_o - t_r} = 0.15$

$t_m = t_r + 0.15\,(t_o - t_r) = 24 + 0.15\,(33 - 24)$

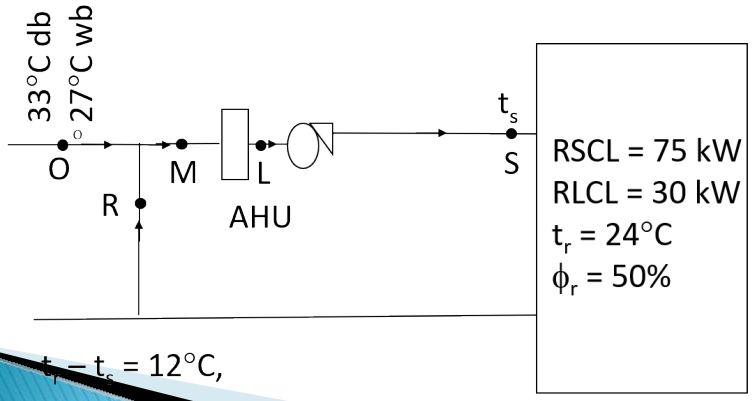
$= 25.35°C$

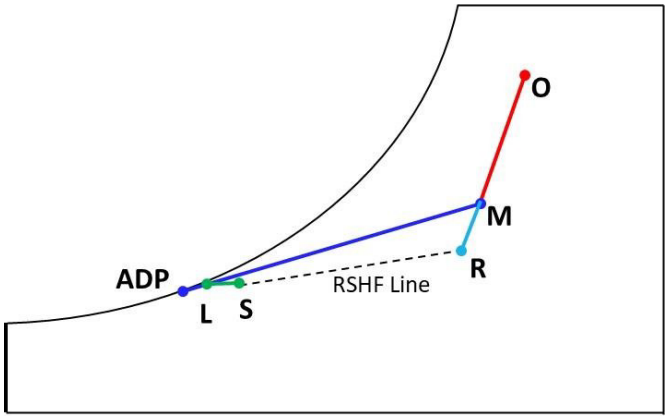



Plot Point O and Point R.

Draw line to join Point O and R.

Plot $t_m$ to intersect with line O-R to obtain Point M.




$t_r - t_s = 12°C$,

$t_s - t_l = 1°C$,

$\frac{\dot{m}_o}{\dot{m}} = = 0.15$

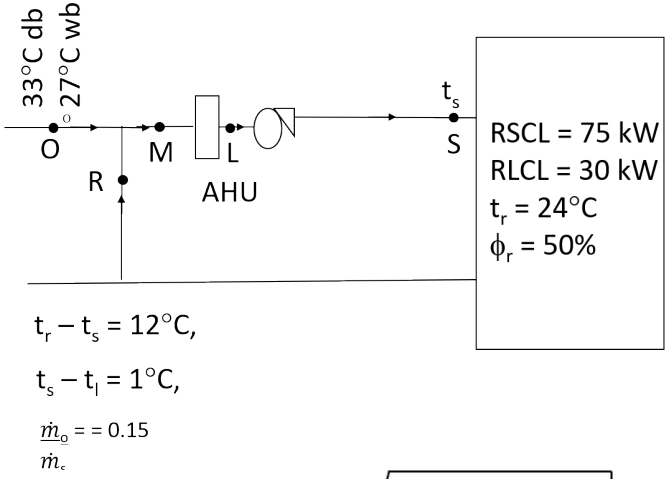
# Example 5.2

(iv) $g_s$

Given : $t_r - t_s = 12°C$

$\therefore \quad t_s = t_r - 12 = 24 - 12 = 12°C$

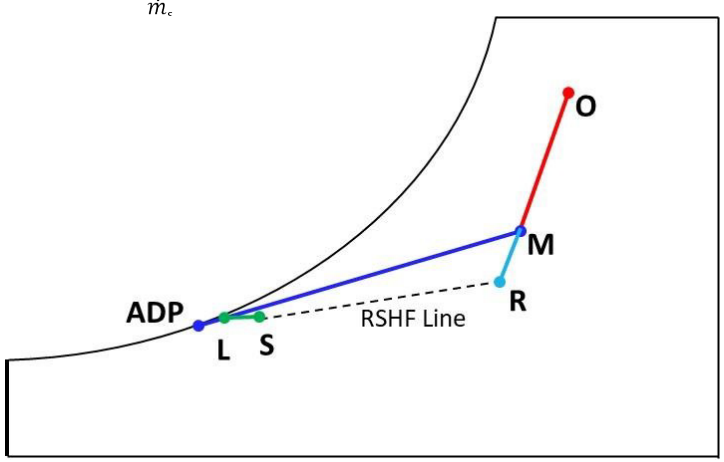
$$RSHF = \frac{SCL}{SCl + LCL} = \frac{75}{75 + 30} = 0.714$$




Plot $t_s$.

Plot RSHF through Point R to intersect with $t_s$ to obtain Point S.

From plot, $g_s$ = 0.0076 kg/kg




(v) ADP and Bypass Factor

Given : $t_s - t_l = 1°C$

Off-coil air temperature, $t_l = t_s - 1.0 = 12 - 1.0 = 11°C$

# Example 5.2

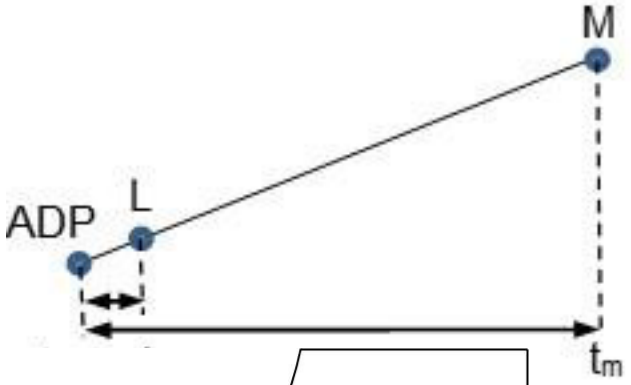
Plot $t_l$.

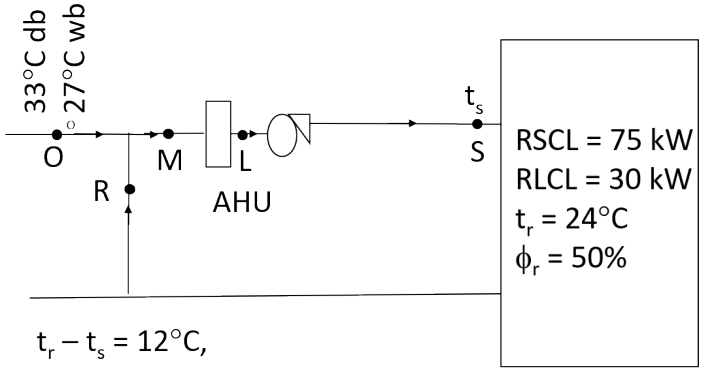
Plot a constant moisture content line through Point S to intersect with $t_l$ to obtain Point L.

Draw line from Point M to Point L. Extend line to intersect with 100% saturation line to obtain ADP.

From plot, ADP is 8.5°C

$$BF_1 = \frac{t_L - t_{ADP}}{t_m - t_{ADP}} = \frac{11 - 8.5}{25.35 - 8.5} = 0.148$$

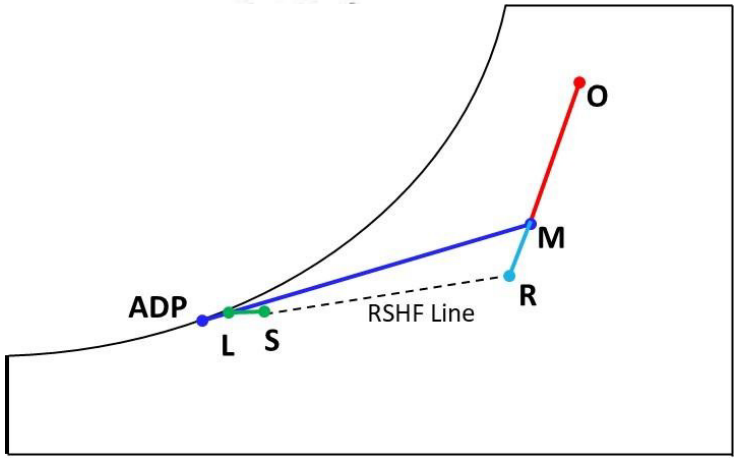



$t_r - t_s = 12°C$,

$t_s - t_l = 1°C$,

$\frac{\dot{m}_o}{\dot{m}_s} = = 0.15$



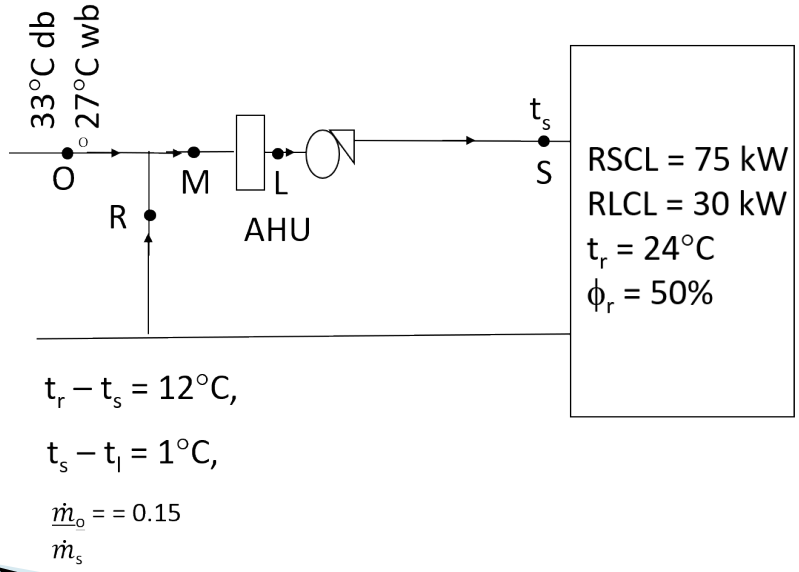
# Example 5.2

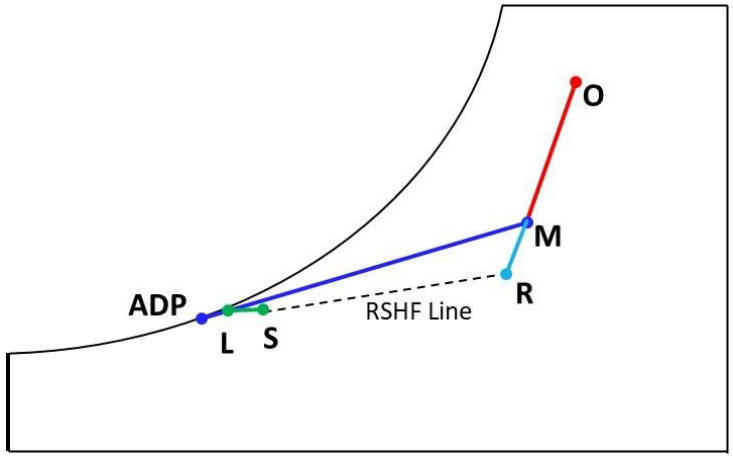
(vi) Peak Cooling Capacity of AHU

Peak cooling capacity of AHU = $\dot{m}_s(h_m - h_l)$ kW

From plot, $h_m$ = 54.0 kJ/kg, $h_l$ = 30.0 kJ/kg

Peak cooling capacity of AHU = 6.18(54 – 30) = 148.32 kW



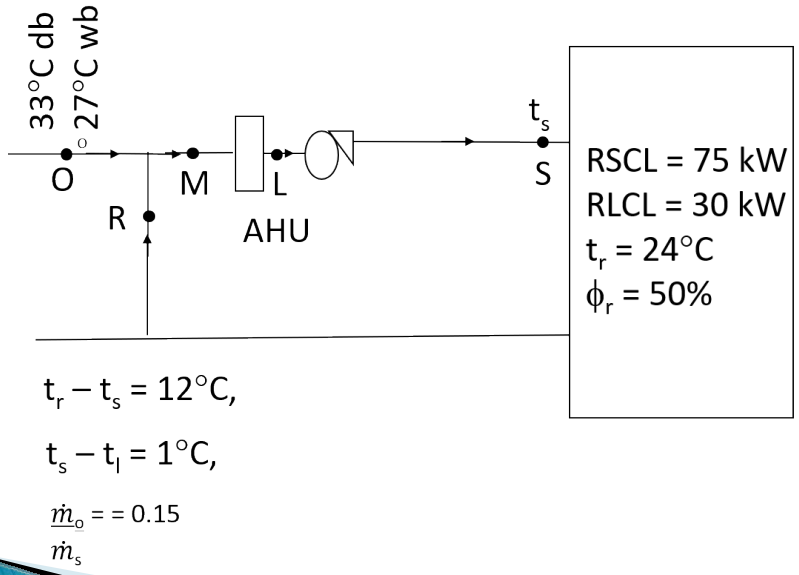
# Example 5.2

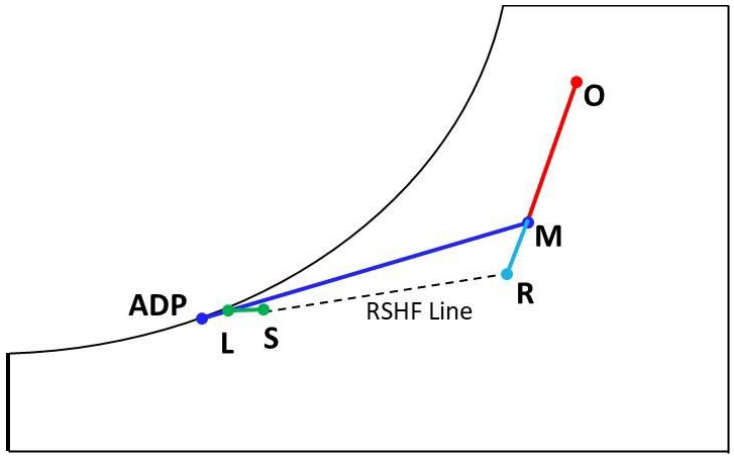
(vii) Rate of condensate formation

Rate of condensate formation = $\dot{m}_s(g_m - g_l)$ kg/s

From plot, $g_m = 0.0111$ kg/kg, $g_l = 0.0076$ kg/kg

Rate of condensate formation = $6.18(0.0111 - 0.0076)$
$= 0.0216$ kg/s



# Example 5.2

(viii) $t_{s,p}$

$$SCL_p = \frac{(100 - 30)}{100} \times 75 = 52.5 \text{ kW}$$

$$SCL_p = \dot{m}_{s,p} \times c_{p,a} \times (t_{r,p} - t_{s,p})$$

$$t_{r,p} - t_{s,p} = \frac{SCL_p}{\dot{m}_{s,p} \times c_{p,a}}$$

$$t_{s,p} = t_{r,p} - \frac{SCL_p}{\dot{m}_{s,p} \times c_{p,a}} = 24 - \frac{52.5}{6.18 \times 1.012} = 15.6°C$$

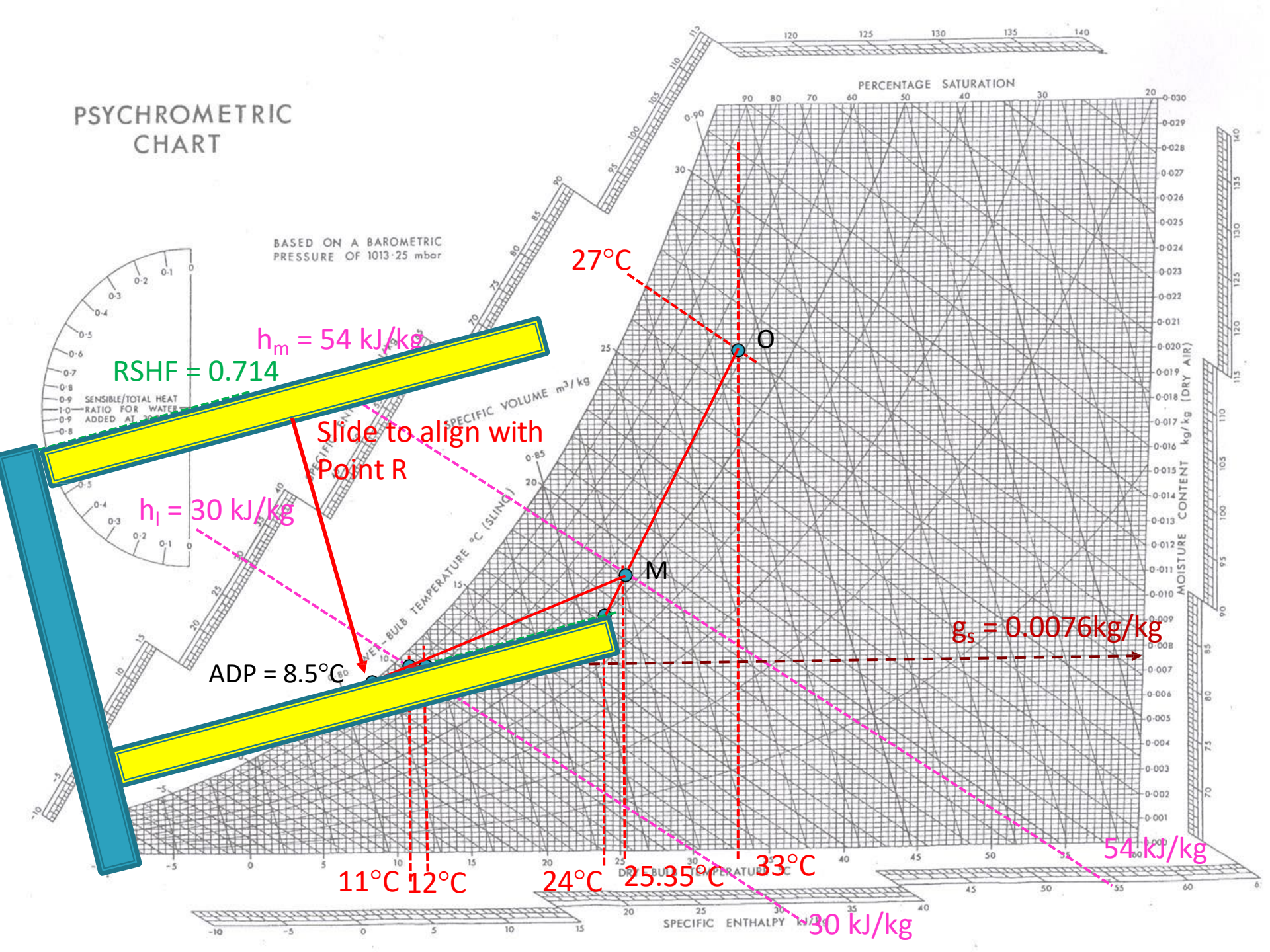
(viii) $t_{s,p}$

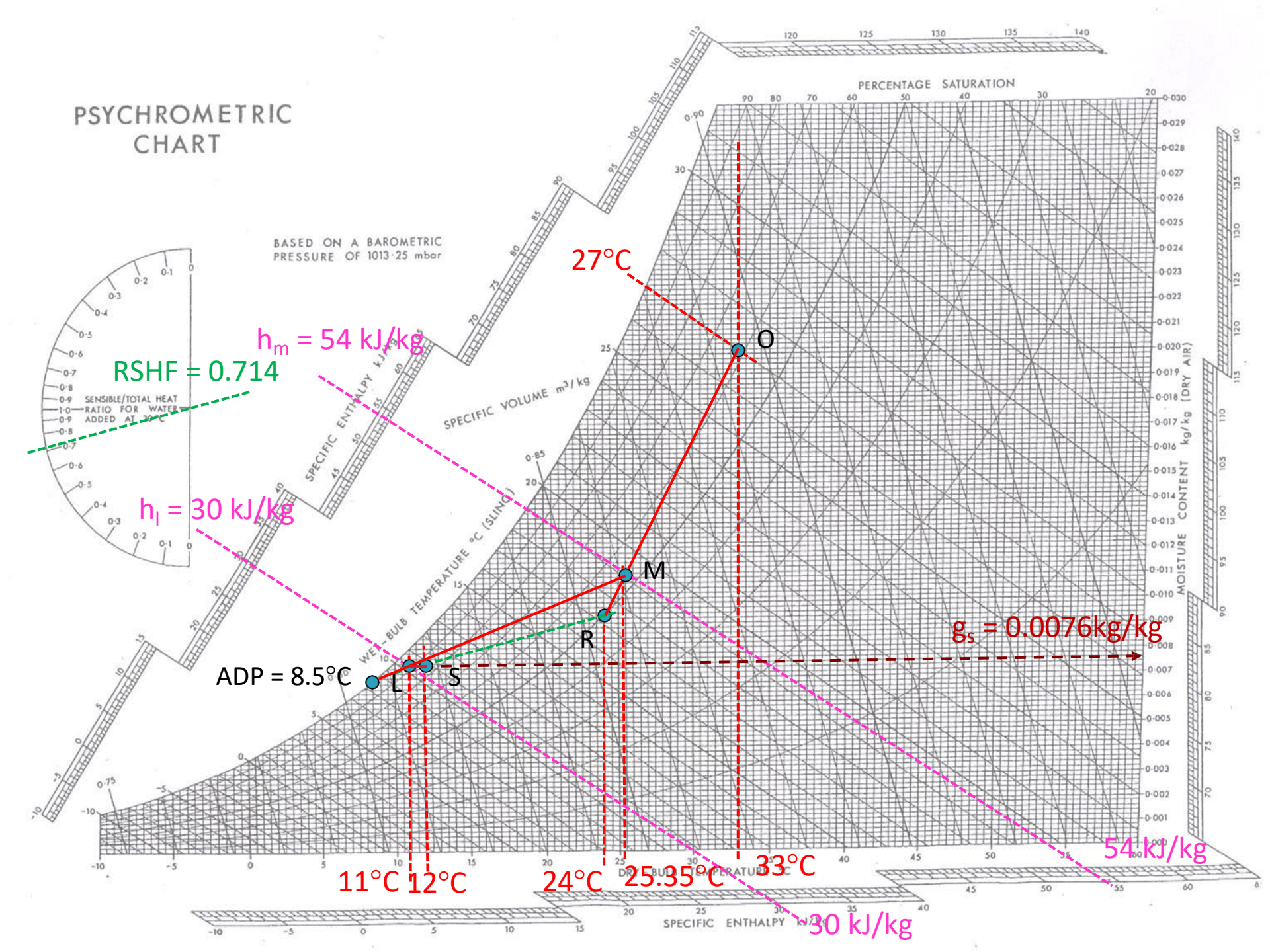
$SCL_p = \frac{(100 - 30)}{100} \times 75 = 52.5$ kW

$SCL_p = \dot{m}_{s,p} \times c_{p,a} \times (t_{r,p} - t_{s,p})$

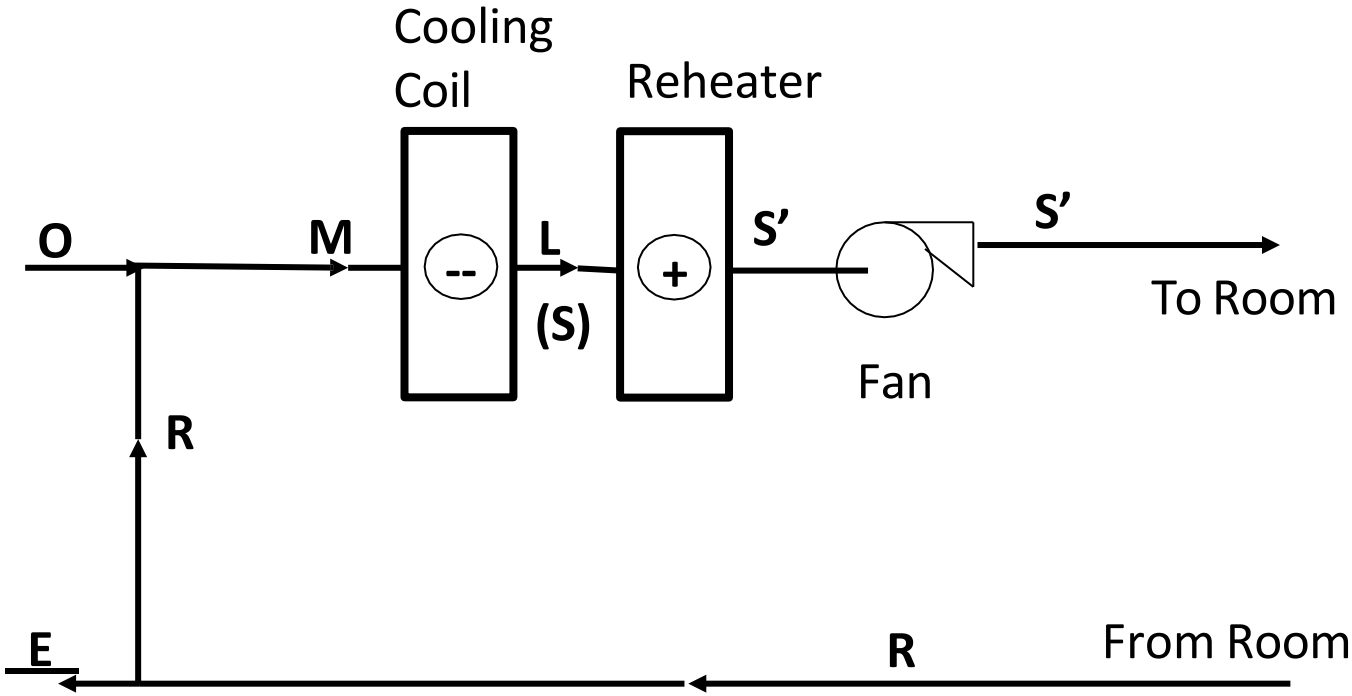
$t_{r,p} - t_{s,p} = \frac{SCL_p}{\dot{m}_{s,p} \times c_{p,a}}$

$t_{s,p} = t_{r,p} - \frac{SCL_p}{\dot{m}_{s,p} \times c_{p,a}} = 24 - \frac{52.5}{6.18 \times 1.012} = 15.6°C$


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE °C (SLING)
MOISTURE CONTENT kg/kg (DRY AIR)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
$h_m$ = 54 kJ/kg
RSHF = 0.714
Slide to align with Point R
$h_l$ = 30 kJ/kg
ADP = 8.5°C
27°C
O
M
$g_s$ = 0.0076kg/kg
11°C
12°C
24°C
25.35°C
33°C
54 kJ/kg
30 kJ/kg


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SPECIFIC VOLUME m³/kg
SPECIFIC ENTHALPY kJ/kg
WET-BULB TEMPERATURE °C (SLING)
MOISTURE CONTENT kg/kg (DRY AIR)
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30°C
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
27°C
O
$h_m$ = 54 kJ/kg
RSHF = 0.714
$h_l$ = 30 kJ/kg
M
R
$g_s$ = 0.0076kg/kg
ADP = 8.5°C
L
S
11°C
12°C
24°C
25.35°C
33°C
54 kJ/kg
30 kJ/kg

# Reheat



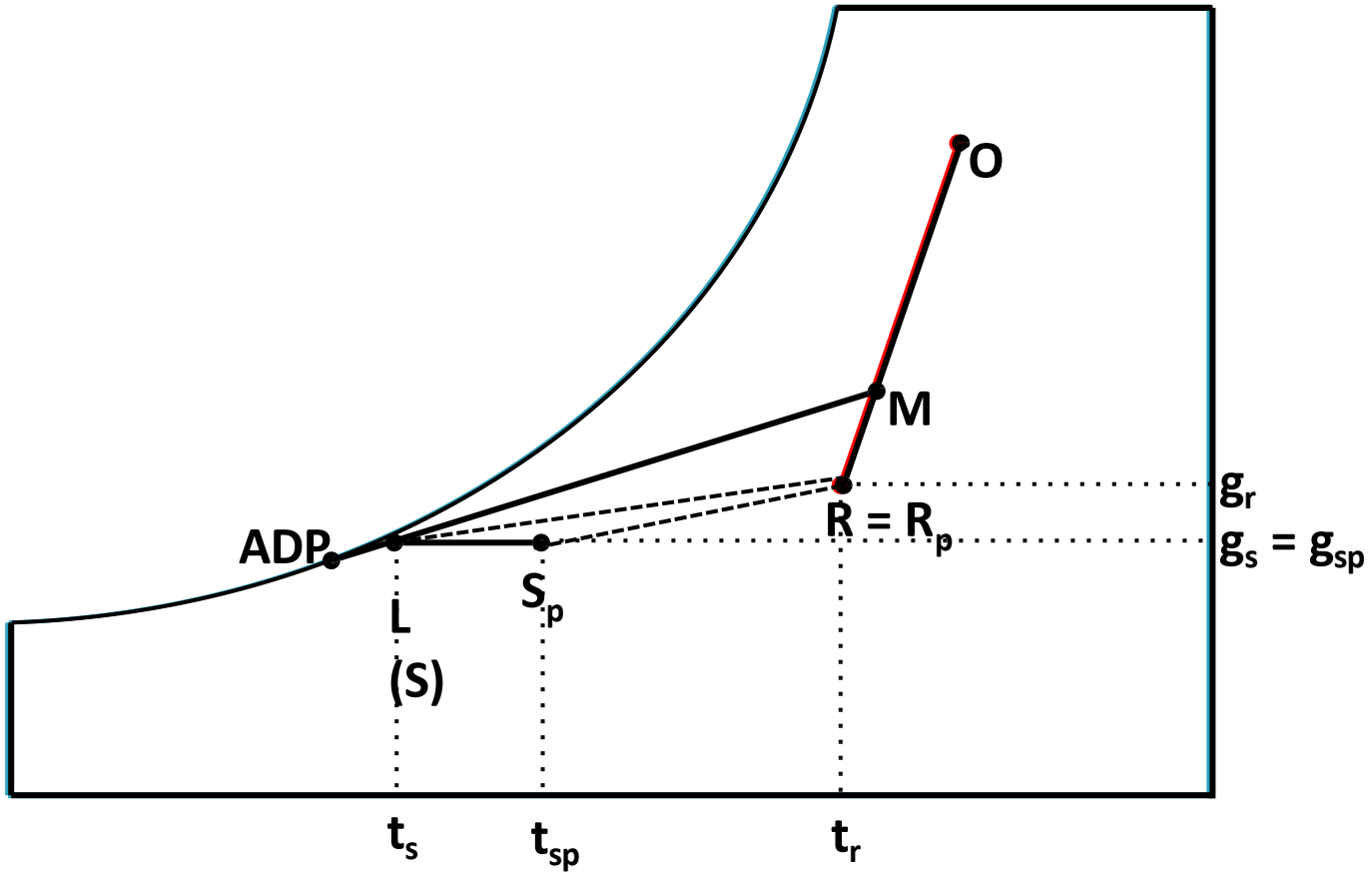

- Artificial sensible load to compensate for reduction in sensible load.
- After being cooled by cooling coil, air will then be passed through a reheater coil to increase its temperature.

# Reheat

- No reduction of cooling requirement during partial load, but an increase consumption of energy required by reheat coil (or heater).

**More energy is consumed during partial load.**

# Reheat




- Except for reheat, all the other three methods are <u>not</u> suitable for close humidity control.

# Zoning Requirements

Take a simple floor with east and west exposures.

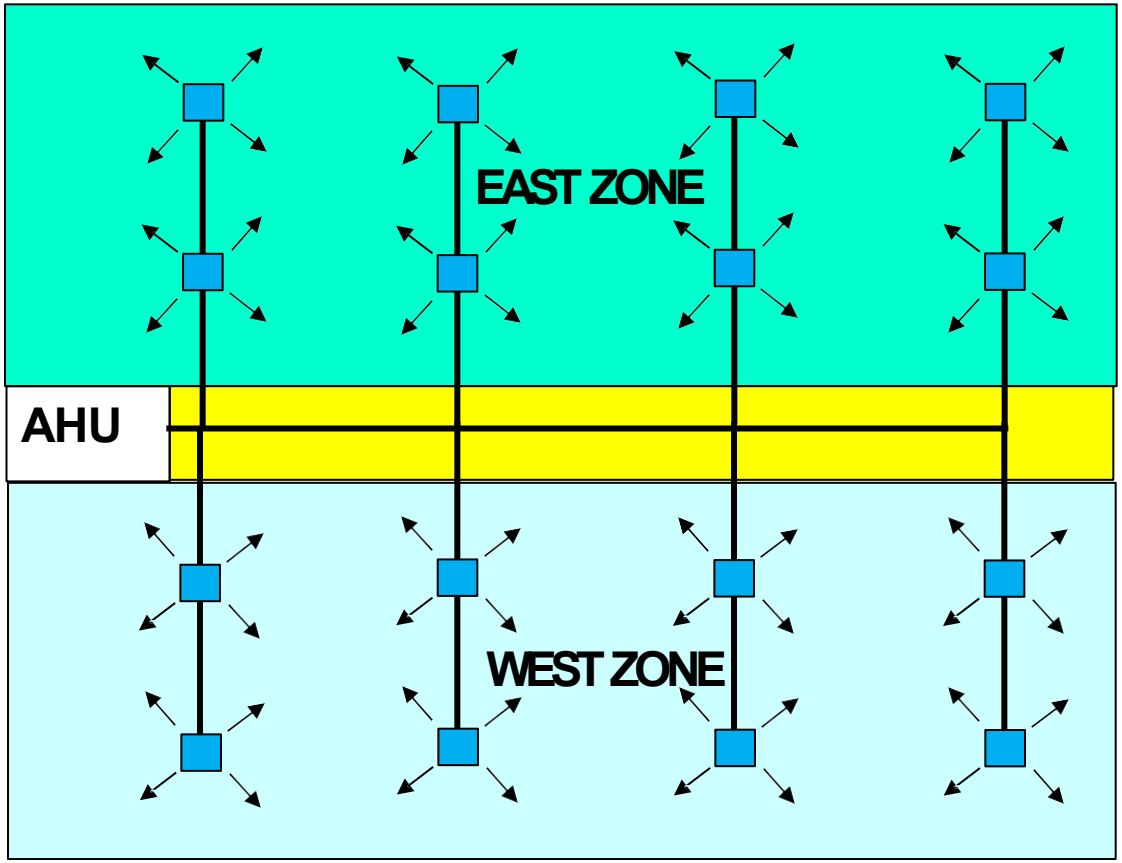
Sensible Cooling Load, kW

| Time / Zone | 10 am | 12 noon | 2 pm | 4 pm |
|---|---|---|---|---|
| **East** | 11.58 | 8.10 | 8.10 | 7.61 |
| **West** | 6.95 | 6.95 | 11.20 | 14.71 |
| **Total** | 18.53 | 15.05 | 19.30 | 22.32 |

# Zoning Requirements

- CAV single-zone system - common duct installed to serve both east and west areas as a single space.

| Zone \ Time | 10 am | 12 noon | 2 pm | 4 pm |
|---|---|---|---|---|
| East | 11.58 | 8.10 | 8.10 | 7.61 |
| West | 6.95 | 6.95 | 11.20 | 14.71 |
| Total | 18.53 | 15.05 | 19.30 | 22.32 |

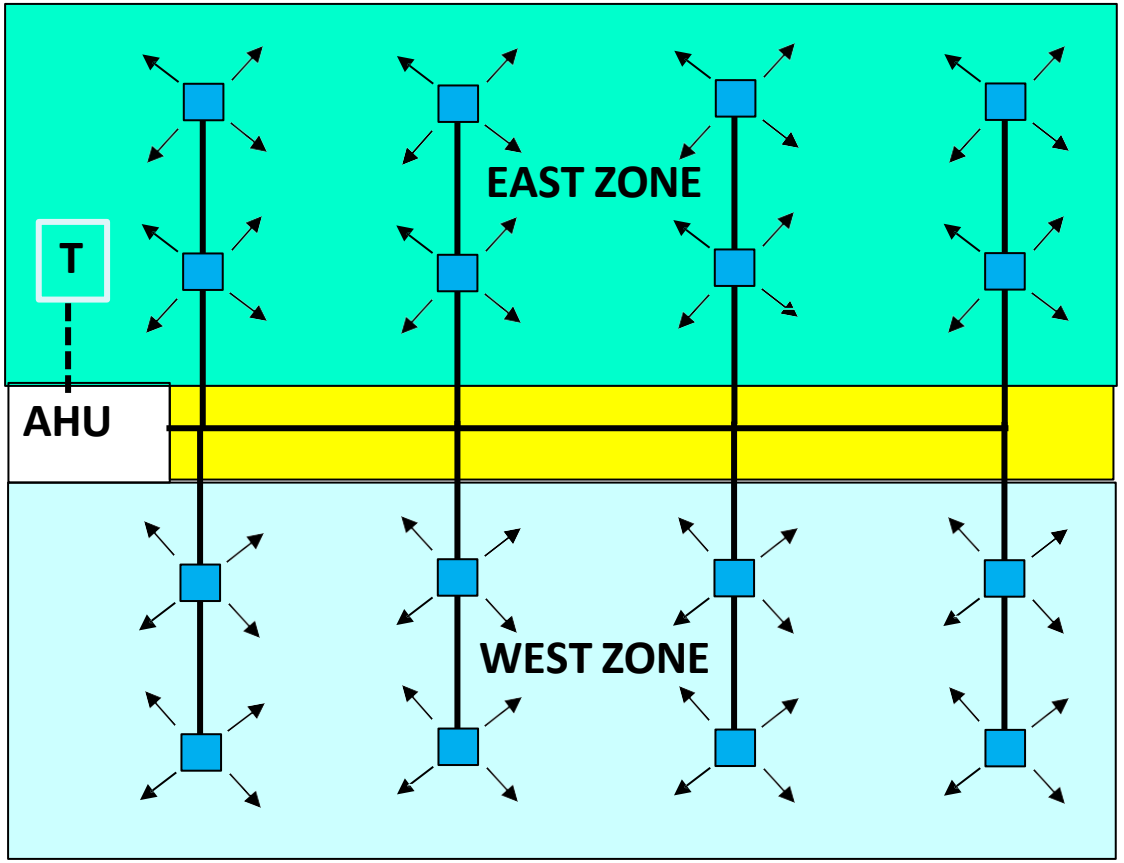


# Zoning Requirements

- Thermostat in East section

  West zone overcooled in morning and too warm in afternoon.

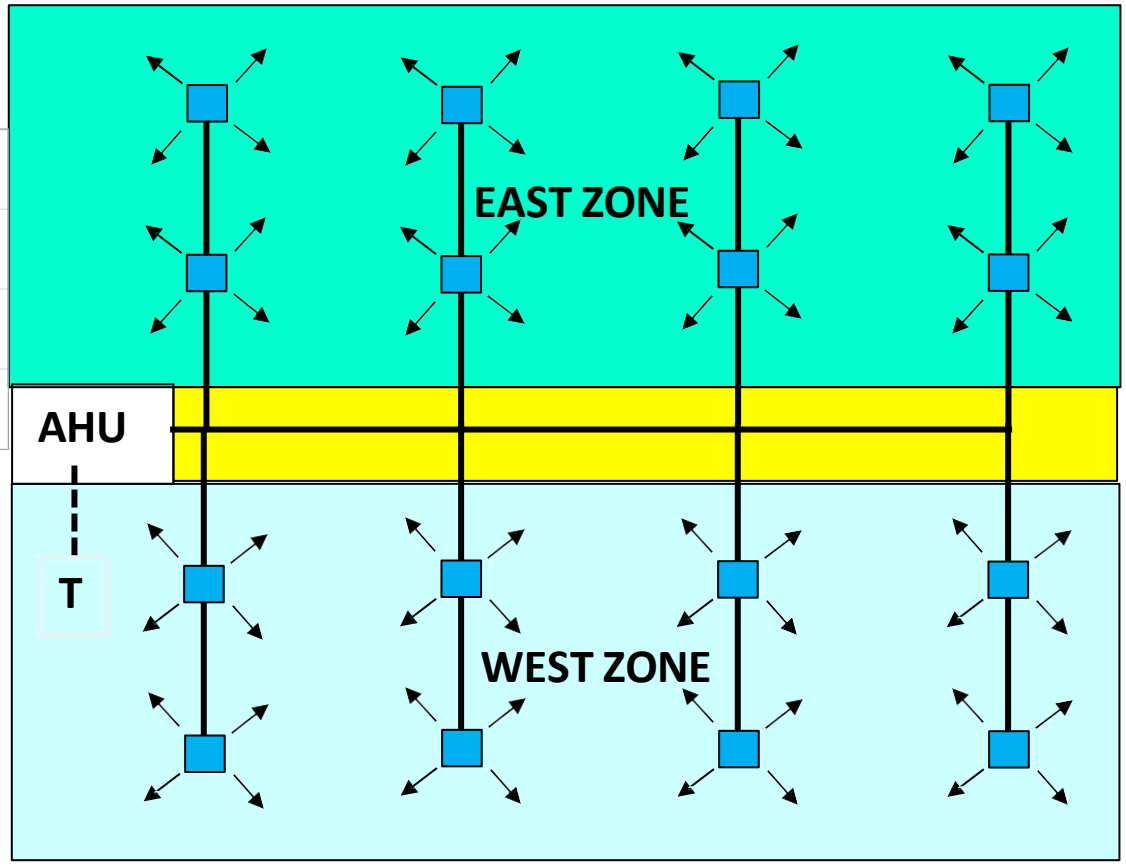
| Time / Zone | 10 am | 12 noon | 2 pm | 4 pm |
|---|---|---|---|---|
| East | 11.58 | 8.10 | 8.10 | 7.61 |
| West | 6.95 | 6.95 | 11.20 | 14.71 |
| Total | 18.53 | 15.05 | 19.30 | 22.32 |



# Zoning Requirements

- Thermostat in West section
  - East zone too warm in morning and too cold in afternoon.

| Time / Zone | 10 am | 12 noon | 2 pm | 4 pm |
|---|---|---|---|---|
| East | 11.58 | 8.10 | 8.10 | 7.61 |
| West | 6.95 | 6.95 | 11.20 | 14.71 |
| Total | 18.53 | 15.05 | 19.30 | 22.32 |



# Zoning Requirements

- East zone must be controlled independently of West zone.
- A zone is defined as a space or a group of spaces which temperature is controlled by one thermostat.

# Zoning Requirements

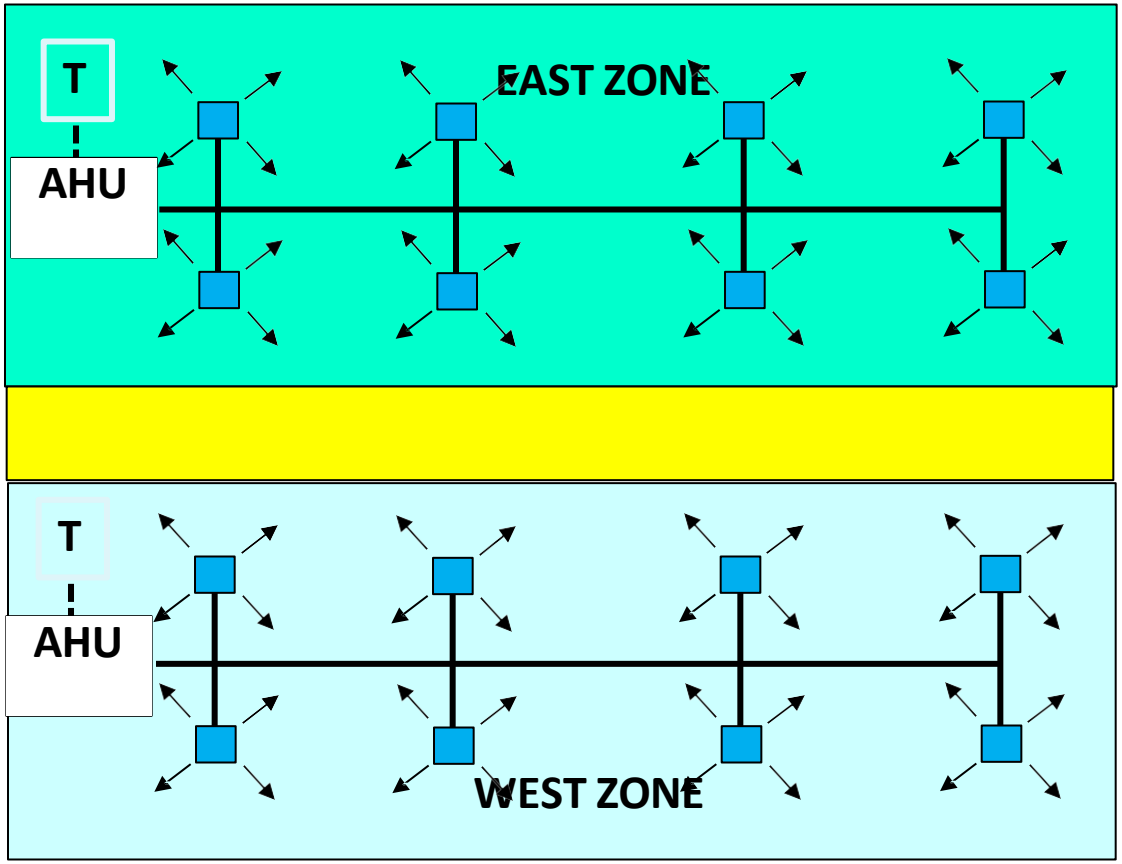
- Significantly different load profiles
- Individual control independent of the rest of the building
- Maintained at different indoor conditions
- Different operating hours
- Cross contamination need to be prevented.

# Multiple CAV Systems

- More than one zone - a multiple CAV system comprises a central refrigeration plant and multiple AHUs can supply air to different zones at different temperatures.
- AHUs can be fitted with partial load control features.

# Multiple CAV Systems

| Time / Zone | 10 am | 12 noon | 2 pm | 4 pm |
|---|---|---|---|---|
| East | 11.58 | 8.10 | 8.10 | 7.61 |
| West | 6.95 | 6.95 | 11.20 | 14.71 |
| Total | 18.53 | 15.05 | 19.30 | 22.32 |



# VARIABLE AIR VOLUME SYSTEMS

# VAV Systems

- Controls space dry-bulb temp. by varying $\dot{V}_s/\dot{m}_s$
- Basic system maintains constant $t_s$ to all zones throughout year and zonal load fluctuations are matched by varying $\dot{V}_s/\dot{m}_s$

SCL = $\dot{m}_s \cdot c_{p,a} \cdot (t_r - t_s)$

# Advantages

- Individual temperature control
- Saving in capital cost.
- Smaller AHUs - take advantage of diversity in load variations.
- System sized based on TCL at peak load period rather than sum of individual maximum loads.
- Saving in running cost of fan. Max. $V_s$ reduced by about 30%, but average operating air flow rate over whole year can be reduced by 50%.

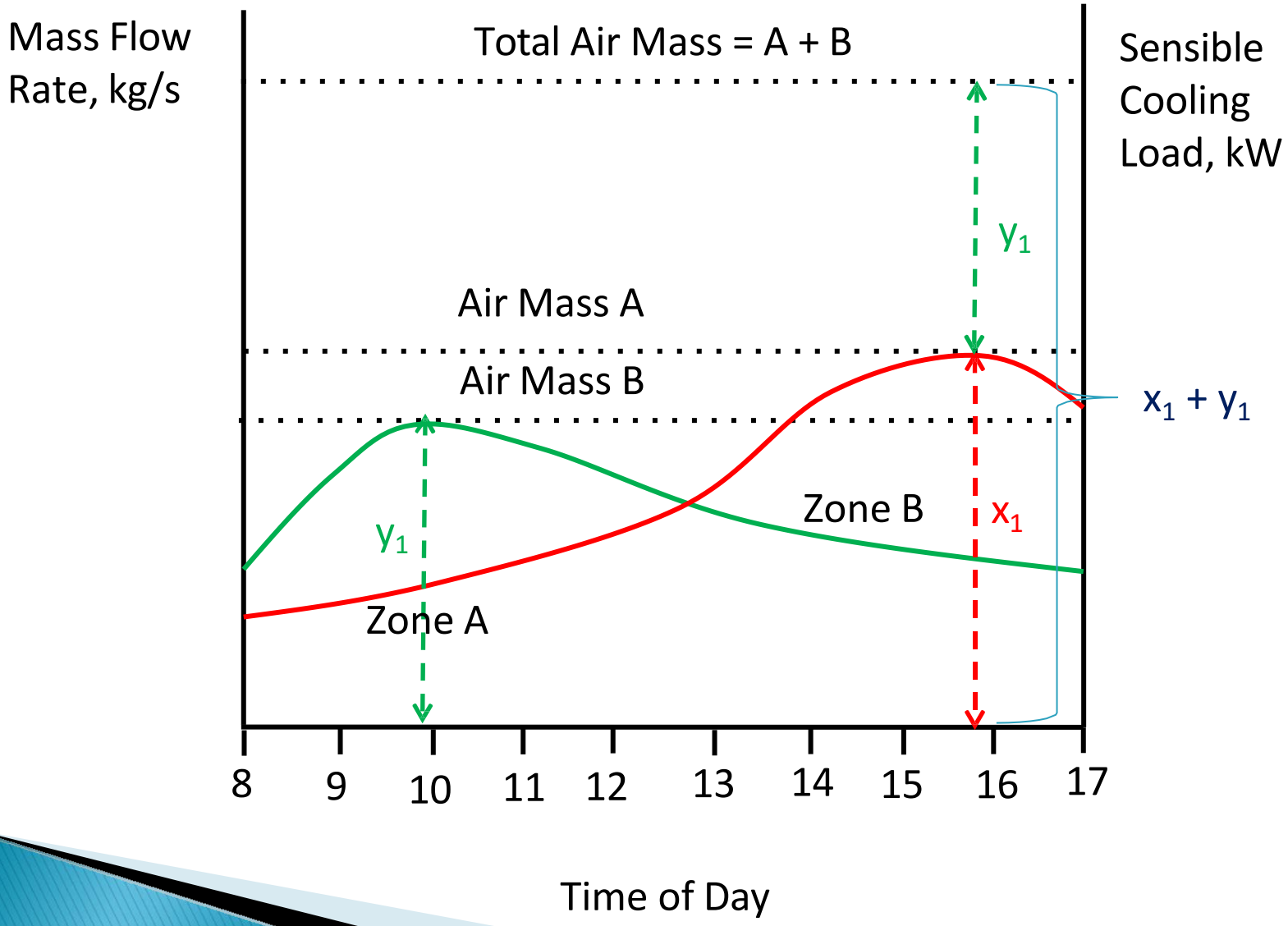
# Advantages

| Time of Day | Sensible Cooling Load (kW) | | | |
|---|---|---|---|---|
| | Zone 1 | Zone 2 | Zone 3 | Simultaneous load |
| 9 – 10 | 50 | 50 | 35 | 135 |
| 10 – 11 | 60 | 55 | 40 | 155 |
| 11 – 12 | 55 | 55 | 45 | 155 |
| 12 –13 | 50 | 55 | 50 | 155 |
| 13 – 14 | 45 | 45 | 45 | 135 |
| 14 – 15 | 40 | 50 | 55 | 145 |
| 15 – 16 | 40 | 58 | 65 | 163 |
| 16 – 17 | 35 | 55 | 60 | 150 |
| Highest | 60 | 58 | 65 | |
| | 60 + 58 + 65 = 183 kW | | | |

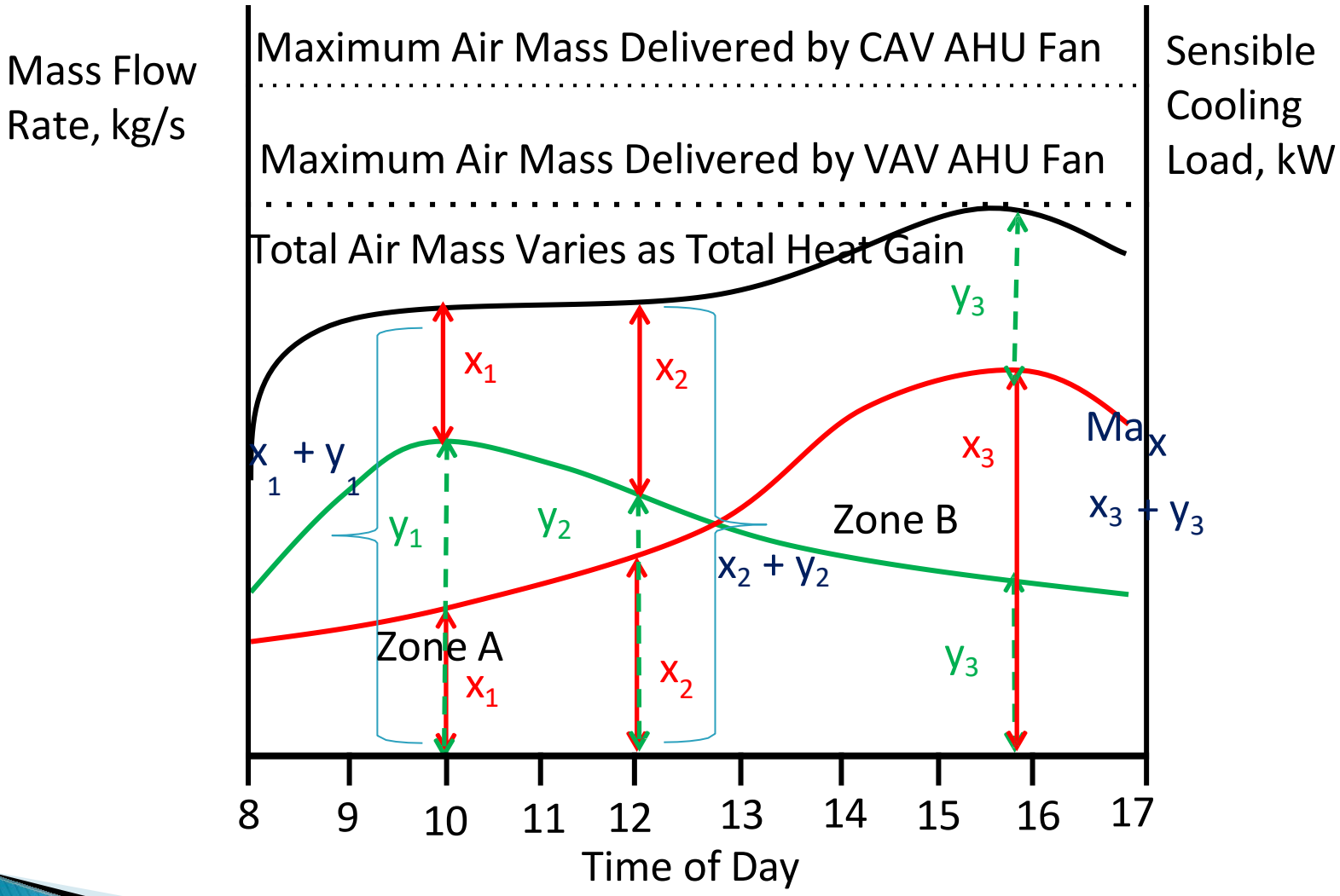
VAV Fan Size based on 163

CAV Fan Size based on 183

# Load Variations - CAV



# Load Variations - VAV



# Disadvantages

- Used when humidity control is not critical, e.g. for comfort purposes
- LCL only changes slightly while SCL may reduce by more than 40%.
- When supply air flow rate is reduced, sensible and latent cooling capacities are reduced together.

Therefore, humidity of space will increase.

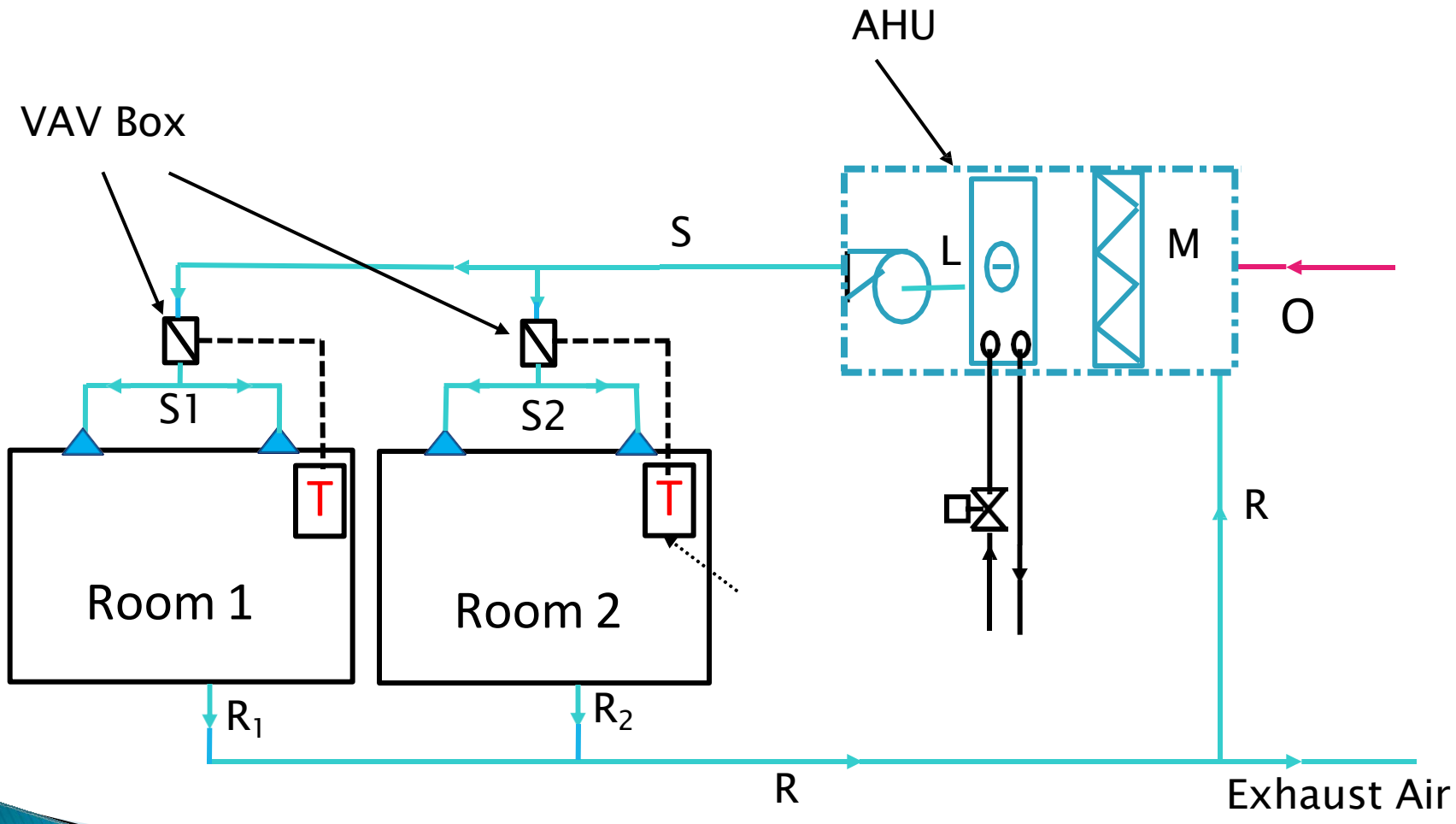
# Disadvantages

- Supply air flow rate is reduced, quantity of outdoor air also decreases. It is possible that reduction in sensible heat load will cause a depletion of outdoor air.
- VAV box normally provides a minimum set point about 50% to 40% to avoid this problem. It means the full potential of energy saving cannot be achieved.

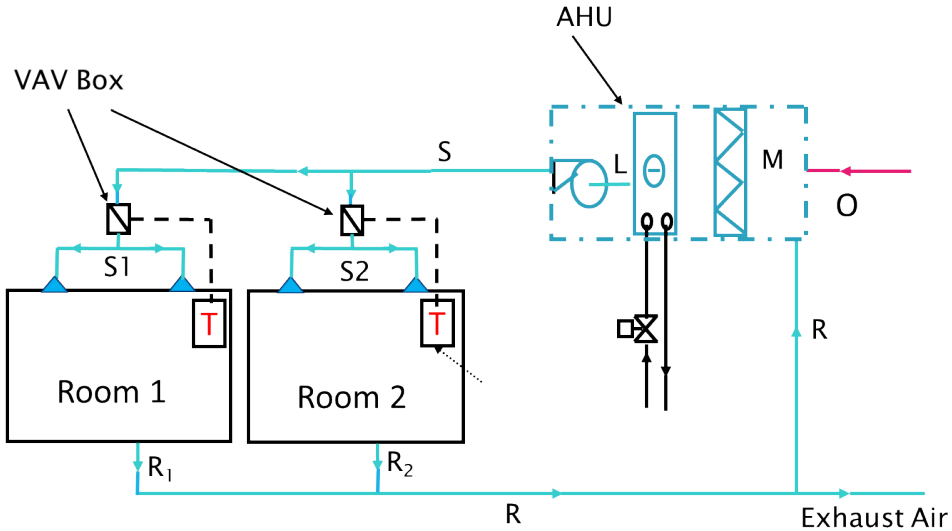
# Disadvantages

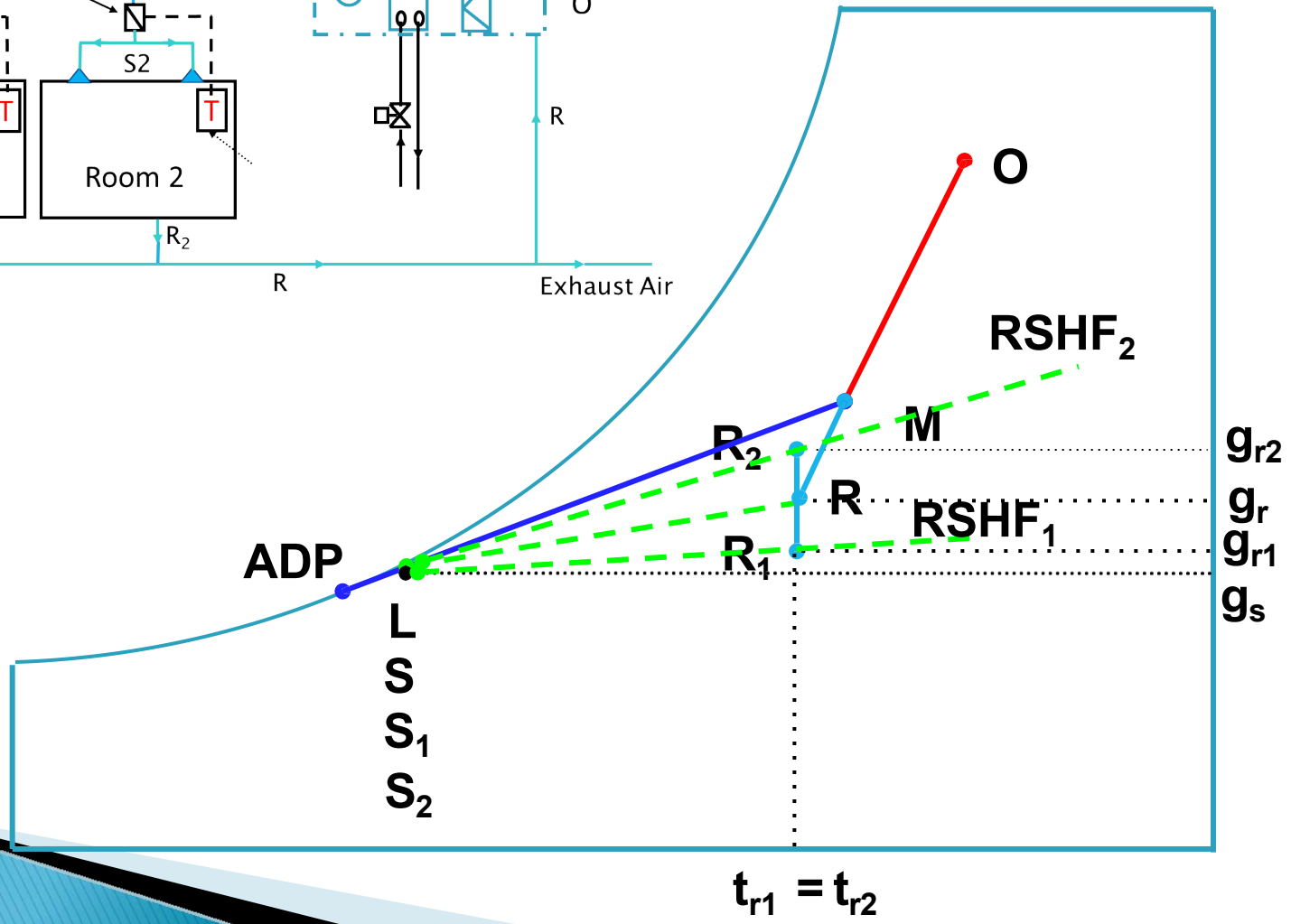
- If supply air flow rate is reduced beyond a certain point, air distribution and air movement in room may deteriorate.
- Effect depends on efficiency of air terminal device.
- A return fan is required to provide good flow rate control.

# VAV System Schematic



# VAV System Schematic



# Example 5.3

An office space is conditioned by a draw through VAV air handling unit (AHU). The space is maintained at 24°C dry bulb temperature (dbt) and 55% saturation. The peak sensible and latent space cooling loads for the space are 126 kW and 54 kW respectively. The outdoor air is at 32°C dbt and 26°C wet bulb temperature.

The supply air temperature is 14°C. Heat gains due to the air handling unit fan and at the supply duct increase the air temperature by 1.0°C. The mass ratio of the outside-to-supply air is 0.2.

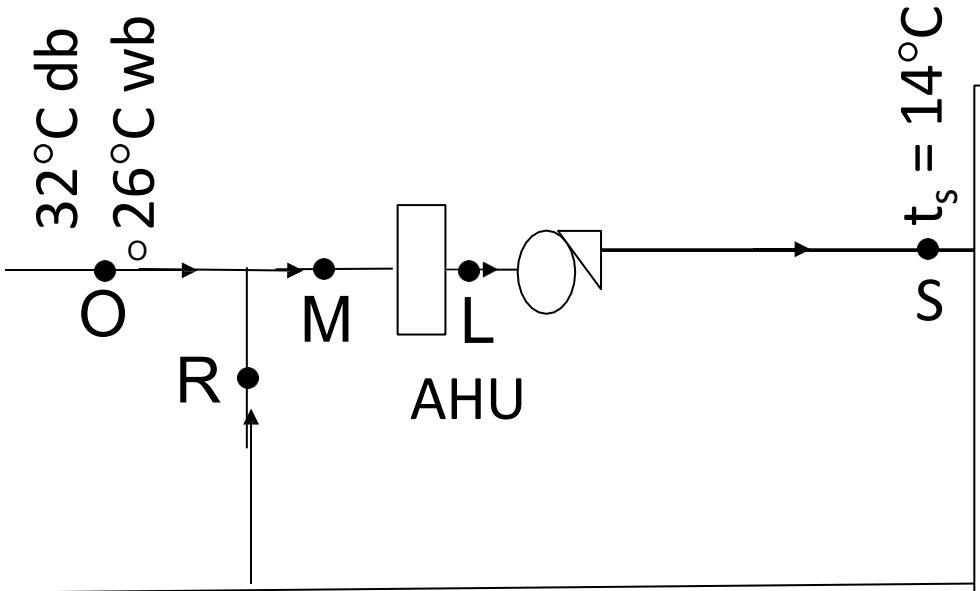
# Example 5.3

(i) Plot the air-conditioning processes onto the psychrometric chart.

Determine the following:

(ii) supply air mass flow rate at peak load.
(iii) on-coil air dry bulb temperature and dew point
(iv) off-coil air dry bulb temperature
(v) room sensible heat factor.
(vi) cooling coil apparatus dew point.
(vii) peak cooling coil load.
(viii) supply air mass flow rate when the sensible cooling load reduces by 30%.

# Example 5.3




SCL = 126 kW

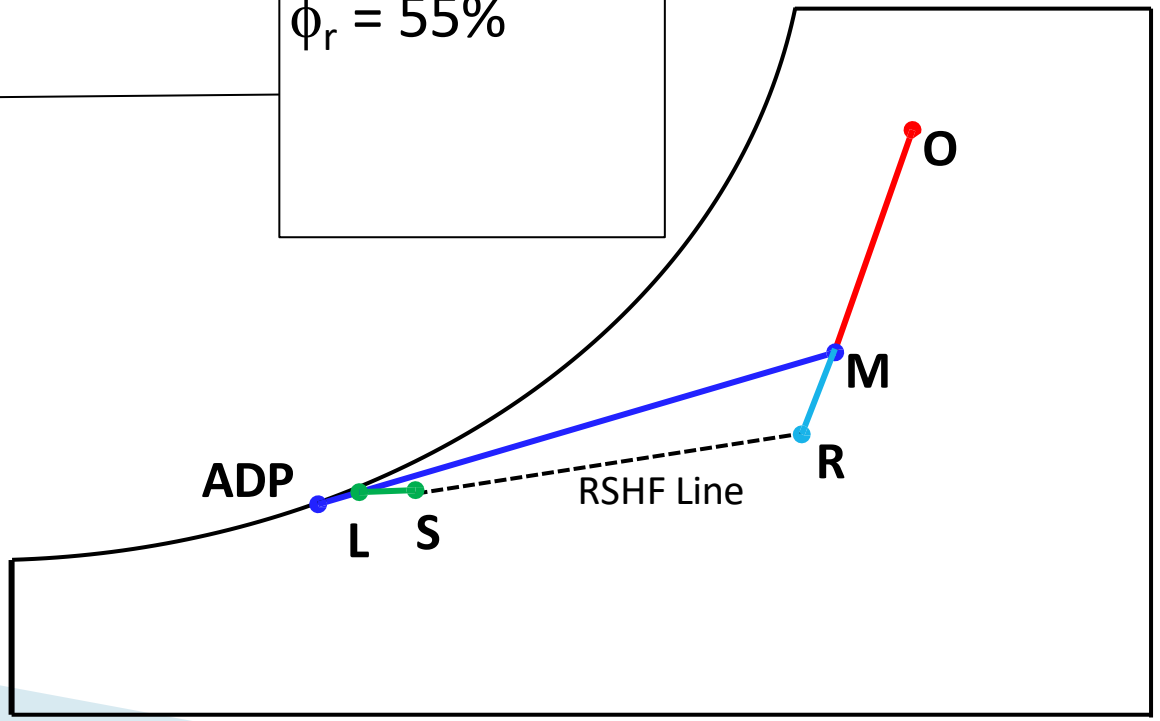
LCL = 54 kW

$t_r$ = 24°C

$\phi_r$ = 55%

$t_s - t_l$ = 1°C,

$\frac{\dot{m}_o}{\dot{m}_s}$ = 0.2

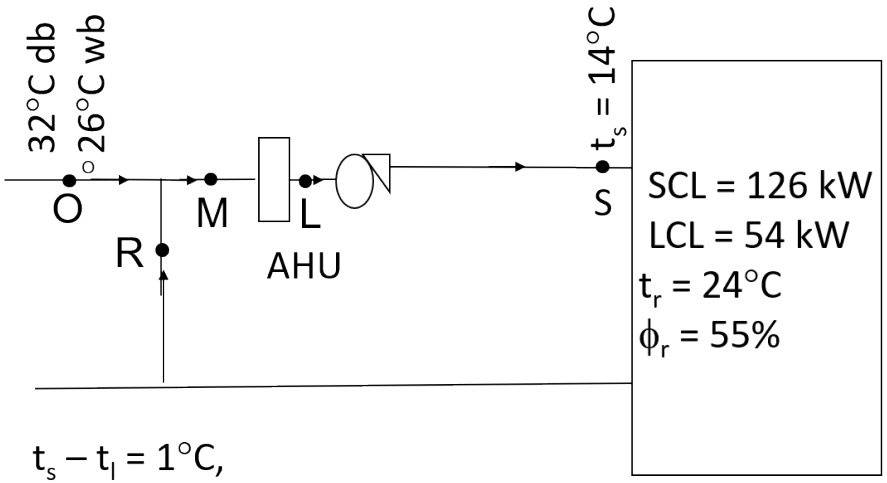


# Example 5.3

(ii) $\dot{m}_s$

$$RSCL = \dot{m}_s . c_{p,a} . (t_r - t_s)$$

$$\dot{m}_s = \frac{RSCL}{c_{p,a} \times (t_r - t_s)} = \frac{126}{1.012(24 - 14)} = 12.45 \text{ kg/s}$$




$t_s - t_l = 1°C$,

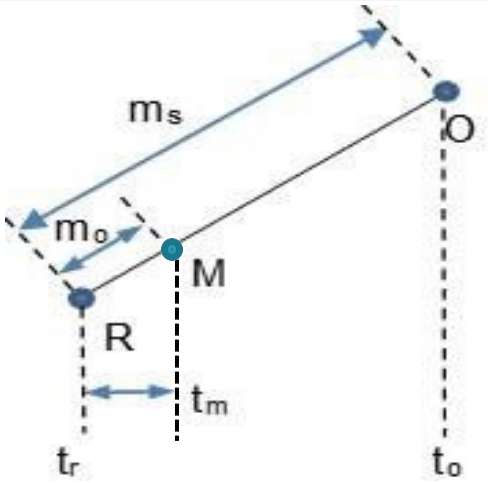
$\frac{\dot{m}_o}{\dot{m}_s}$ = = 0.2

# Example 5.3

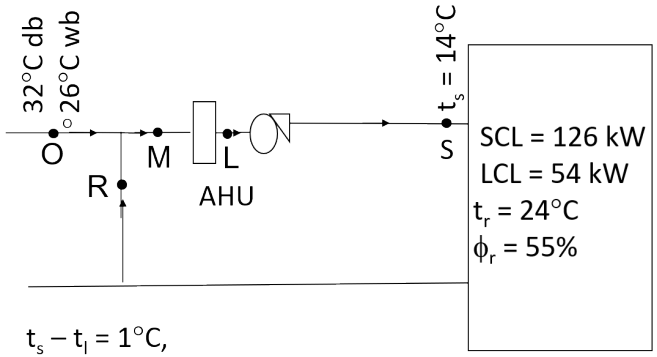
(iii) $t_m$ and dew point

$$\frac{\dot{m}_o}{\dot{m}_s} = \frac{t_m - t_r}{t_o - t_r} = 0.2$$

$$t_m = t_r + 0.2\,(t_o - t_r) = 24 + 0.2\,(32 - 24)$$
$$= 25.6°C$$

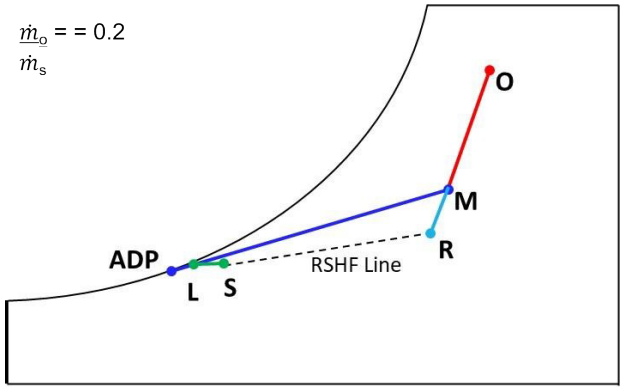



Plot Point O and Point R.
Draw line to join Point O and R.
Plot $t_m$ to intersect with line O-R to obtain Point M.
From the psychrometric chart, dew point is 16.8°C




$t_s - t_l = 1°C$,

$\frac{\dot{m}_o}{\dot{m}_s} = = 0.2$



# Example 5.3

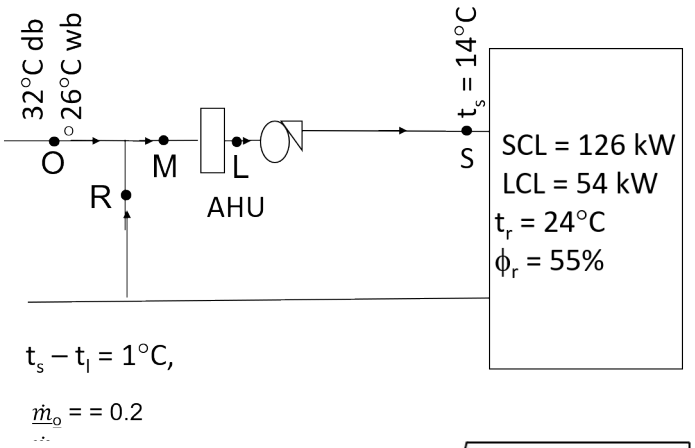
(iv) $t_l$

Given : $t_s - t_l = 1°C$

Off-coil air temperature, $t_l = t_s - 1.0 = 14 - 1.0 = 13°C$

(v) RSHF

$$RSHF = \frac{SCL}{SCl + LCL} = \frac{126}{126 + 54} = 0.7$$




$t_s - t_l = 1°C$,

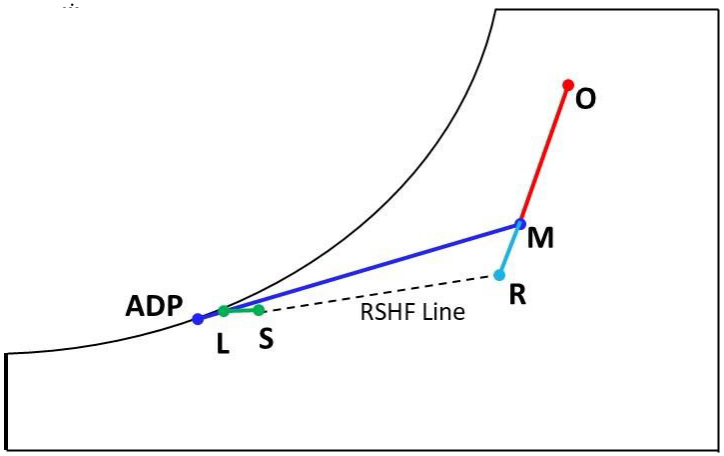
$\frac{\dot{m}_o}{} = = 0.2$

(vi) ADP

Plot $t_s$.

Plot RSHF through Point R to intersect with $t_s$ to obtain Point S.

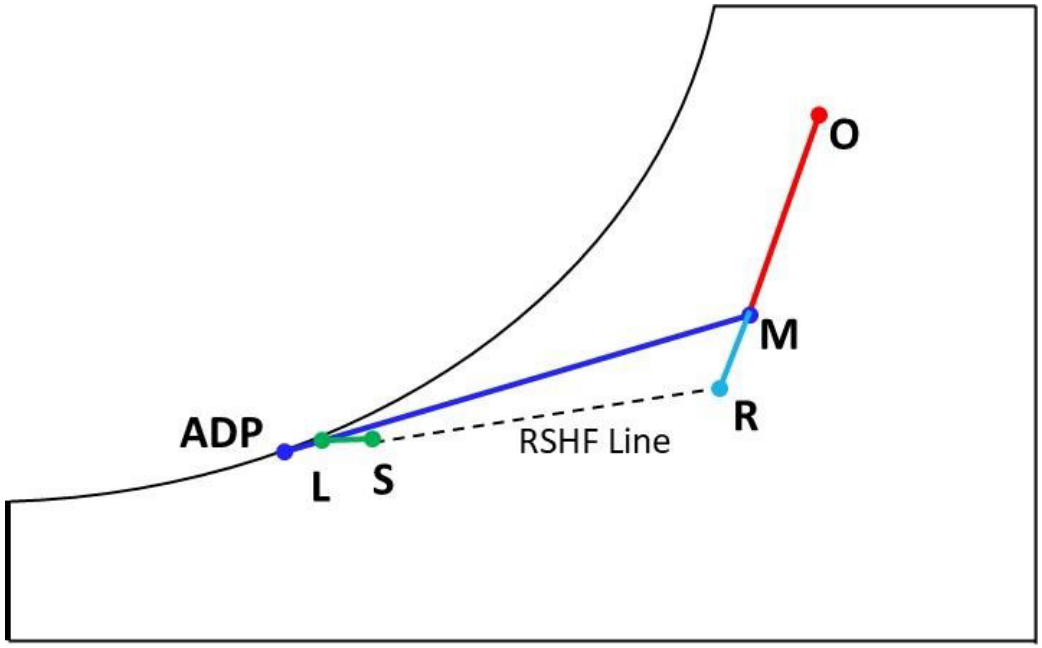
Plot $t_l$.



# Example 5.3

Plot a constant moisture content line through Point S to intersect with $t_l$ to obtain Point L.
Draw line from Point M to Point L. Extend line to intersect with 100% saturation line to obtain ADP.

From plot, ADP is 11°C



# Example 5.3

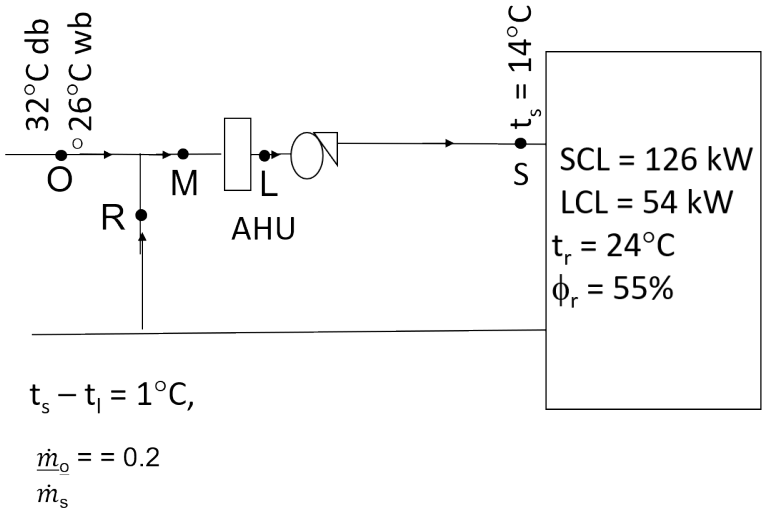
(vi) Peak Cooling Capacity of AHU

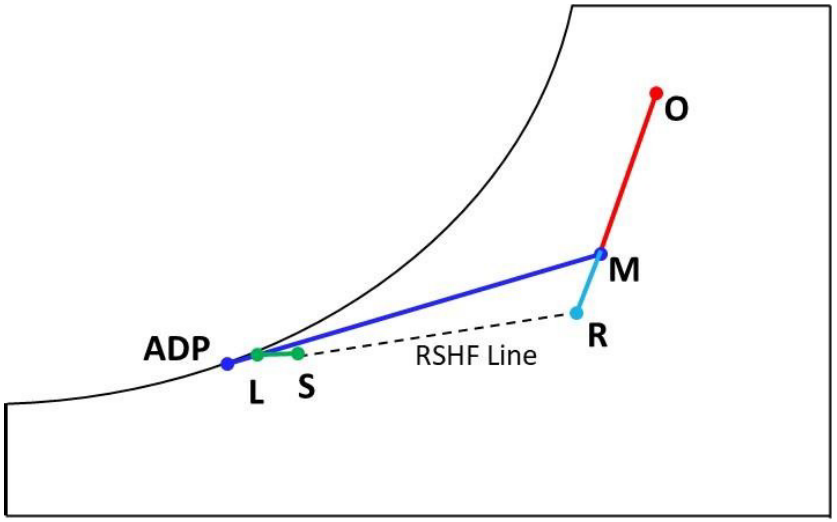
Peak cooling capacity of AHU = $\dot{m}_s(h_m - h_l)$ kW

From plot, $h_m$ = 56.0 kJ/kg, $h_l$ = 35.0 kJ/kg

Peak cooling capacity of AHU = 12.45(56.0 – 35.0)

= 261.5 kW



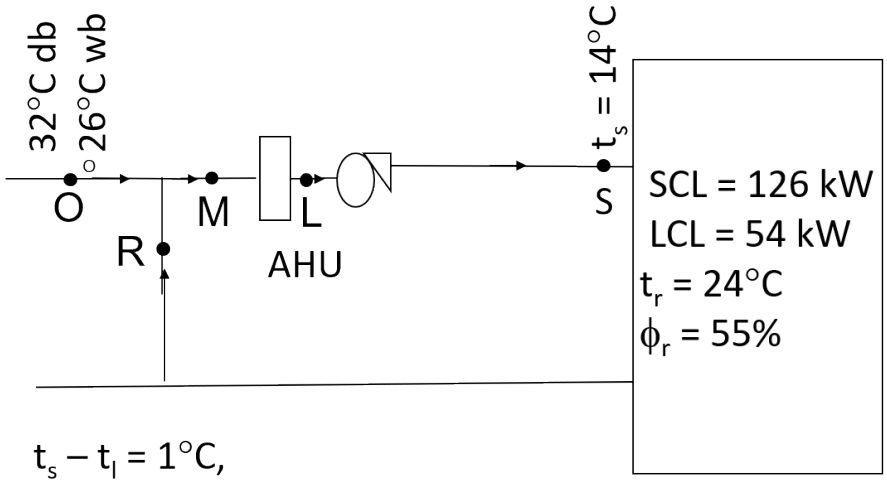
# Example 5.3

(viii) $\dot{m}_{s,p}$

$$SCL_p = \frac{(100 - 30)}{100} \times 126 = 88.20\text{kW}$$

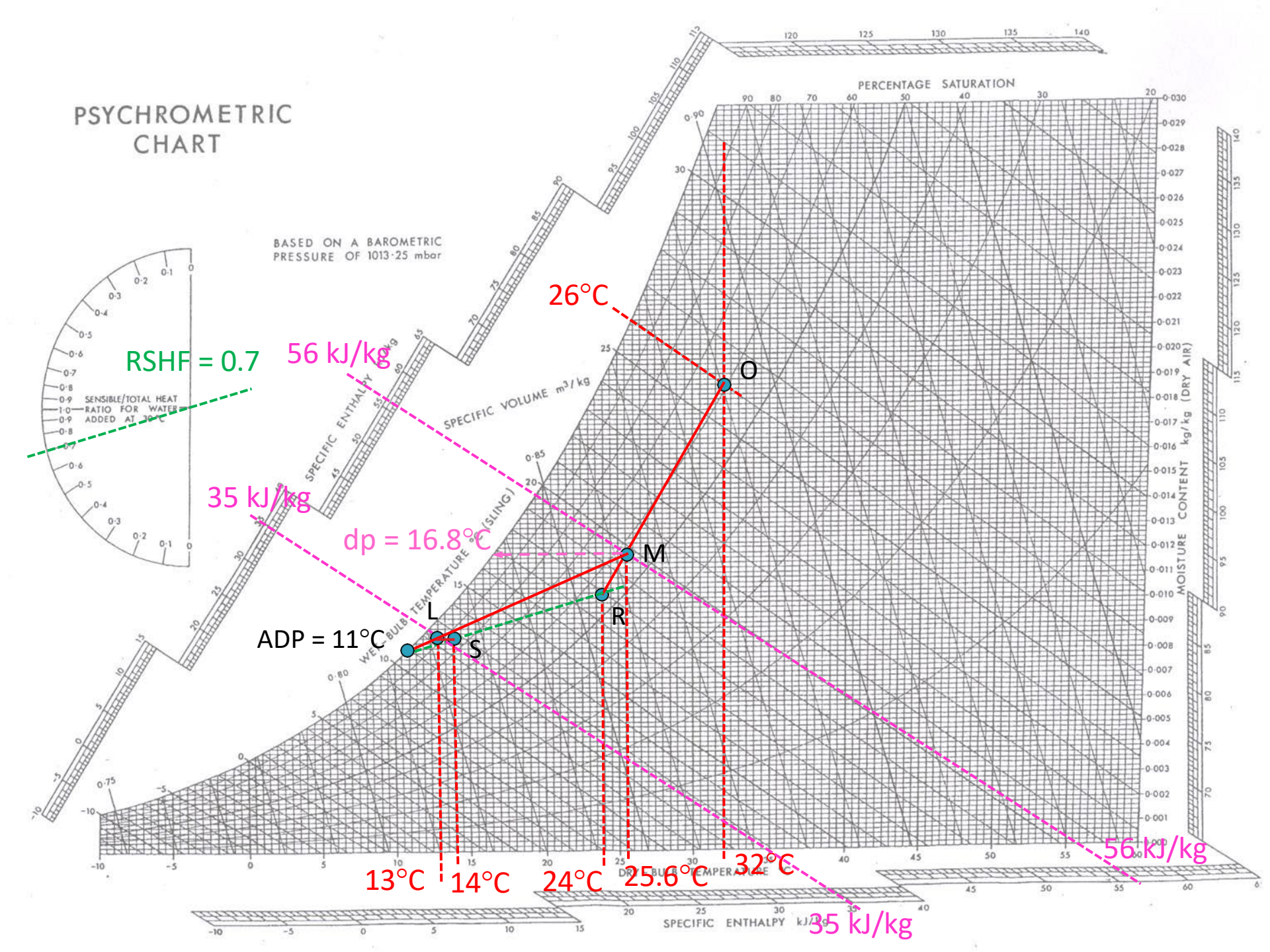
$$SCL_p = \dot{m}_{s,p} \times c_{p,a} \times (t_{r,p} - t_{s,p})$$

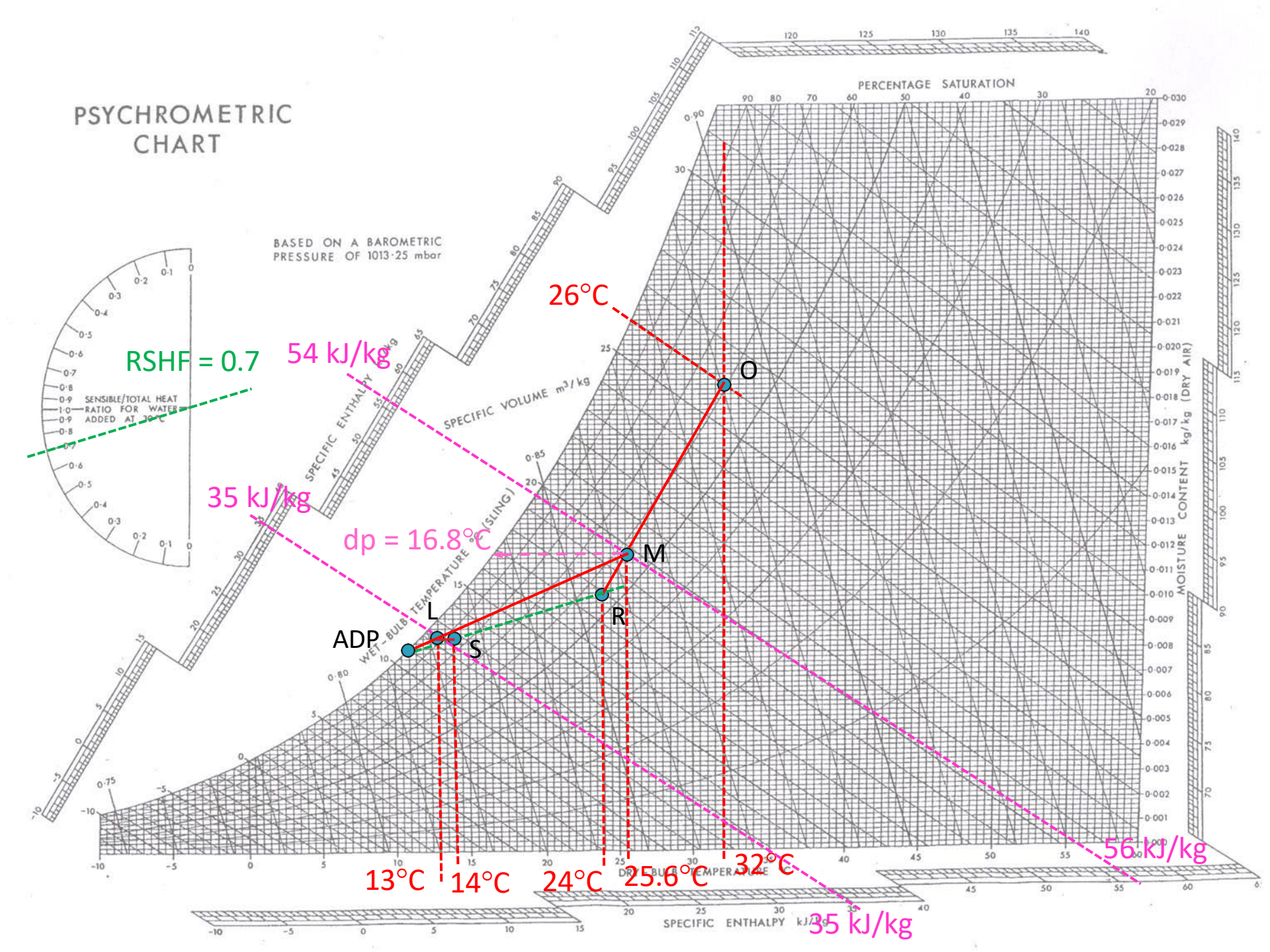
$$\dot{m}_{s,p} = \frac{SCL_p}{c_{p,a}(t_{r,p} - t_{s,p})} = \frac{88.20}{1.012(24 - 14)} = 8.72 \text{ kg/s}$$




$t_s - t_l = 1°C$,

$\frac{\dot{m}_o}{\dot{m}_s}$ = = 0.2


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SPECIFIC VOLUME m³/kg
SPECIFIC ENTHALPY kJ/kg
WET BULB TEMPERATURE °C (SLING)
DRY BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
MOISTURE CONTENT kg/kg (DRY AIR)
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30°C
RSHF = 0.7
26°C
56 kJ/kg
35 kJ/kg
dp = 16.8°C
ADP = 11°C
O
M
R
L
S
13°C
14°C
24°C
25.6°C
32°C
56 kJ/kg
35 kJ/kg


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SPECIFIC VOLUME m³/kg
SPECIFIC ENTHALPY kJ/kg
WET-BULB TEMPERATURE °C (SLING)
MOISTURE CONTENT kg/kg (DRY AIR)
SPECIFIC ENTHALPY kJ/kg
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30°C
RSHF = 0.7
54 kJ/kg
35 kJ/kg
dp = 16.8°C
26°C
O
M
R
L
S
ADP
13°C
14°C
24°C
25.6°C
32°C
56 kJ/kg
35 kJ/kg

# Quiz 4

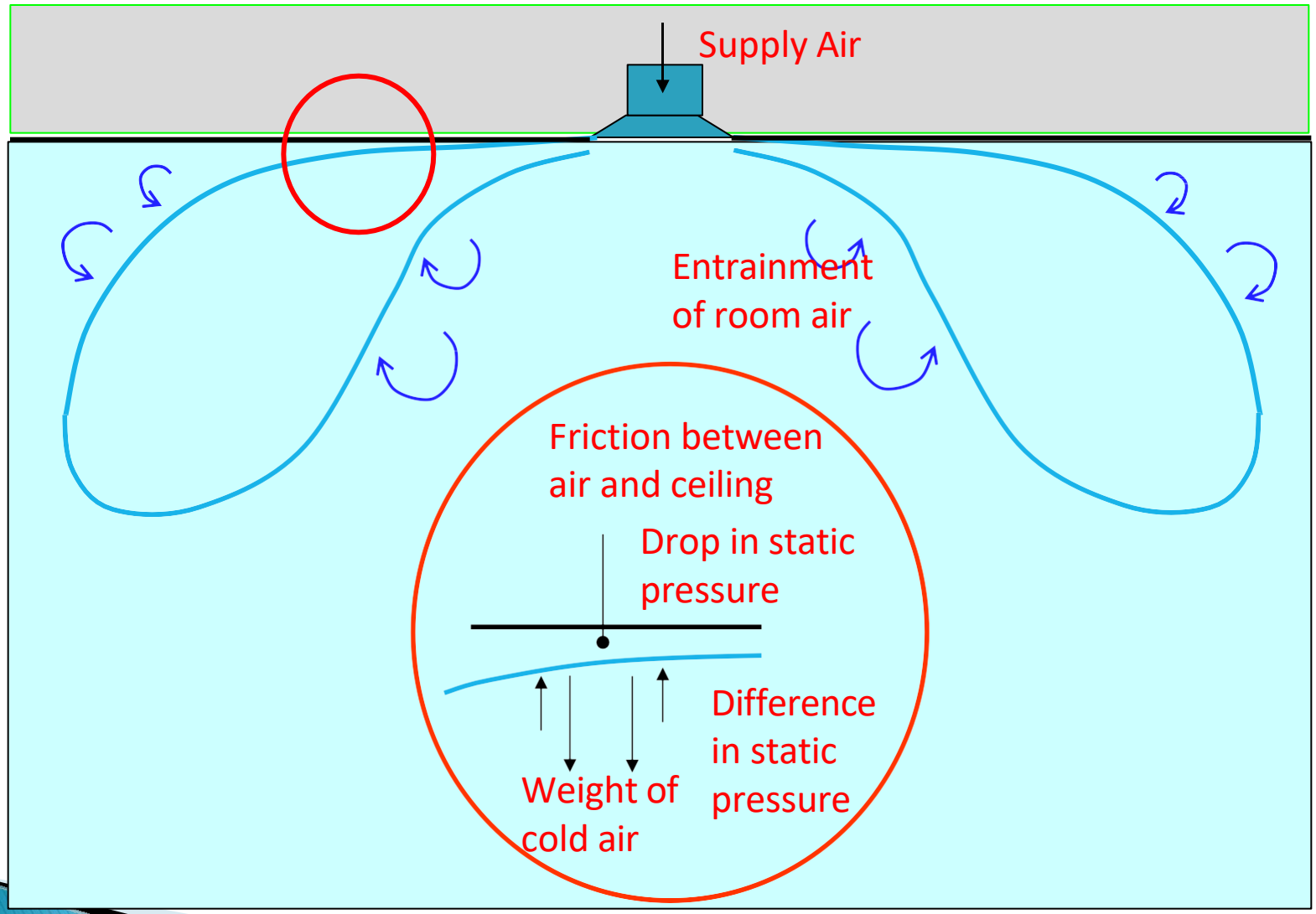
| Quiz 4 |
| --- |
| Psychrometry |
| 18 questions amounting to 18 points |
| Three attempts. No backtracking, Auto-submission after 60 minutes |
| Attempt with highest final point will be taken as the final point. |
| Available from 9.00am Friday, 20 Jun to 2359 Monday, 23 Jun 25 |

Official Open

# Tutorial AMVS 4

- Answer Questions 3, 4 and 5.
- All answers shall be hand-written.
- Submission can be by hardcopy or through POLITEMall. For POLITEMall submission, the answers shall be scanned in only **ONE pdf file**
- Submission deadline : 2359, Wednesday 25 Jun 2025

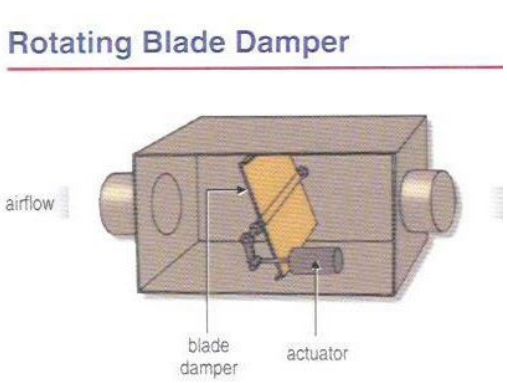
# Coanda Effect

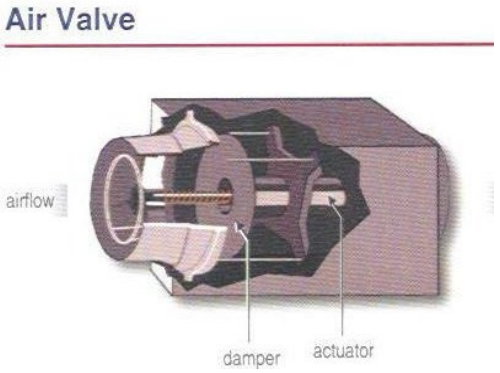


# Air Distribution

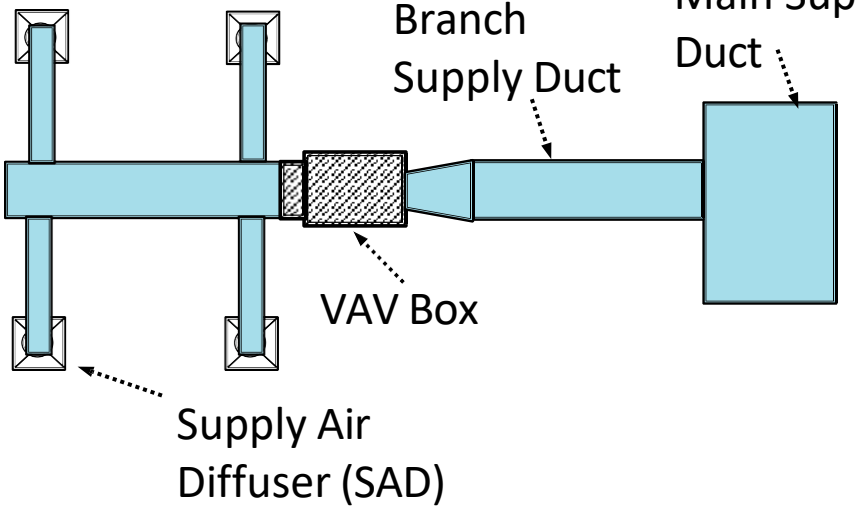
- If initial velocity of air stream is not sufficient, separation occurs when air stream is still much colder than room air. Cold air will sink into occupied zone and cause local discomfort. This is called **dumping**.
- Dumping will not occur if VAV units are chosen properly. Vital to select correct terminals to give good air distribution over range of expected duties .

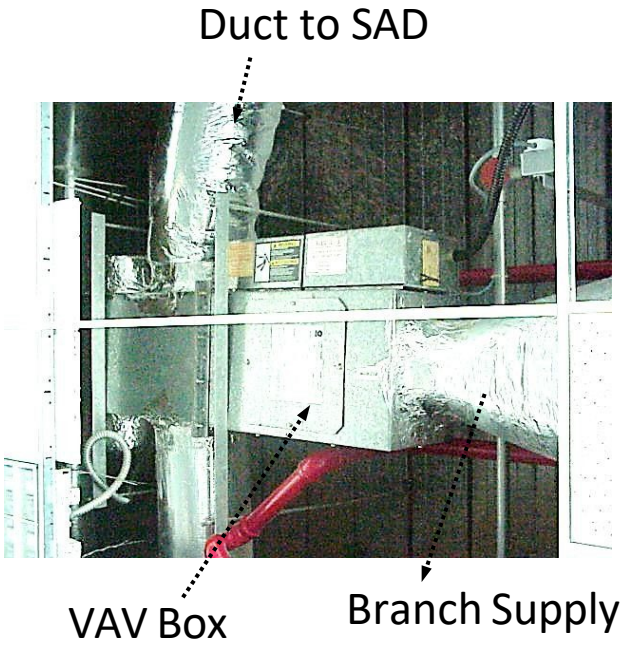
# VAV Box




## VAV Boxes without Reheat



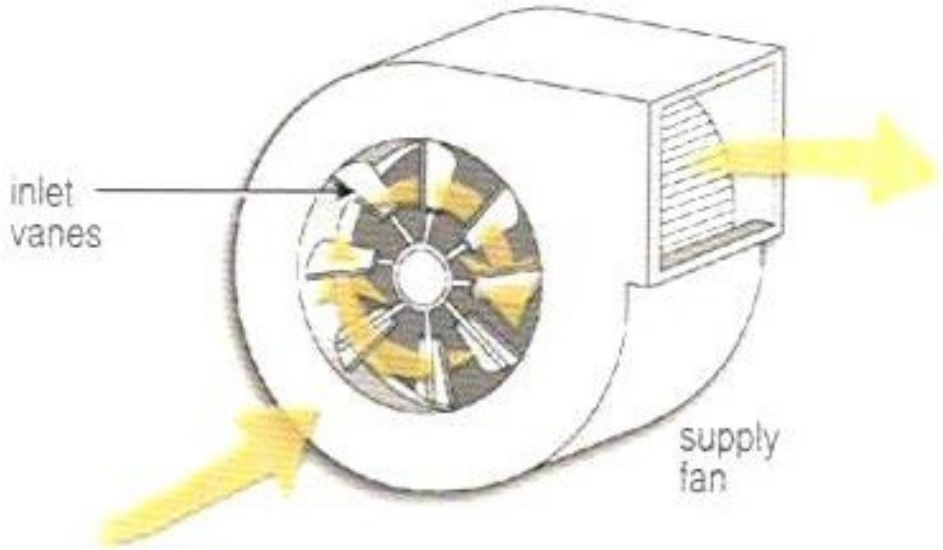

## VAV Box and SAD Installation

# Fan Control Methods

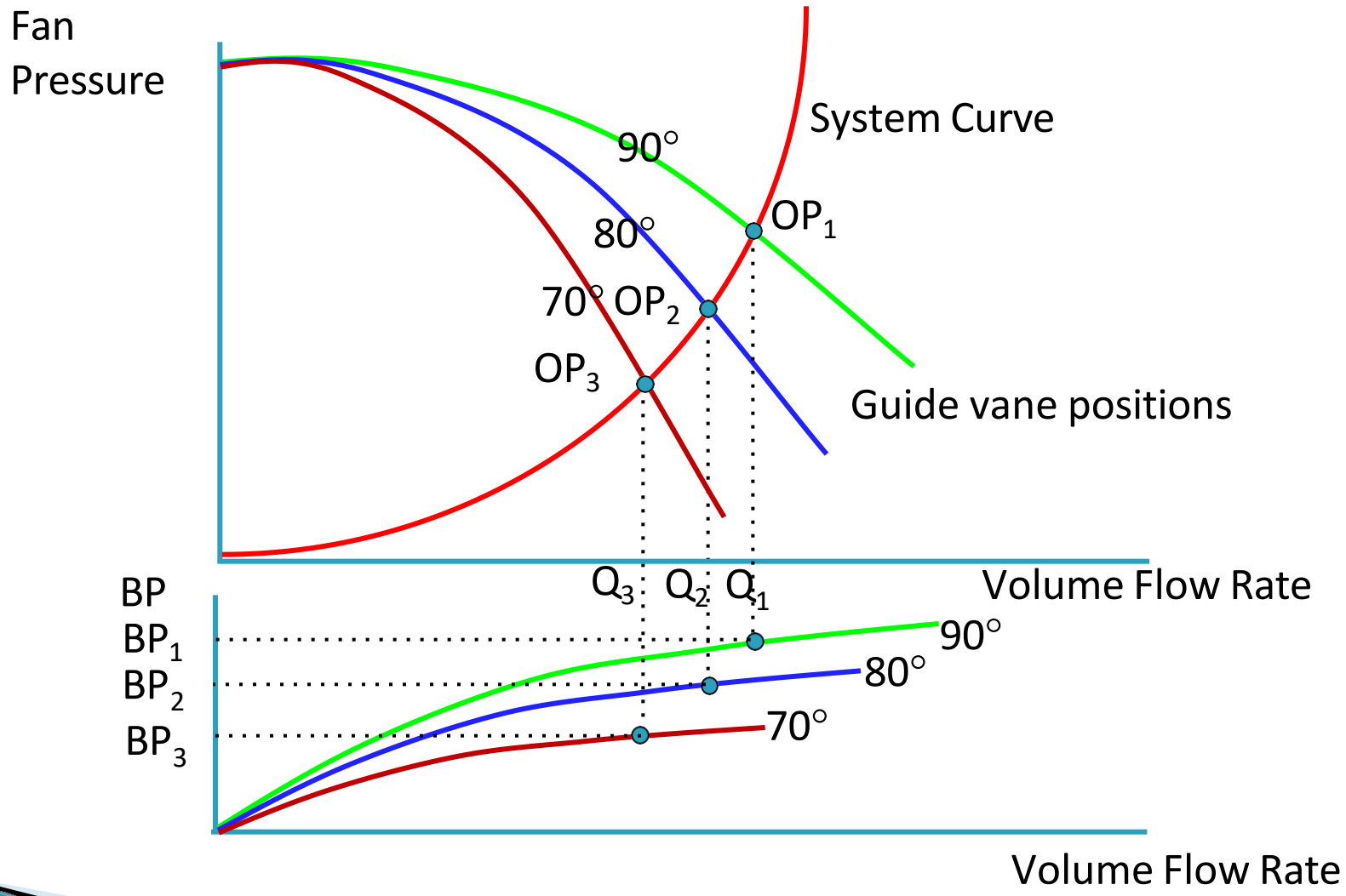
- As volume flow rates of different zones reduce, total volume flow of fan should also reduce.
  - Adjustable Inlet guide vanes at eye of centrifugal fan impeller
  - Adjustable pitch impeller blades on axial flow fans
  - Variable speed drive

# Variable Inlet Guide Vanes

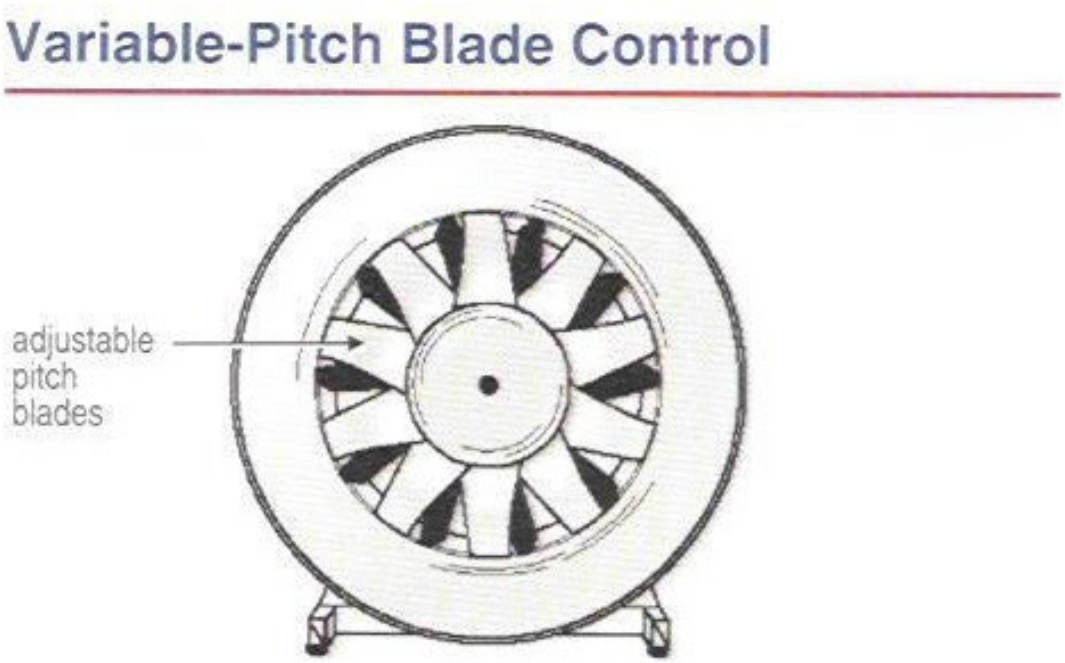
## Fan Inlet Vanes



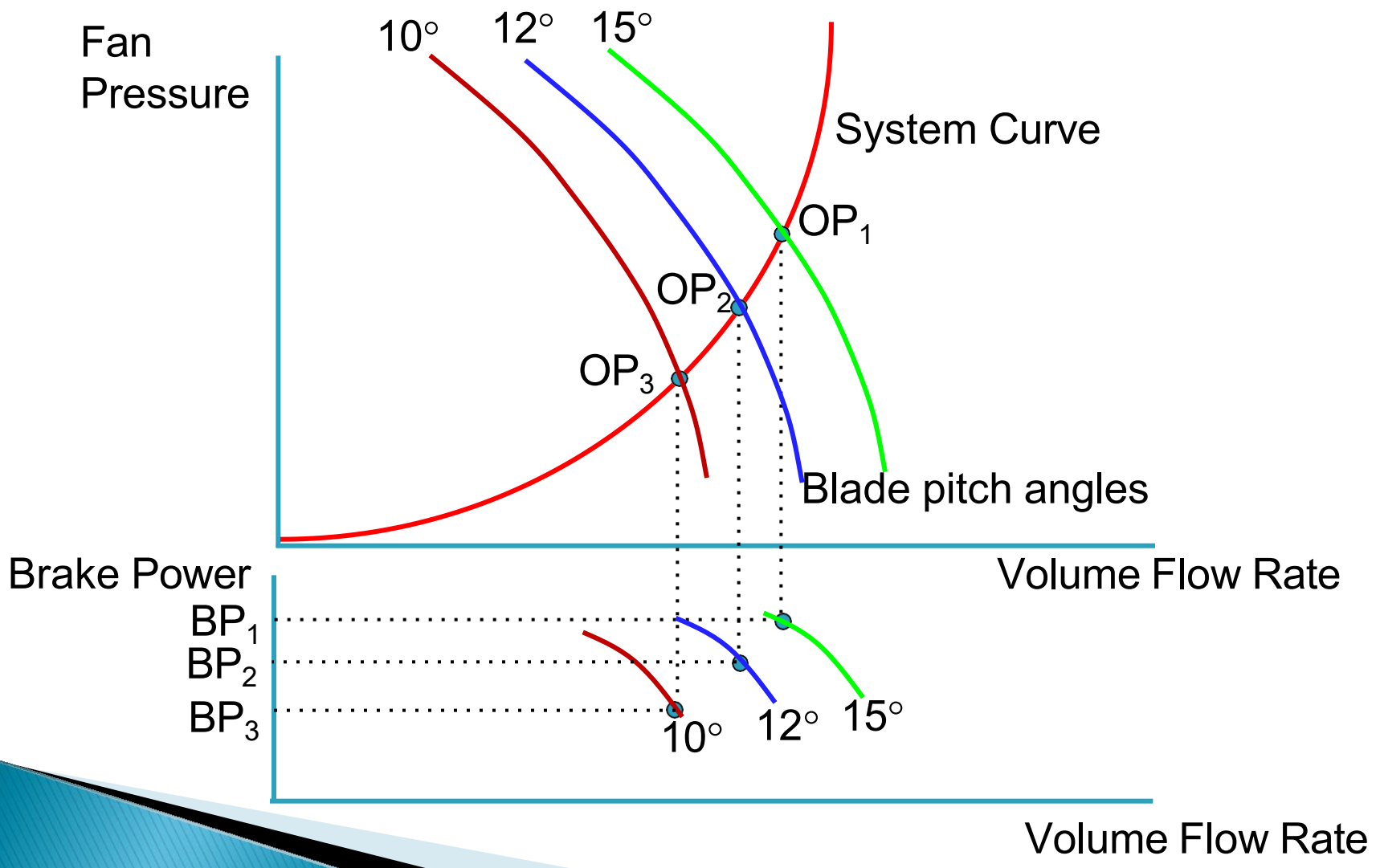
# Variable Inlet Guide Vanes



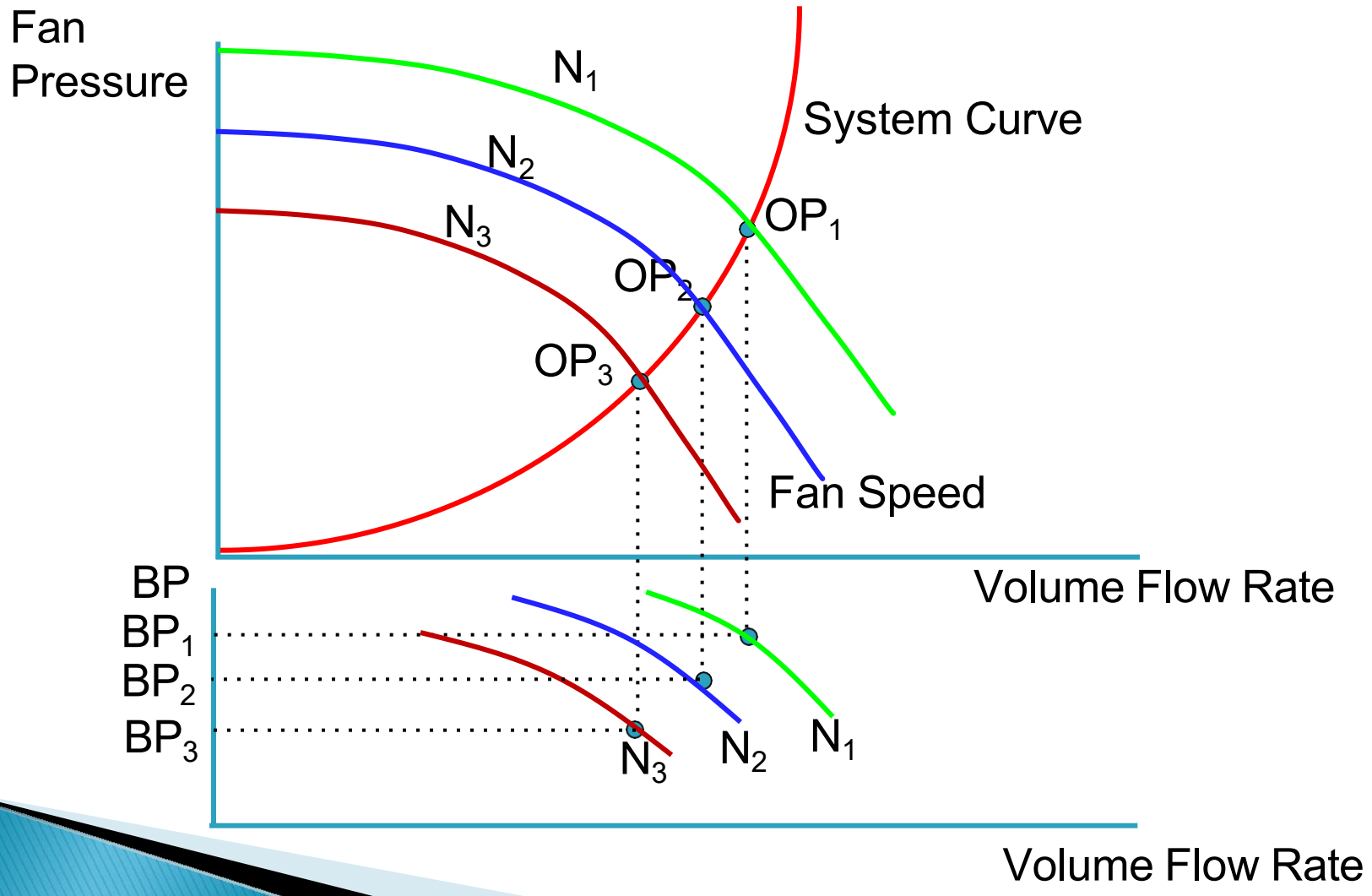
# Variable Pitch – Axial Fan



# Variable Pitch – Axial Fan

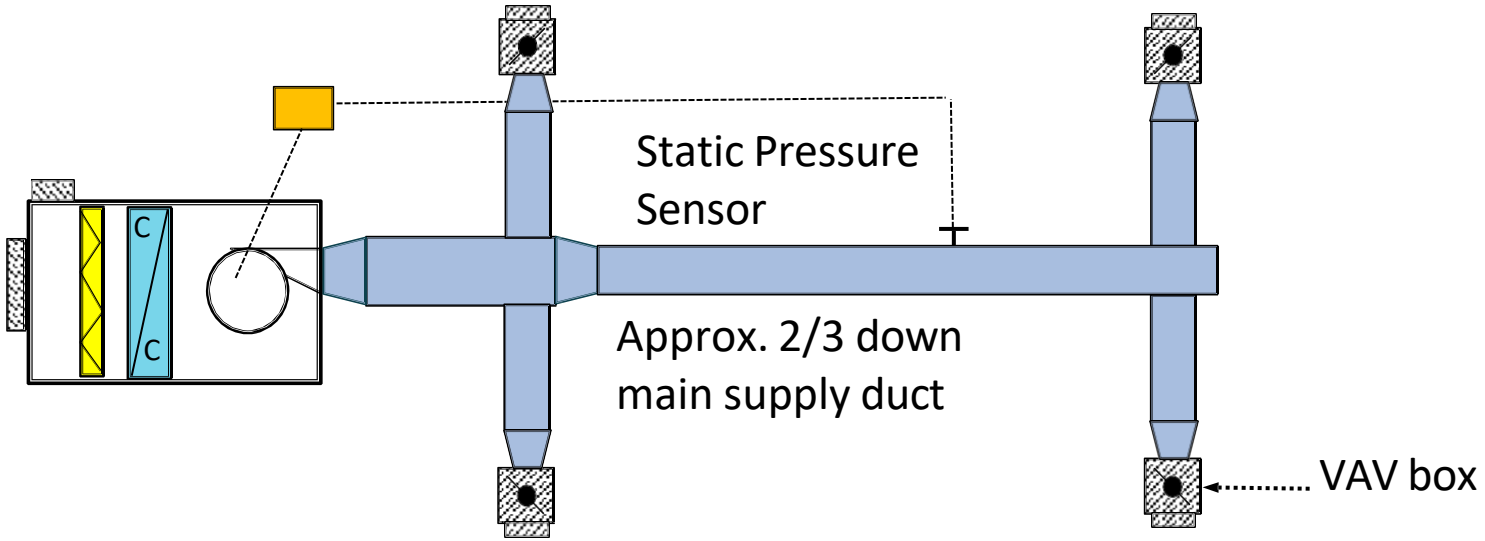


# Variable Speed Drive



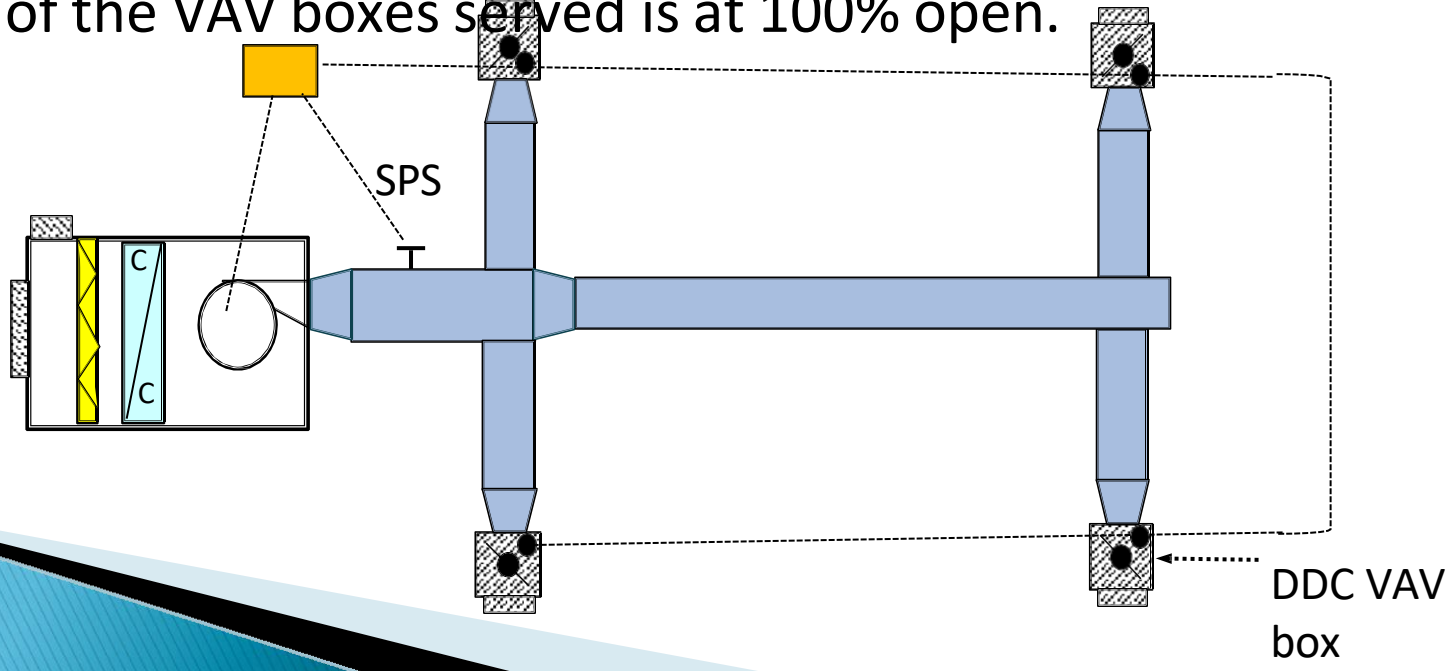
# Static Pressure Control

- SPS installed in supply air duct. When $V_s$ to zones æreduced by VAV boxes, static pressure in main supply air duct will rise, SPS will send signals to reduce supply fan capacity.



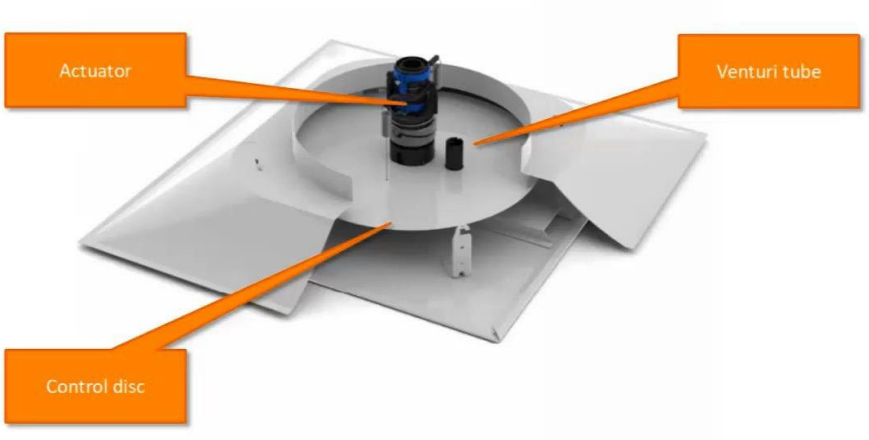
# Static Pressure Reset

- Control strategy to reduce fan energy consumed.
- DDC enables position of every VCD of boxes to be monitored.
- When BMS detects that all VAV boxes served by VAV AHU are partially closed, BMS will signal AHU fan to reduce speed to lower static pressure in main duct system until at least 1 one of the VAV boxes served is at 100% open.



# VAV Diffusers

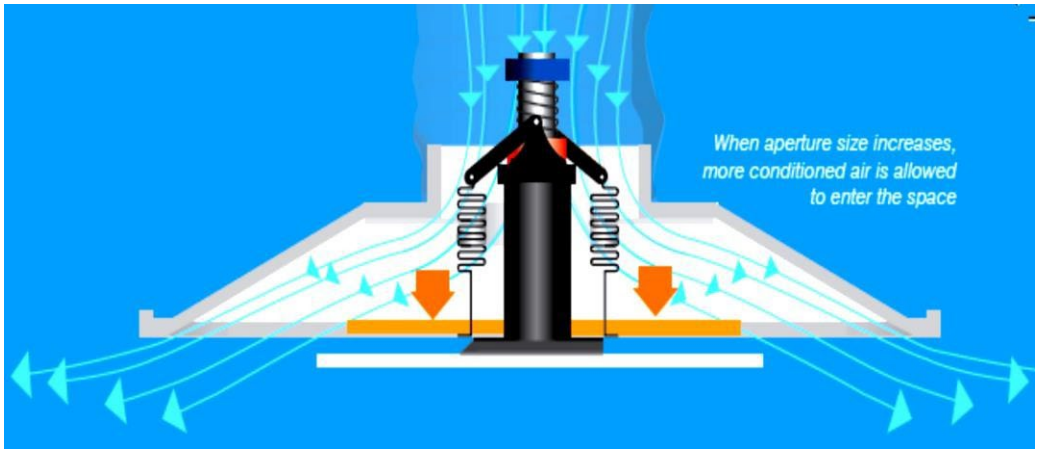


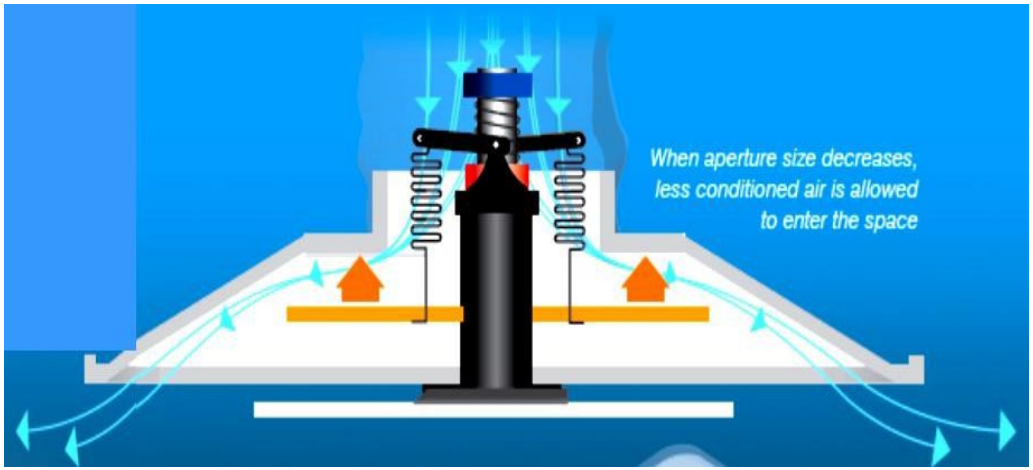



- In-built VCD, thermostat and sometimes a $CO_2$ sensor.
- Delivers only supply air flowrate to space by adjusting in-built VCD based on signals from in-built thermostat or wall-mounted thermostat.

# VAV Diffusers

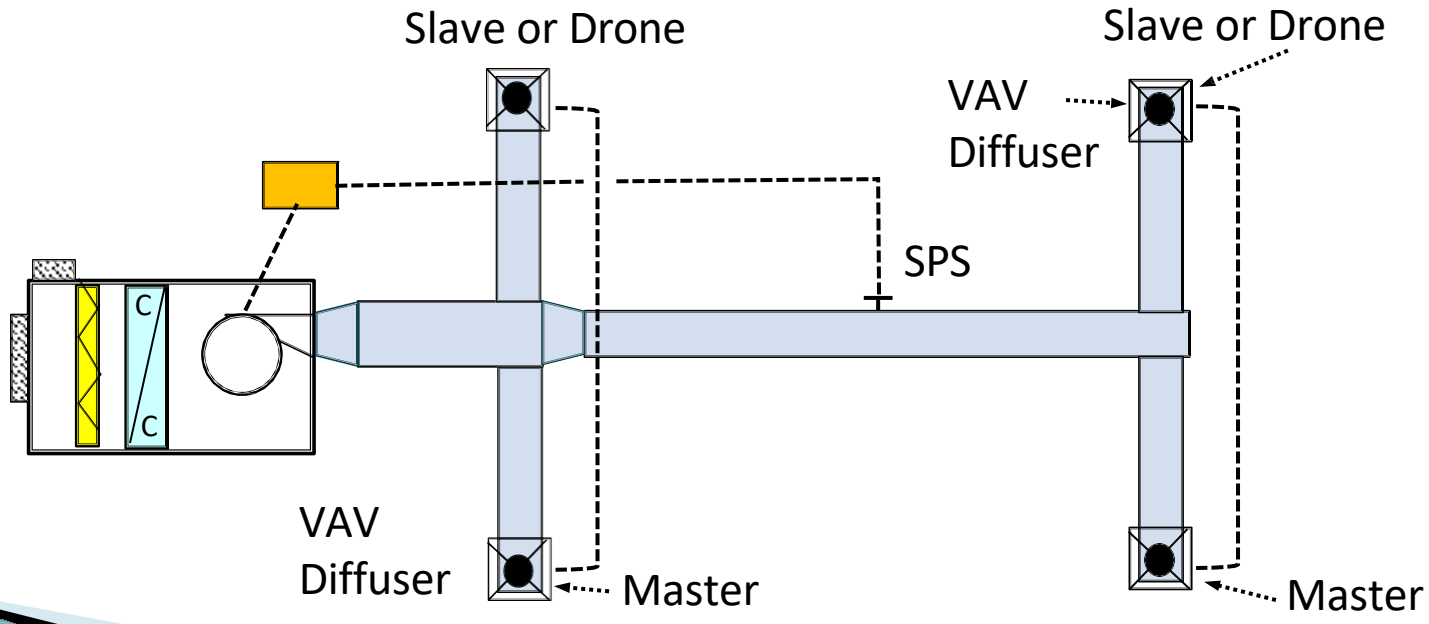
- In-built VCD creates a variable opening
- Near-constant discharge velocity from fully open to minimum positions.
- Good room air movement throughout its operation.



# VAV Diffusers

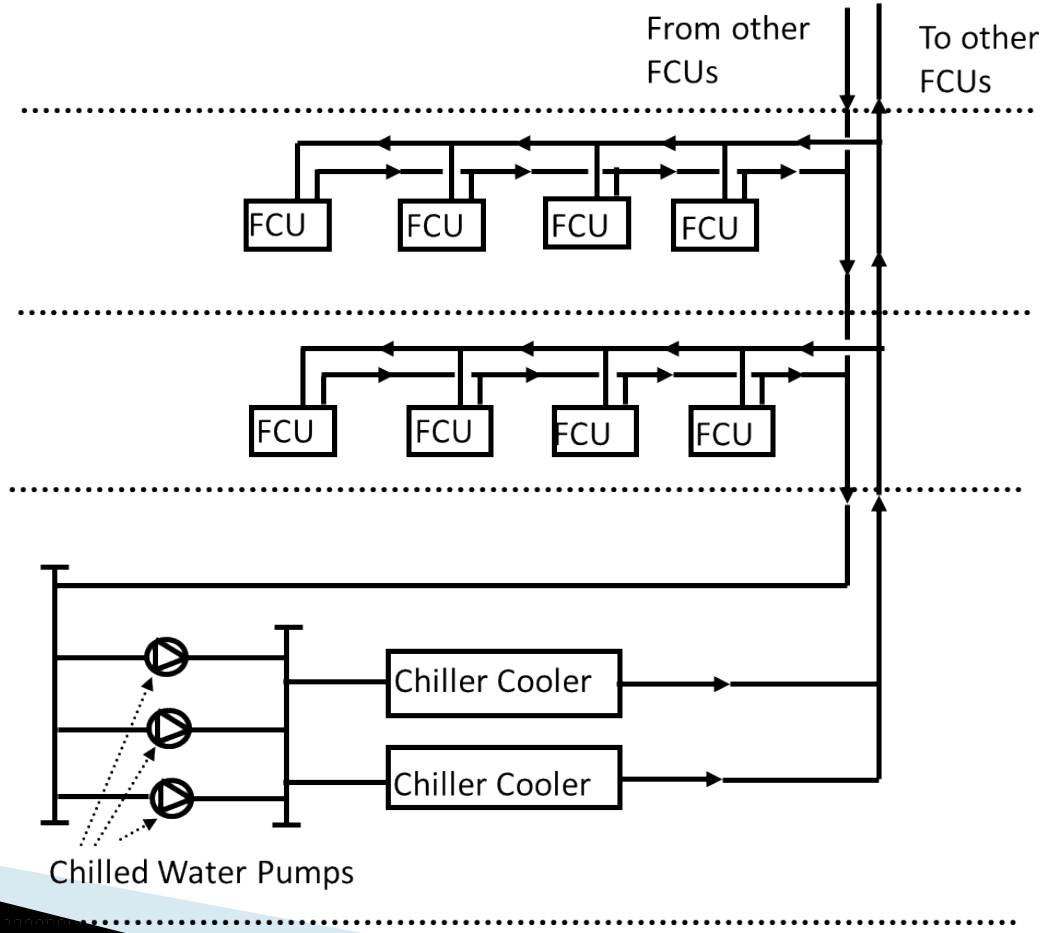
- Standalone basis
- Master and slave arrangement.
- Thermal or mechanical, digital wired and digital wireless with wireless thermostats.



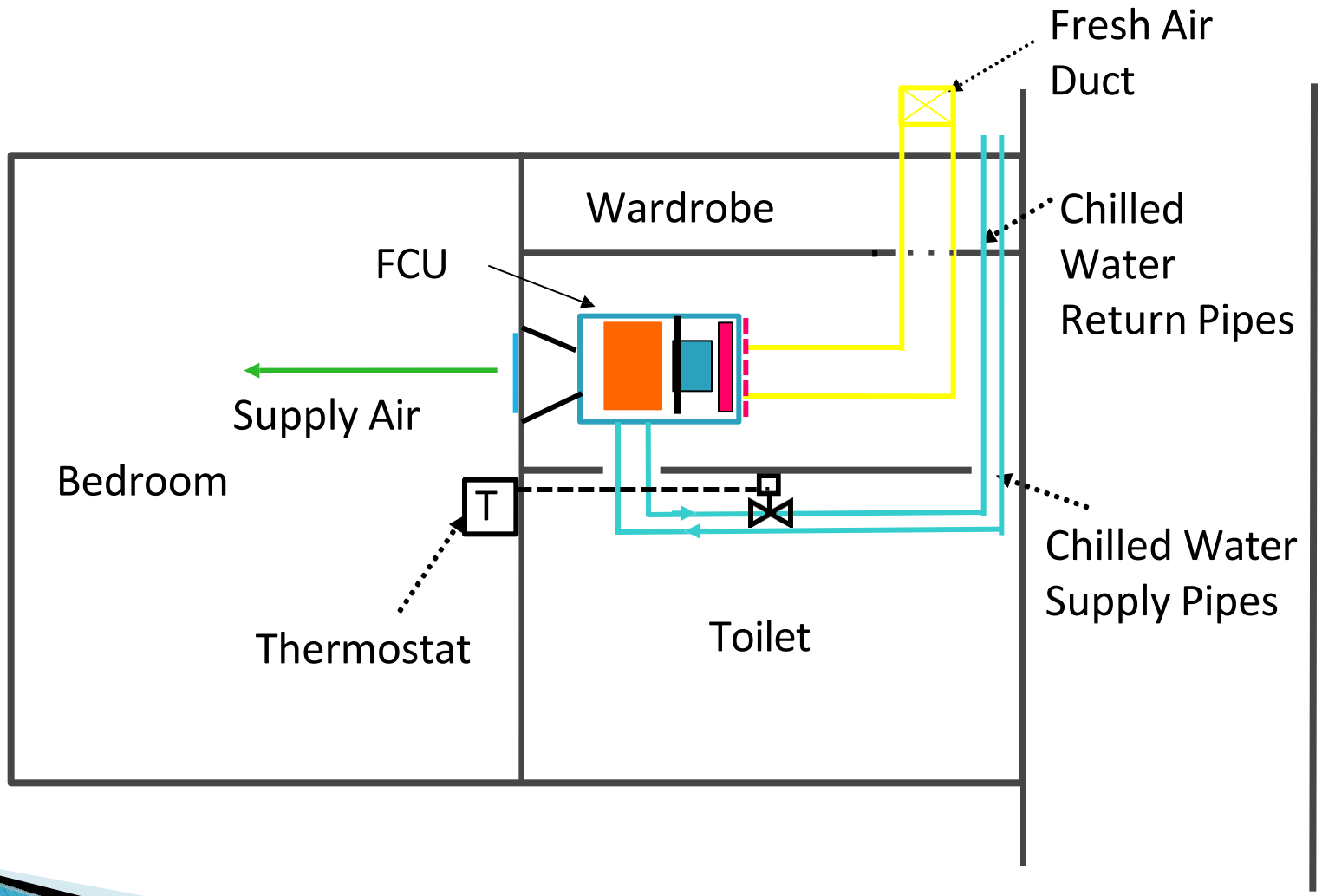
# FAN COIL UNIT SYSTEMS

# System Description

- Chilled water circulation plant (air-cooled or water-cooled) that supplies chilled water to fan coil units (FCU)
- Untreated or Pre-treated fresh air for FCUs.

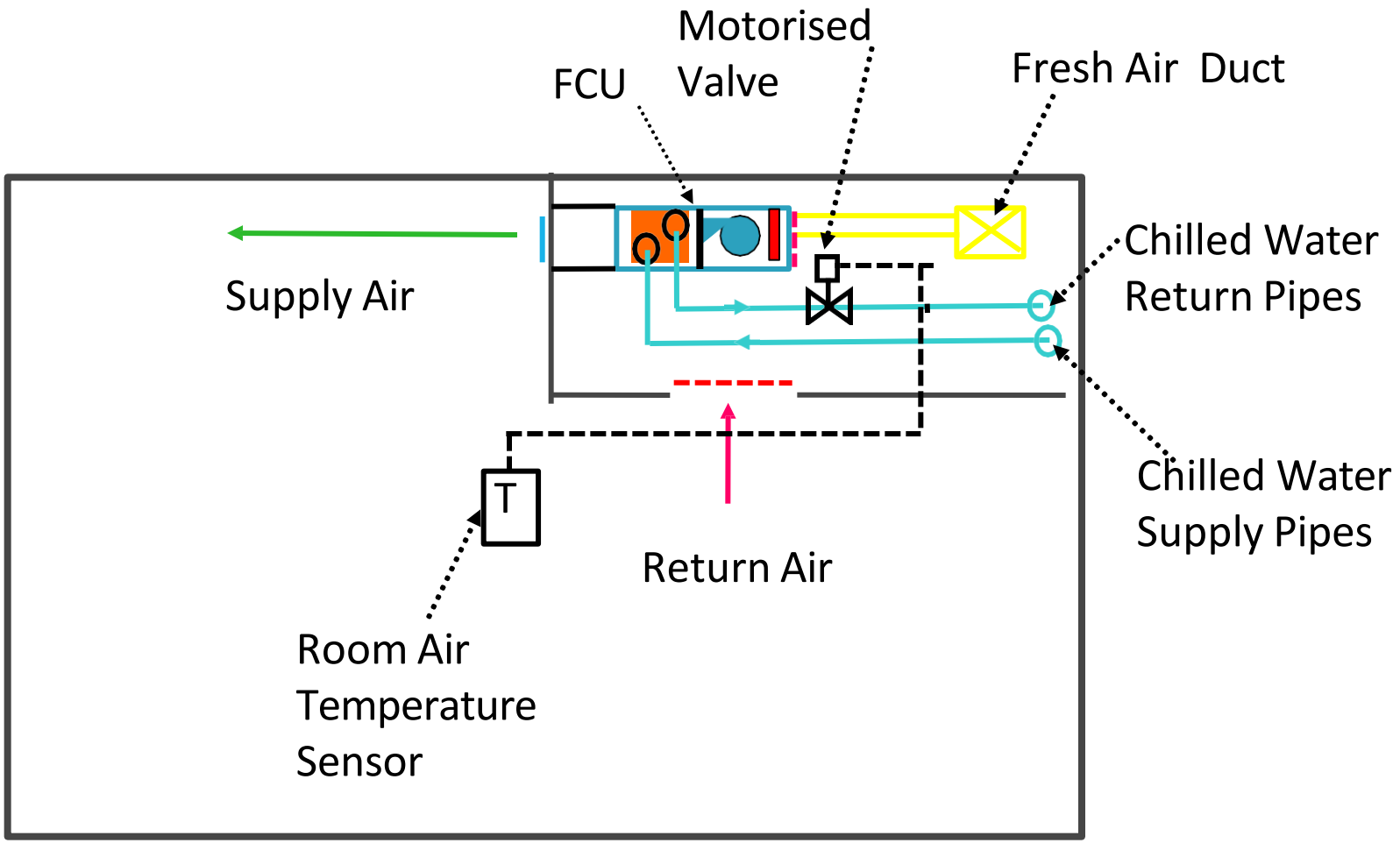


# Fan Coil Unit System




Video

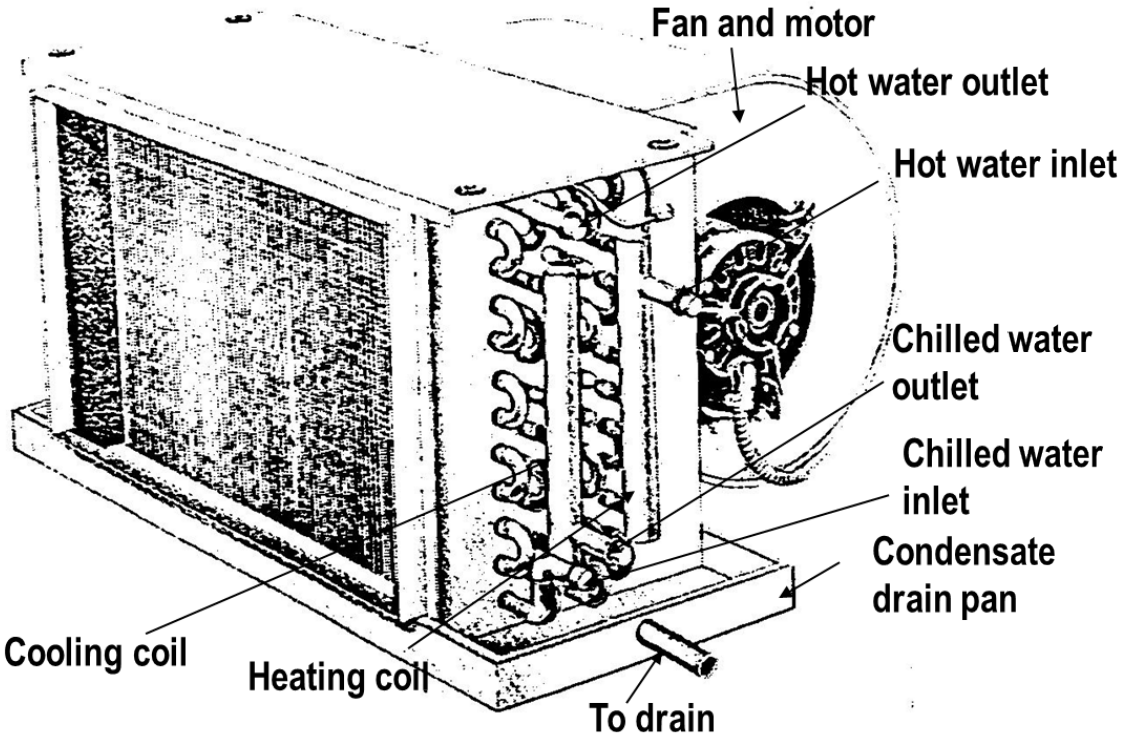
# Fan Coil Unit System




Video

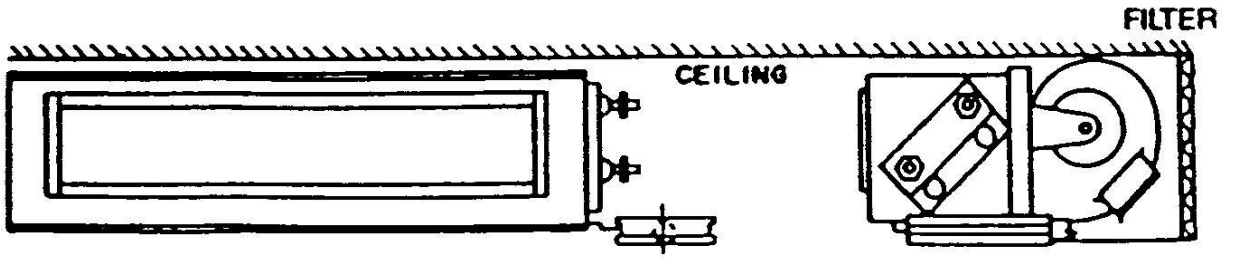
# FCU Construction

- Consists of a finned tube water coil, drip pan, fan and motor assembly, filter, cabinet and controls.

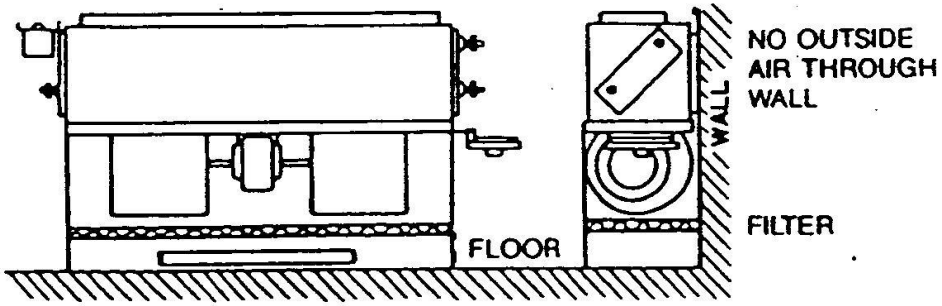


# Ceiling Mounted FCU

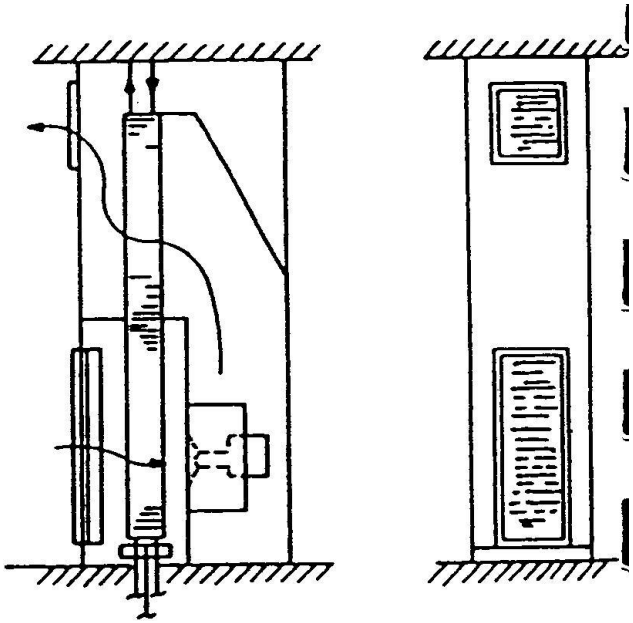
- Available for horizontal (ceiling), vertical (floor) or chase-enclosed (column) mounting; and for exposed or concealed installation.




HORIZONTAL MODEL - CEILING MOUNTED




VERTICAL MODEL - FLOOR MOUNTED



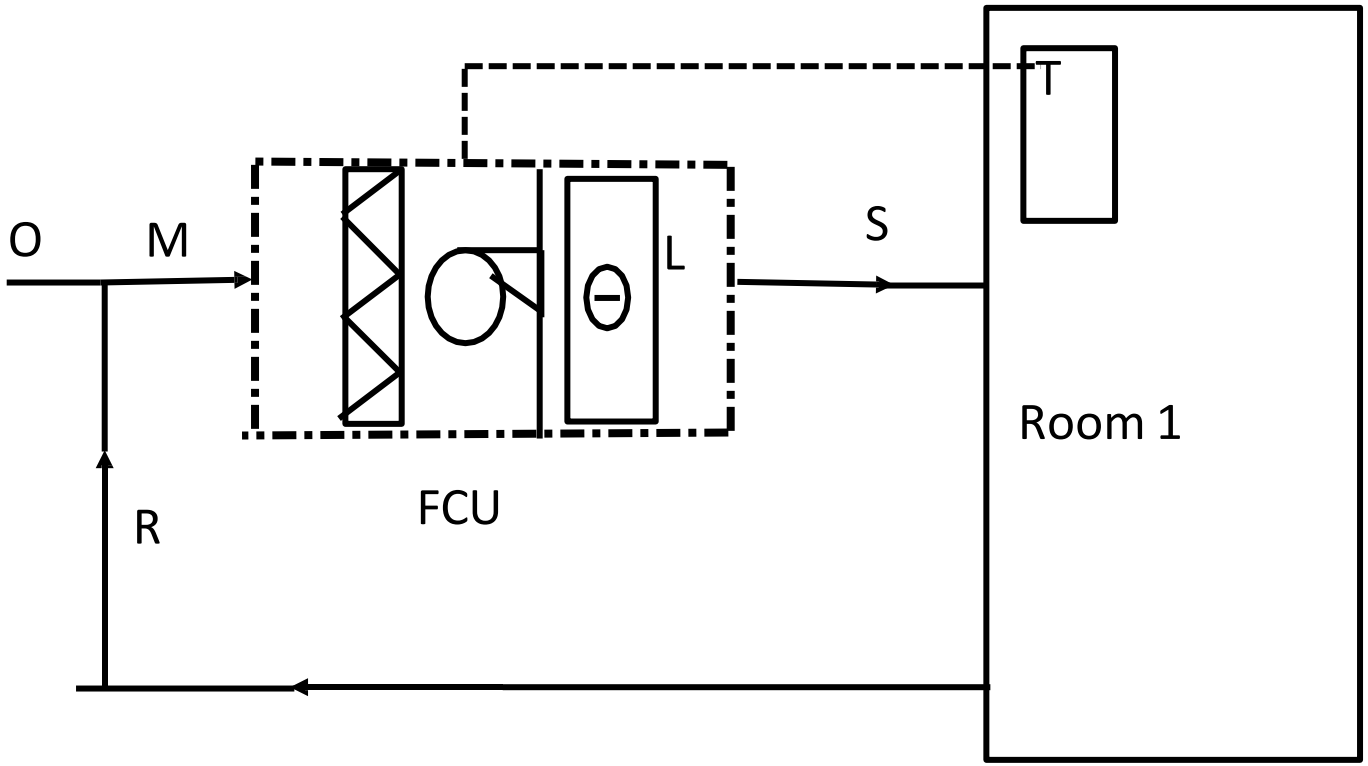
CHASE-ENCLOSED MODEL WITH FINNED-RISER HEAT EXCHANGER

(Place units back-to-back—one for each room. Do not use one unit for two rooms.)

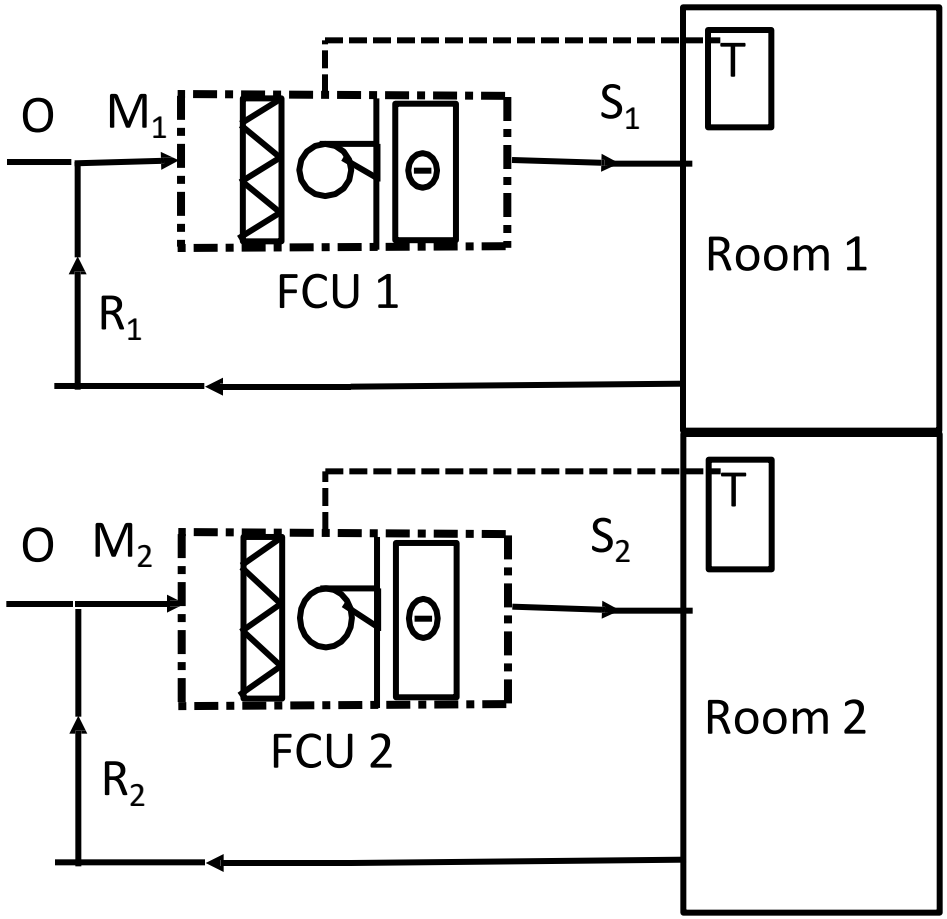
# Outdoor Air for Ventilation

- Untreated outoor Air
- Treated outdoor air from central AHU supplied independently to space
- Treated outdoor air from central AHU supplied to FCUs return air plenum

# Untreated Outdoor Air

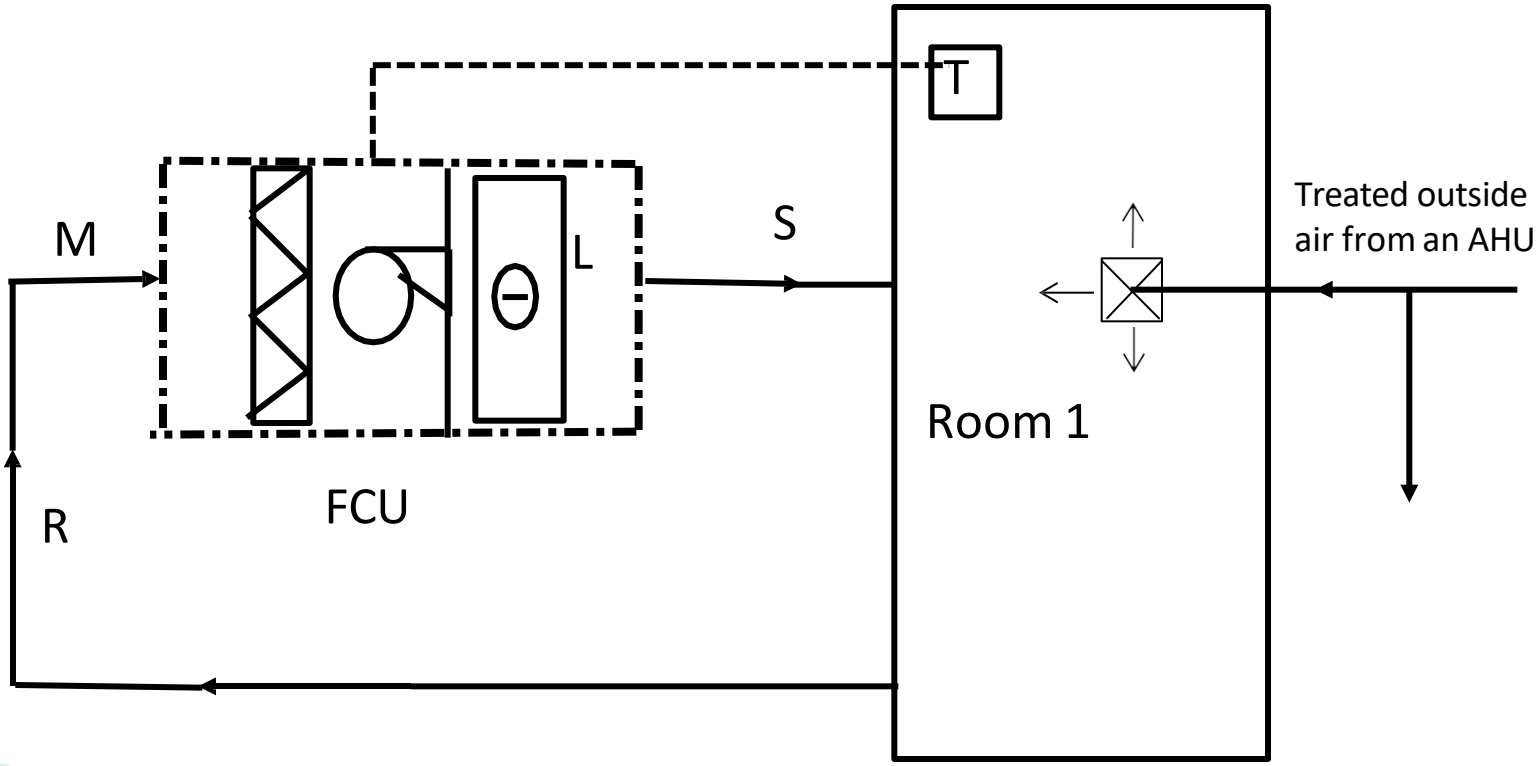


# Untreated Outdoor Air



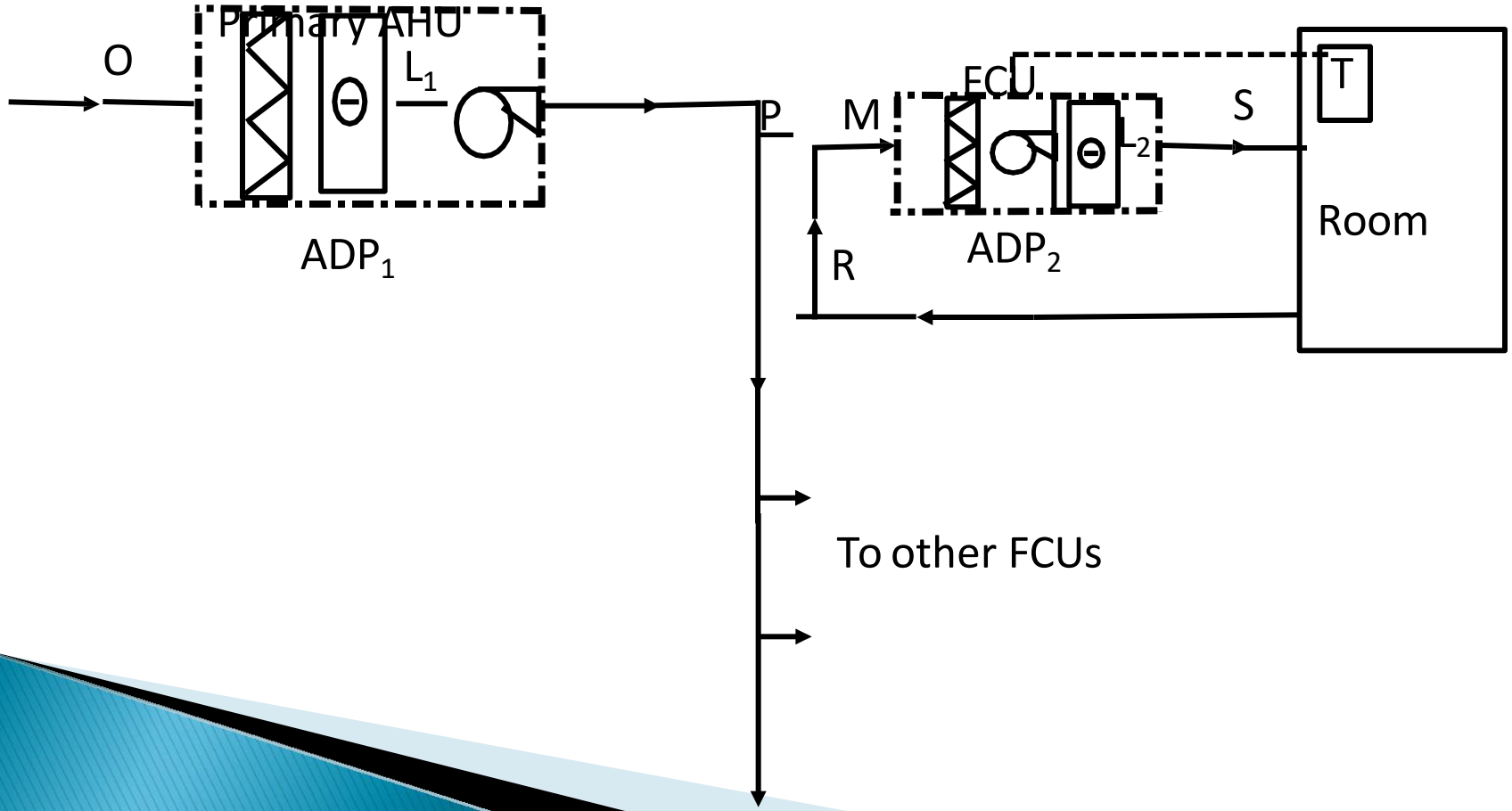
# Treated Outdoor Air – Central AHU

- Treated outside air from a central AHU introduced independently through air terminal devices

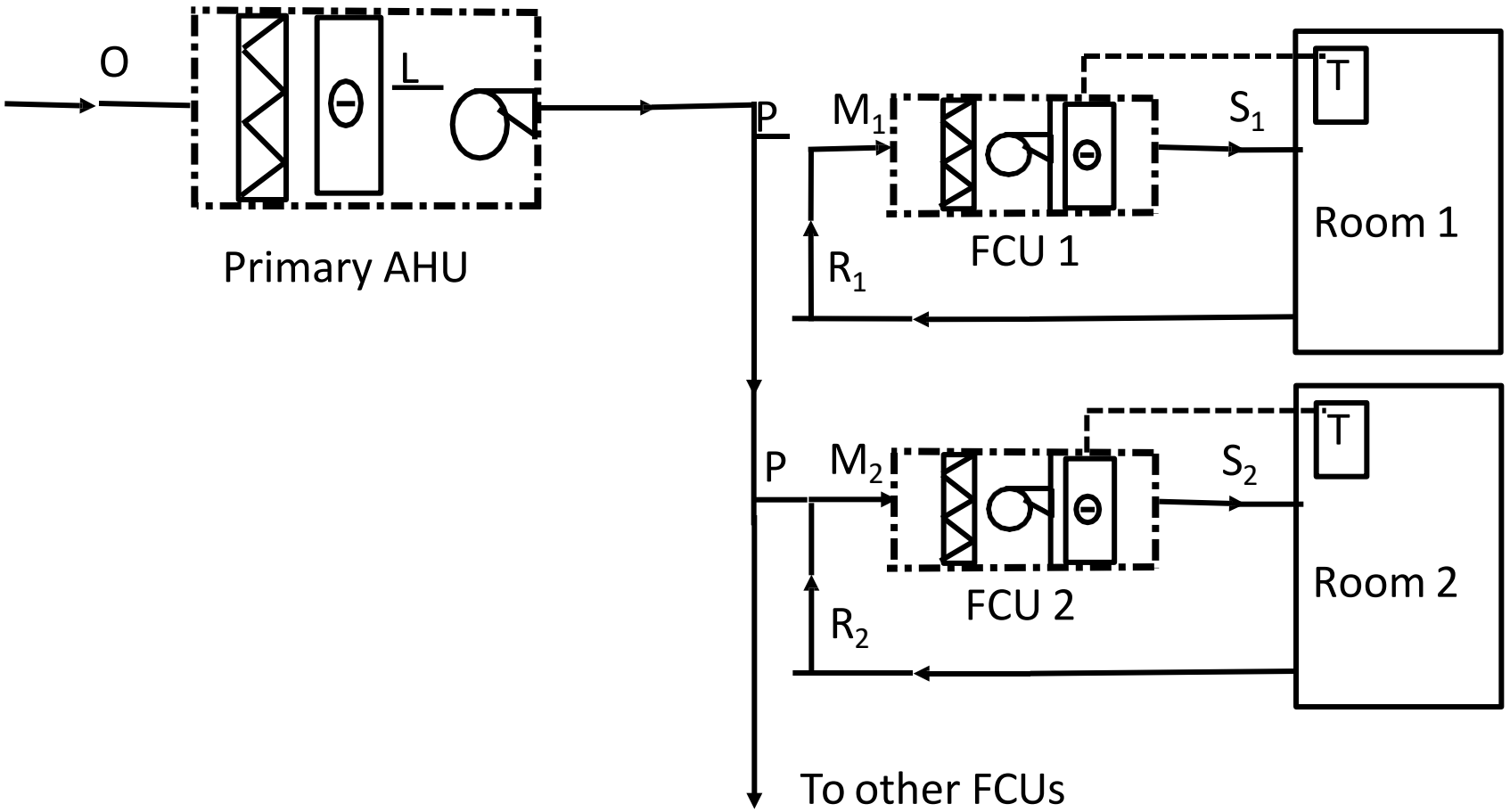


# Treated Outdoor Air – Primary AHU

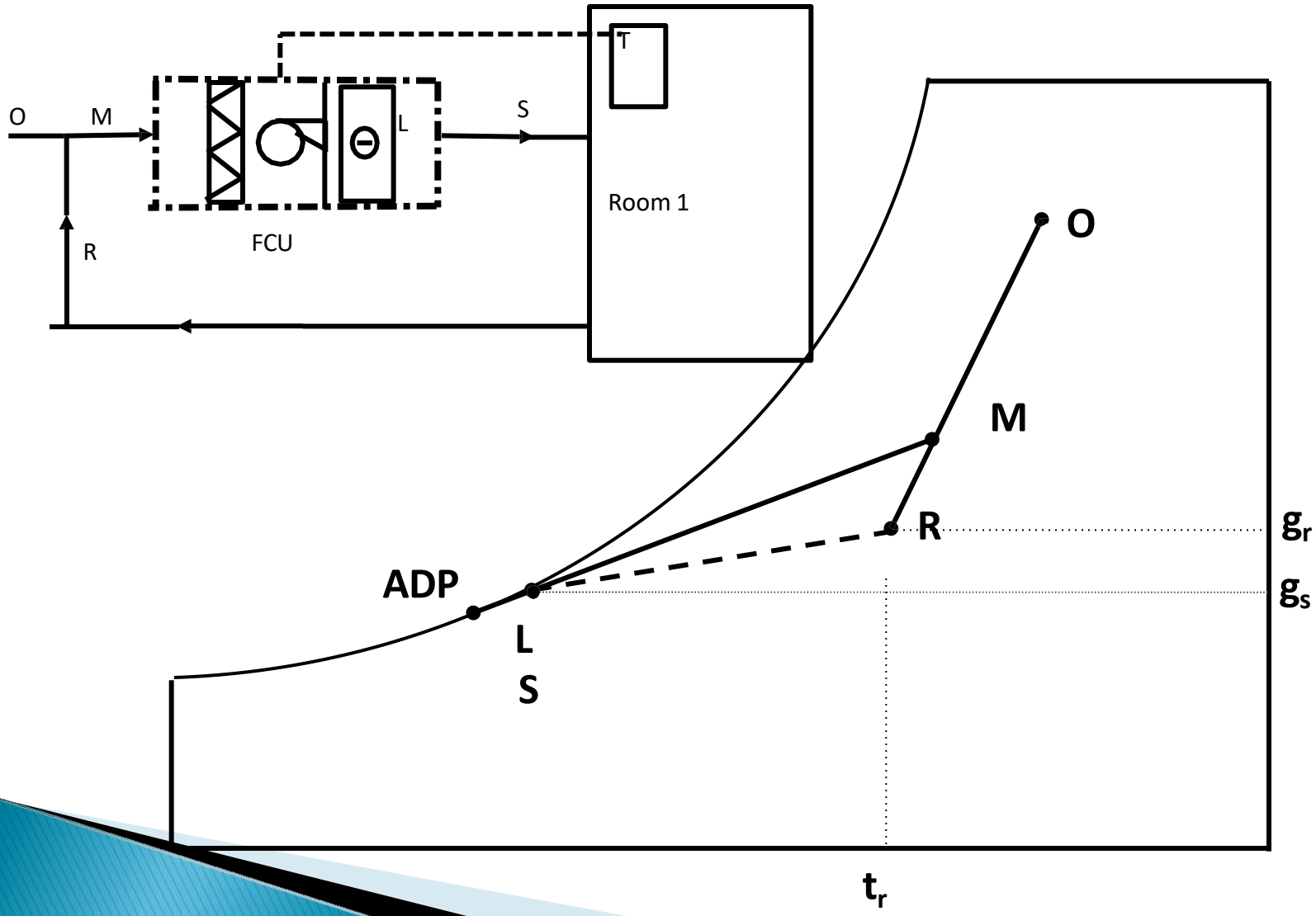
- Treated outside air from a primary AHU and distribute to the FCUs' return air plenums.



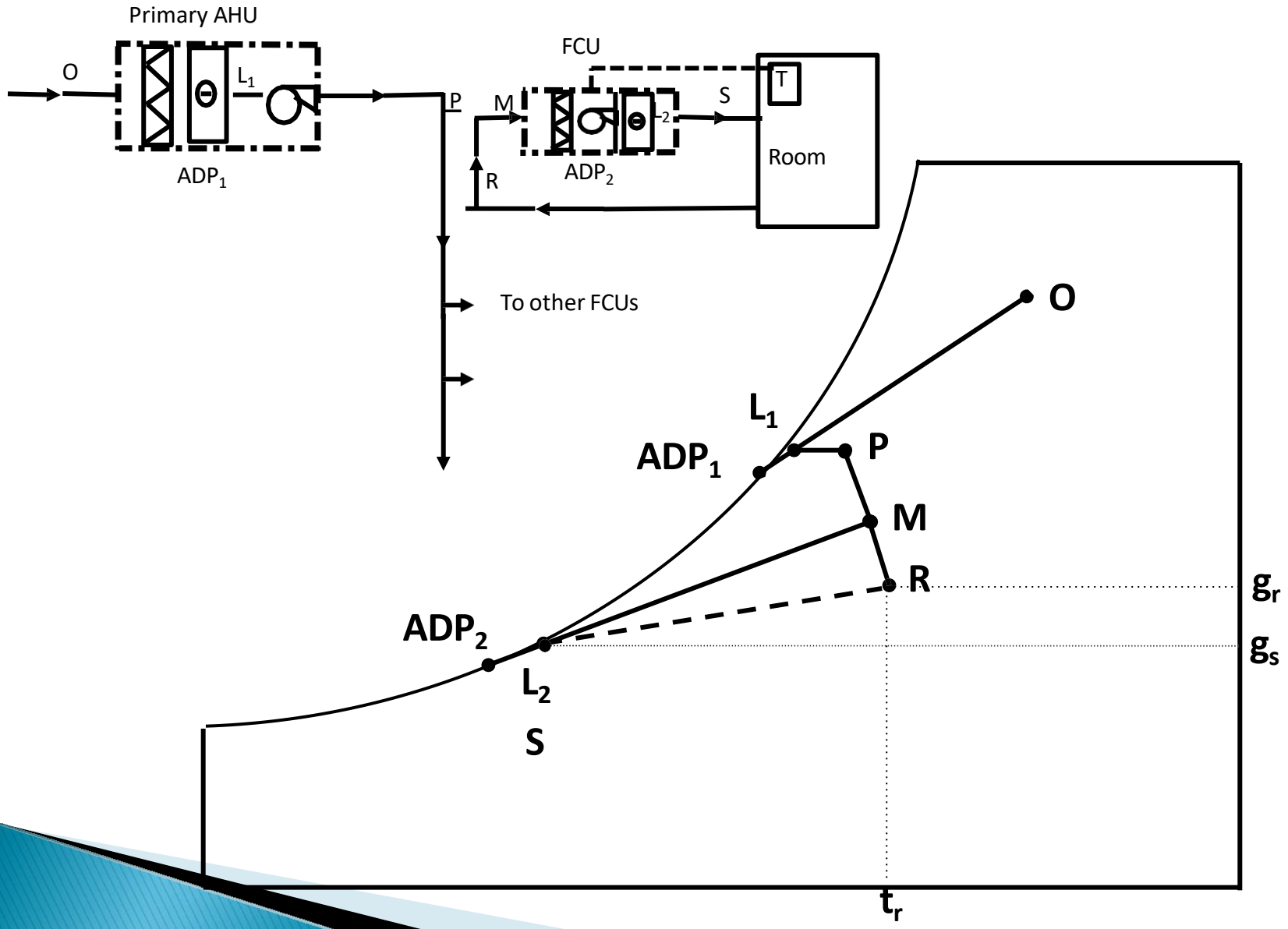
# Treated Outdoor Air – Primary AHU



# Untreated Outside Air



# Treated Outdoor Air – Primary AHU



# Example 5.4

Fifty identical guestrooms in a hotel are served by a fan coil unit air-conditioning system. The primary air handling unit (AHU) located on the roof treats outside air and distributes it equally to the return air plenums of the fan coil units (FCUs) located in the guestrooms.

At the primary AHU, outdoor air enters the cooling coil at 33°C dry bulb temperature and 27°C wet bulb temperature and leaves at 17°C dry bulb temperature and 16.5°C wet bulb temperature. Heat gains due to ducts and primary AHU fan results in a 1K rise in air temperature.

# Example 5.4

The mass ratio of the treated outdoor air to supply air is 0.15 at each fan coil unit.

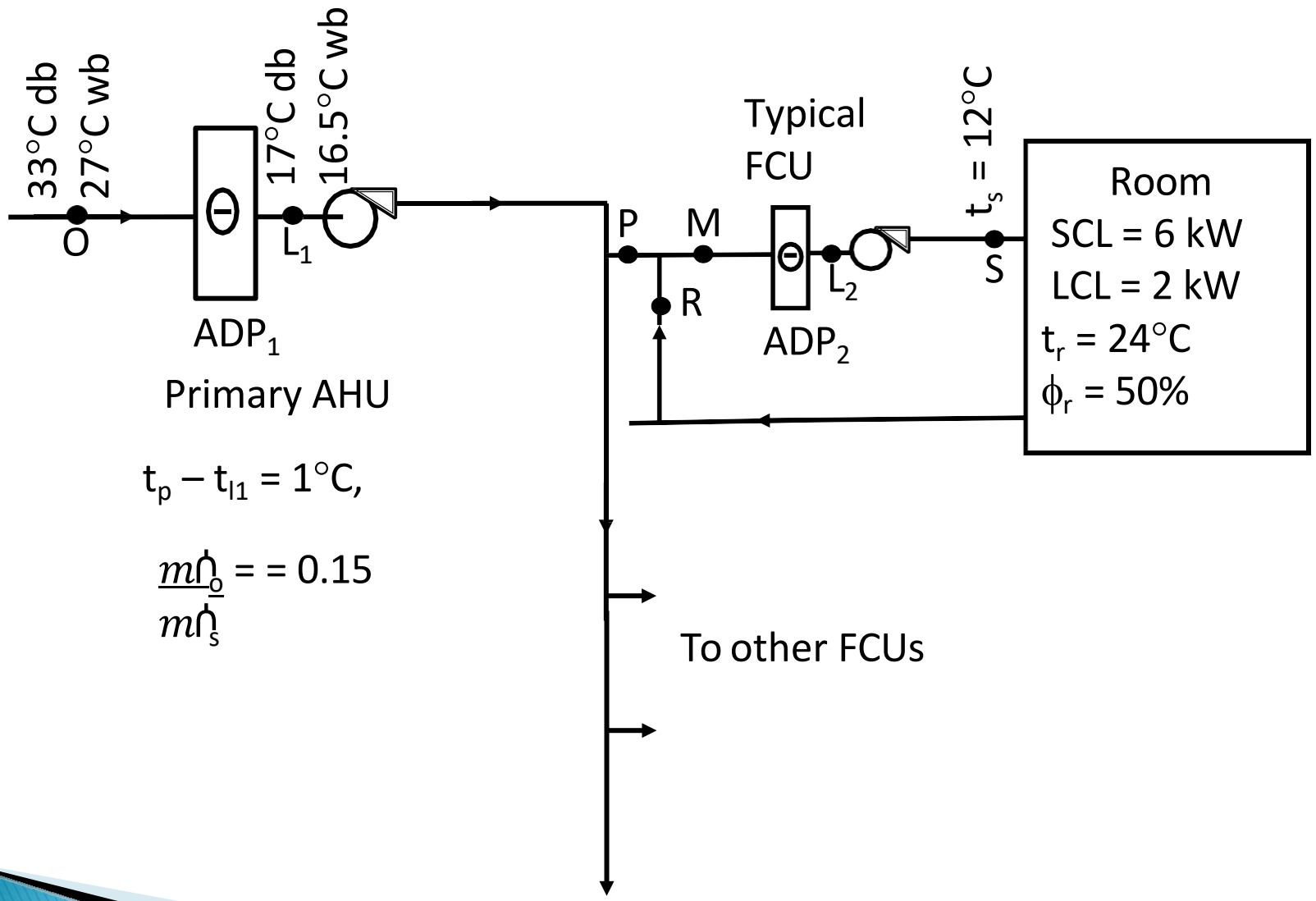
The fan coil units are sized to maintain the guestrooms at 24°C dry bulb temperature and 50% saturation. Each room has sensible and latent cooling loads of 6 kW and 2 kW respectively. The supply air temperature is 12°C. Fan and duct heat gains at the fan coil unit can be assumed to be negligible.

# Example 5.4

Determine:

(i) the apparatus dew point and bypass factor of the primary AHU cooling coil,
(ii) on-coil temperature of air at the FCU,
(iii) mass flow rate of supply air at the FCU,
(iv) moisture content of supply air and
(v) cooling capacities of the primary AHU cooling coil and each FCU cooling coil.

# Example 5.4


33°C db
27°C wb
O
17°C db
16.5°C wb
L1
ADP1
Primary AHU
Typical
FCU
P
M
R
L2
ADP2
ts = 12°C
S
Room
SCL = 6 kW
LCL = 2 kW
tr = 24°C
ϕr = 50%
tp – tl1 = 1°C,
= = 0.15
To other FCUs

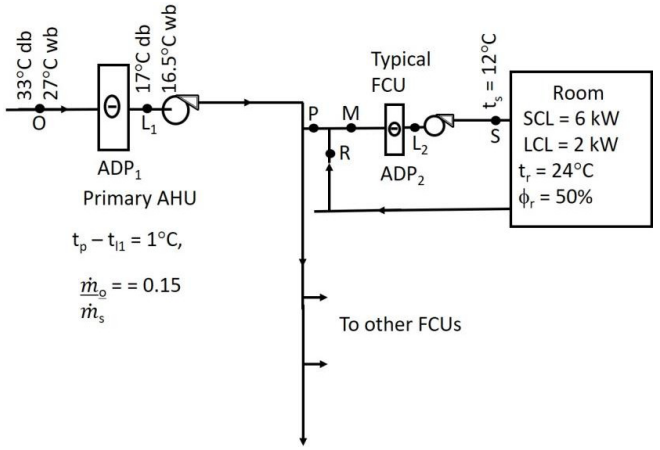
# Example 5.4

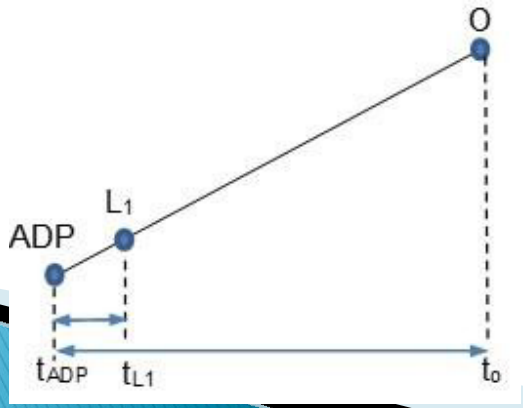
(i) Primary AHU $ADP_1$ and $BF_1$

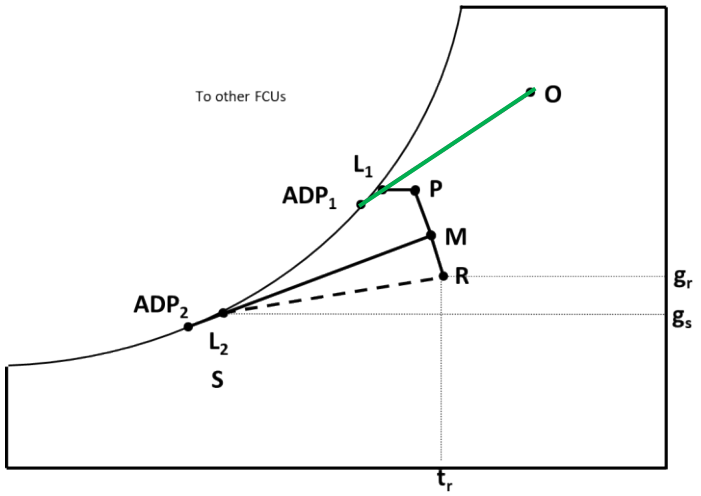
Plot points O and $L_1$.
Draw line to join Point O and $L_1$. Extend line to intersect with 100% saturation line to obtain $ADP_1$.

From plot, $ADP_1$= 14°C

$$BF_1 = \frac{t_{L1} - t_{ADP1}}{t_o - t_{ADP1}} = \frac{17 - 14}{33 - 14} = 0.158$$



# Example 5.4

(ii) On-coil air temperature at each typical FCU, $t_m$

$t_p - t_{l1} = 1.0\ K \Longrightarrow t_p = t_{l1} + 1$

$= 17 + 1 = 18°C$

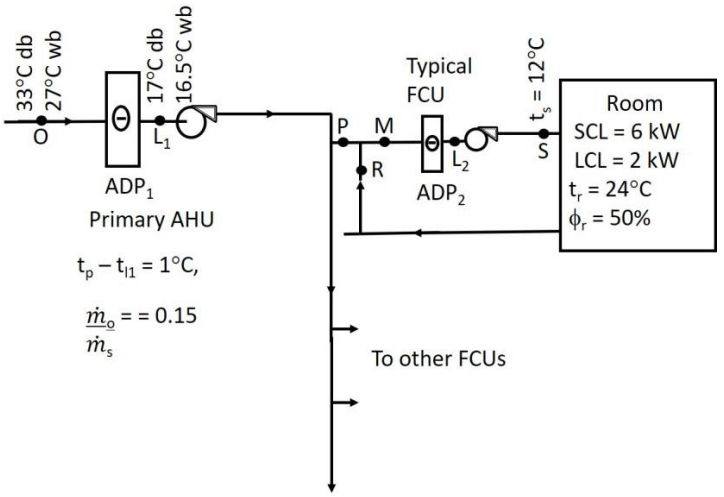
Plot $t_p = 18°C$

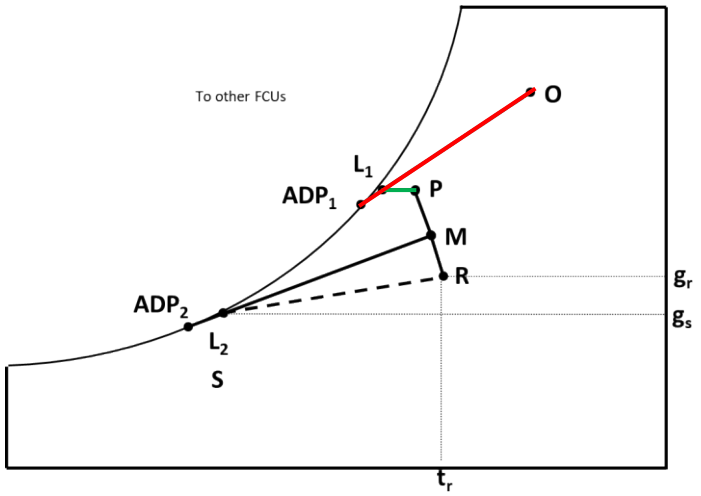
Plot a constant moisture content line through Point $L_1$ to intersect with $t_p$ to obtain Point P.

$$\frac{\dot{m}_o}{\dot{m}_s} = \frac{t_m - t_r}{t_p - t_r}$$

$$t_m = t_r + \frac{\dot{m}_o}{\dot{m}_s}(t_p - t_r)$$

$= 24 + 0.15(18 - 24) = 23.1°C$



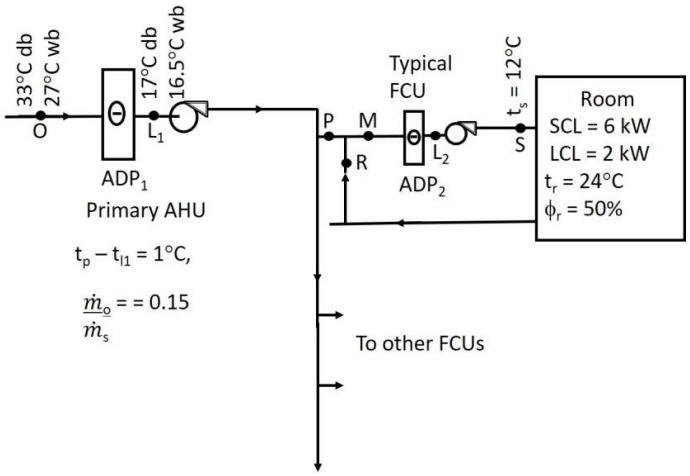



# Example 5.4

Plot Point R.

Draw line to joint Point R and Point P.

Plot $t_m$= 23.1°C to intersect with line joining R to P to obtain point M.

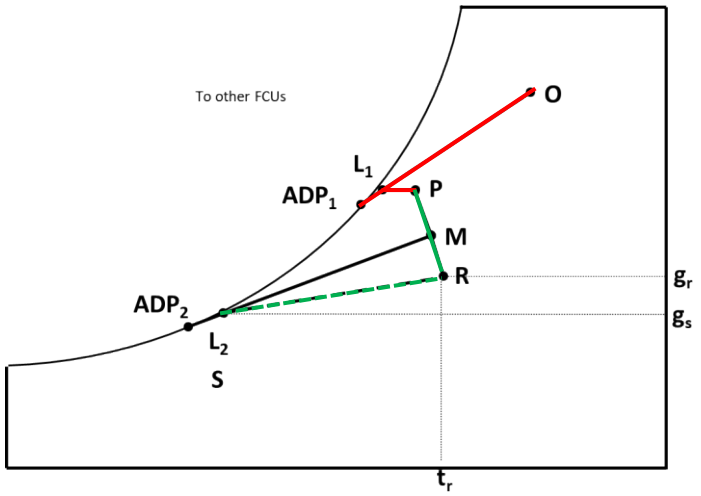



(iii) Supply air mass flow rate

$$SCL = \dot{m}_s \times c_{p,a} \times (t_r - t_s)$$

$$\dot{m}_s = \frac{SCL}{c_{p,a}(t_r - t_s)} = \frac{6 \times 10^3}{1.012 \times 10^3 (24 - 12)}.$$

$$= 0.494 \text{ kg/s}$$




(iv) $g_s$

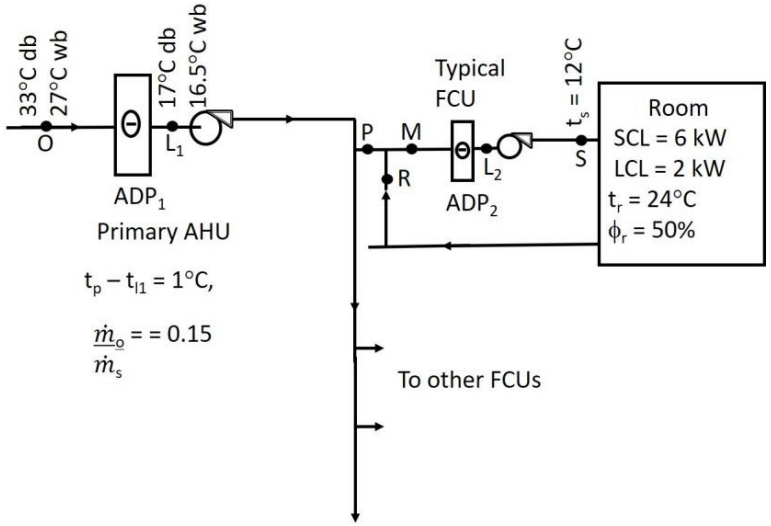
$$RSHF = \frac{SCL}{SCL + LCL} = \frac{6}{6 + 2} = 0.75$$

# Example 5.4

Plot $t_s$ = 12°C.

Plot RSHF through R to intersect with $t_s$ to obtain point S and hence $L_2$

From plot, $g_s$ = 0.0078 kg/kg




(v) <u>Cooling Capacities of FCU and primary AHU</u>

Cooling capacity of FCU = $\dot{m}_s$ ($h_m - h_{L2}$)

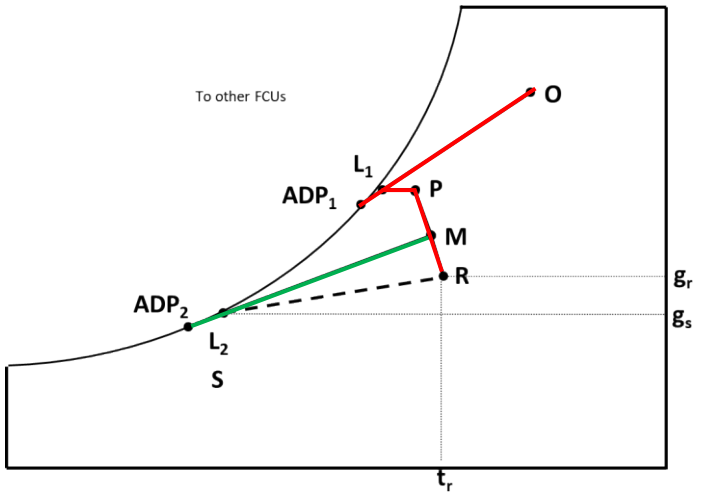
From chart, $h_m$ = 48.0 kJ/kg,

$h_{L2}$ = 31.5kJ/kg

Cooling capacity of FCU

= 0.494 (48 – 31.5)

= 8.151 kW



# Example 5.4

Cooling capacity of primary AHU = $\Sigma(\dot{m}_o)\,(h_o - h_{l1})$

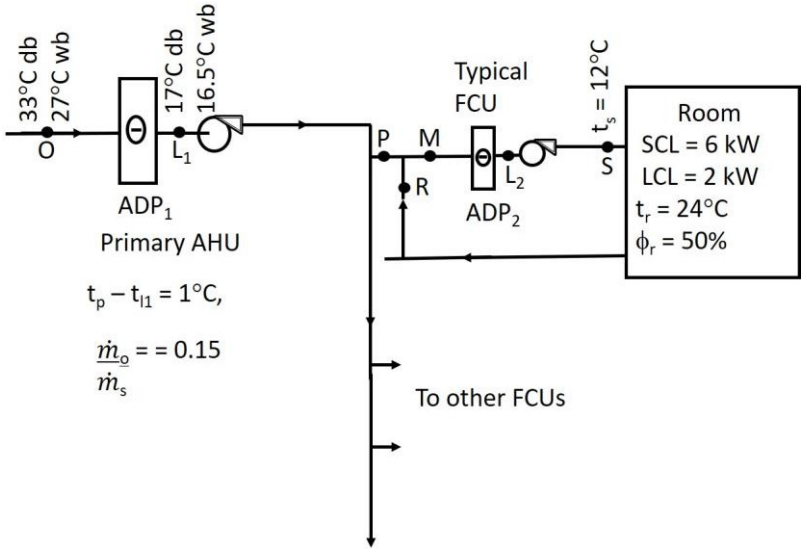
From chart, $h_o$ = 84.5 kJ/kg, $h_{l1}$ = 46.5 kJ/kg

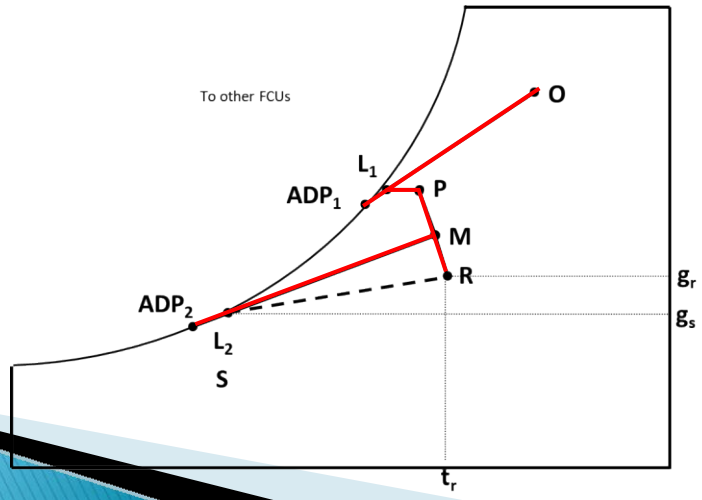
$\dot{m}_o = 0.15\ \dot{m}_s = 0.15 \times 0.494 = 0.0741$ kg/s

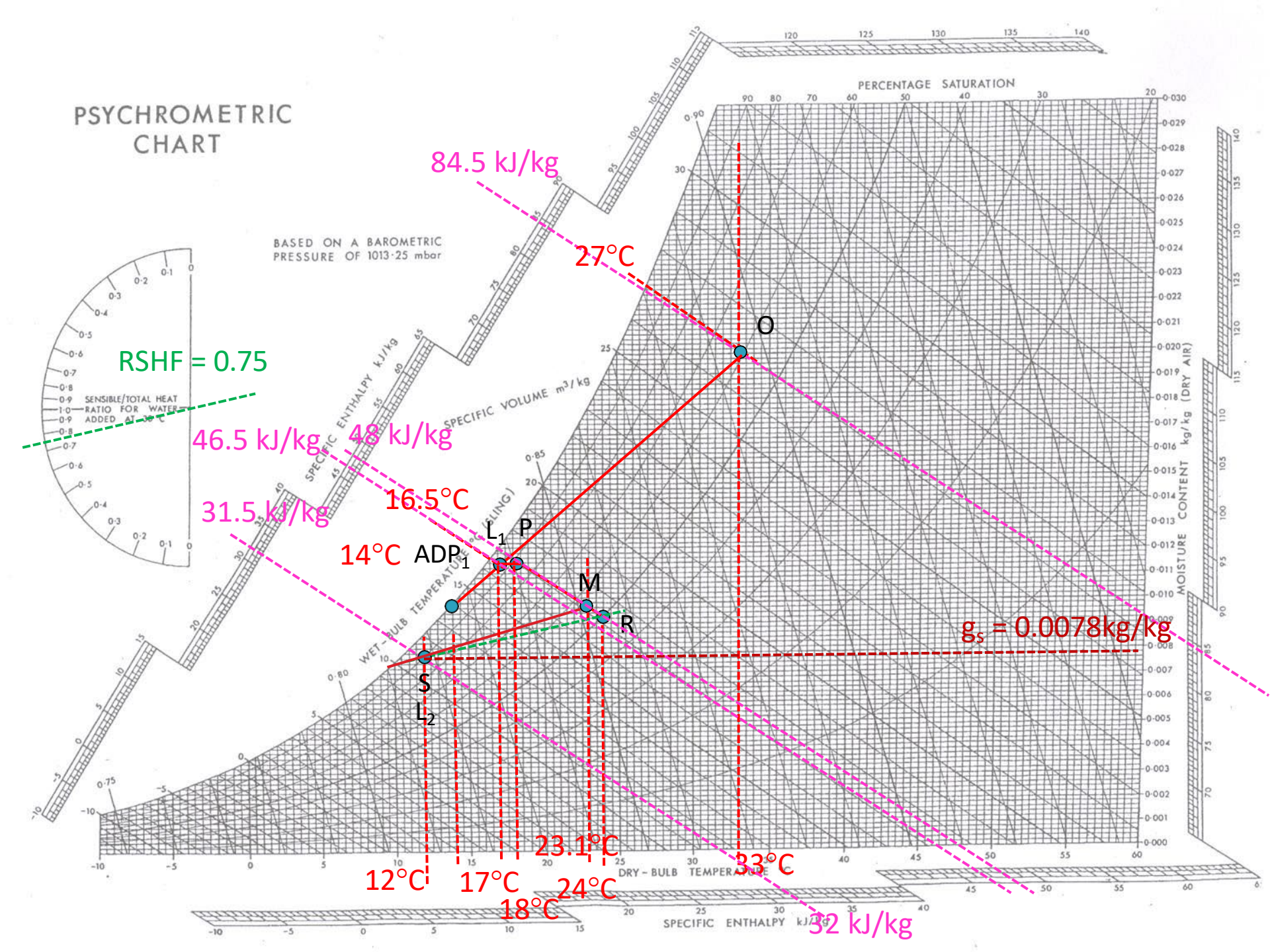
Cooling capacity of primary AHU

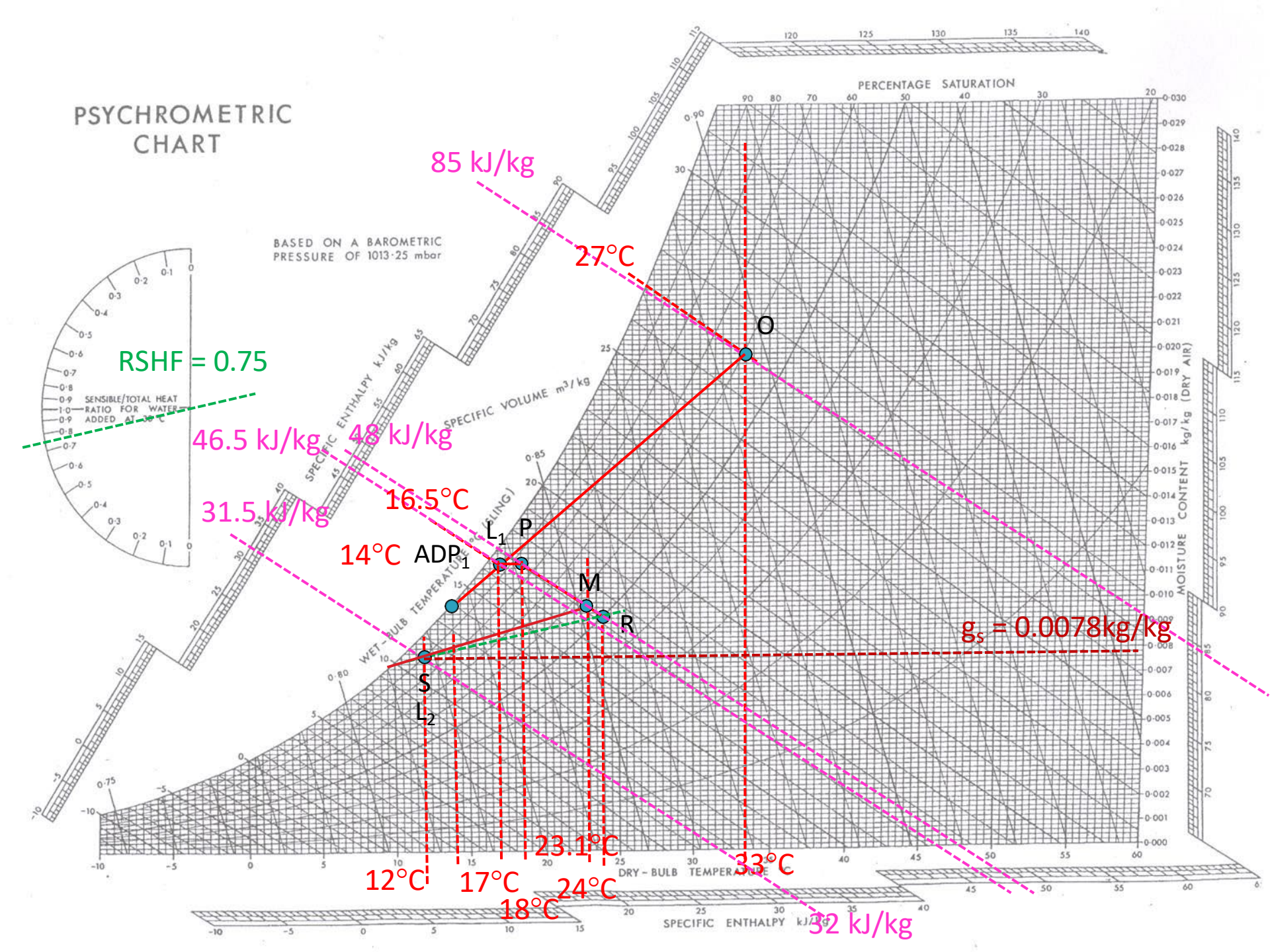
= 50 x 0.0741 (84.5 – 46.5)

= 140.79 kW




PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SPECIFIC ENTHALPY kJ/kg
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE (SLING)
DRY-BULB TEMPERATURE °C
MOISTURE CONTENT kg/kg (DRY AIR)
SPECIFIC ENTHALPY kJ/kg
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 0 °C
84.5 kJ/kg
27°C
O
RSHF = 0.75
46.5 kJ/kg
48 kJ/kg
31.5 kJ/kg
16.5°C
L1
P
14°C
ADP1
M
R
gs = 0.0078kg/kg
S
L2
12°C
17°C
23.1°C
24°C
18°C
33°C
32 kJ/kg


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
PERCENTAGE SATURATION
SPECIFIC ENTHALPY kJ/kg
SPECIFIC VOLUME m3/kg
WET-BULB TEMPERATURE (SLING)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
MOISTURE CONTENT kg/kg (DRY AIR)
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 0°C
RSHF = 0.75
85 kJ/kg
27°C
O
46.5 kJ/kg
48 kJ/kg
31.5 kJ/kg
16.5°C
14°C
ADP1
L1
P
M
R
S
L2
gs = 0.0078kg/kg
12°C
17°C
18°C
23.1°C
24°C
33°C
32 kJ/kg

# Advantages of Using FCU

- Delivery system (more piping vs less ductwork) requires less building space and smaller or no central fan room.
- Can provide temperature control for each room.
- Each room is a separate air system, therefore no cross-contamination of room air.
- FCUs of unoccupied rooms can be switched off to save energy.
- Easier to install for retrofitting of existing buildings.

# Disadvantages of Using FCU

- More moving parts and air filters - more maintenance.
- Cooling coils difficult to clean - Accessibility
- Each FCU requires a condensate drainpipe. Gradient required by drainpipe tends to lower ceiling level. Pipe also difficult to maintain.
- Maintenance work done in occupied areas.

# Disadvantages of Using FCU

- Limited capability for dehumidification. A primary AHU is needed to treat O.A. for good humidity control.
- Filters are small, low in efficiency and require frequent cleaning to maintain flow rate.
- Fan noise can be a problem especially for areas requiring quiet condition.

# Lesson 9

- For those students who are not sitting for the retest, remote learning and Quiz 5 on Duct Systems have been scheduled on Thursday, 10 Jul 2025
- Retest for students will be held from 7.00 to 9.00pm on Thursday, 10 Jul 2025
- Students who are sitting for the retest will do the remote learning and Quiz 5 on Duct Systems on 11 Jul 2025.